

Policy Spaces as Organizational Schemes: A Semantic Theory of Electoral Competition

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Abstract

The distinctions organizing electoral competition depend on who listens and how political appeals are received. This manuscript develops a semantic model in which an understanding records the representations a voter can access and distinguish, together with their assigned meanings. Encounters specify conflict, credibility, cognitive change, and evaluation. The same addition to a speech can attract one audience, leave another's evaluation unchanged, and antagonize a third through meanings it does not admit.

At a common stance, the analysis derives content sufficient to preserve a specified audience's evaluations of a candidate at a finite election horizon. The construction permits active conflict and feedback under either of two incorporation rules, without requiring convergence. Coalition membership combines the relevant dependencies, and feasible representatives preserving all voter scores preserve electoral payoffs and corresponding pure equilibria.

A five-appeal election classifies pure equilibria by audience shares and rejection cost. Increasing that cost creates equilibrium in some compositions and removes it in others; an open region sustains the same unique equilibrium appeal at every nonnegative cost, despite the absence of a common weakly single-peaked ordering. Communication supplies further routes of relevance: an opponent's objection can become valued content for a supporter. Under expansion on a finite field, understandings and evaluations with a fixed initial encounter record eventually become constant, while public reactions can cycle; revision can sustain recurring semantic and electoral phases. The results locate the politically consequential comparisons within a specified representational field and identify conditions supporting both stability and continuing disagreement.

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Contents

1	Introduction.	1
2	Understandings: representations, stances, meanings.	4
2.1	What representations are—and aren’t.	4
2.2	Epistemic stances.	5
2.3	Semantics.	7
2.4	Understandings.	8
2.5	The evaluative anchor.	12
2.6	From understandings to political encounters.	13
3	Encounters and their evaluation.	14
3.1	Conflict structure.	14
3.2	Source credibility.	15
3.3	Acquiring distinctions.	16
3.4	The signed response.	16
3.5	Direct evaluation.	18
3.6	Parameter dependence.	20
4	Appeals and electoral competition.	21
4.1	The electoral setup.	22
4.2	Candidate strategy.	22
4.3	Combining content.	23
4.4	The horizon score.	25
5	Audience and coalition relevance.	26
5.1	The audience table.	27
5.2	Direct relevance.	27
5.3	Public expression.	28
5.4	Relevance through communication.	30
5.5	Feasible reduction.	31
6	Electoral competition among appeals.	33
6.1	The running election.	33
6.2	Isolated audiences.	35
6.3	Majorities and minorities.	36
6.4	Benchmarks.	37

7	Communication.	39
7.1	The augmented state and accumulation.	40
7.2	Settling with standing disagreement.	41
7.3	Recurrence.	42
7.4	Revision.	43
8	Empirical interpretation.	43
8.1	Response components as measurement tasks.	43
8.2	Relevance as a testable invariance.	44
8.3	Combined appeals and observed reactions.	45
8.4	Majority priorities.	45
8.5	Scope and identification.	46
9	Extensions.	46
9.1	Field variation and differential access.	46
9.2	The legislative benchmark.	47
9.3	Source effects.	47
9.4	Incorporation choices.	48
10	Conclusion.	48
	References	50
A	Supporting results and proofs.	52
A.1	Understandings.	52
A.2	Encounters and their evaluation.	53
A.3	Appeals and the electoral game.	55
A.4	Audience and coalition relevance.	58
A.5	Electoral competition among appeals.	60
A.6	Communication.	64
A.7	Extensions.	69

1 Introduction.

Consider a candidate addressing three voters who understand immigration differently. One regards enforcement as effective and desirable, paths to citizenship as unworkable, and immigration’s economic impact as harmful. Another regards enforcement as effective but cruel and citizenship as viable and desirable. A third attaches evaluative meanings to enforcement and citizenship without attaching factual meanings to either. The topics are familiar to all three, but the meanings they assign to them differ.

The candidate supplements a factual account with an evaluative appeal opposing enforcement and welcoming immigration. In the running example developed below, this addition improves the third voter’s evaluation and leaves the second’s unchanged. It lowers the first’s when rejection carries a positive cost. Neither of the first two voters admits the added evaluative meanings; only the first holds meanings with which they conflict. The candidate’s problem therefore involves both the audience the addition attracts and the audience it antagonizes. An appeal can matter through content a listener refuses to take up.

How do voters’ understandings and their responses to political communication organize electoral competition? This manuscript argues that the politically consequential differences between appeals depend on who receives them, how they are received, and which other people’s reactions reach the audience. It derives content sufficient to preserve audience evaluations and uses those evaluations to analyze competition. The running election shows that audience composition and the cost of rejection can support stable appeals, create equilibrium, or remove it, even without a common weakly single-peaked ordering of the alternatives. The politically operative space is thus an object of analysis: which distinctions a political comparison must retain depends on its audience.

This question connects two concerns already present in electoral theory. Downs’s (1957, pp. 115–116) spatial construction assumes an agreed left–right ordering and single-peaked voter preferences. His broader account also examines uncertainty, ideological shortcuts, and strategic ambiguity, including appeals interpreted differently by different constituencies (Downs, 1957, pp. 98–100, 135–136). A common-location spatial benchmark specifies how platforms are understood before comparing their proximity to voters’ ideal points. Interpretive variation raises the question of how those comparisons arise. Stokes (1963) made the organization of electoral choice central to his critique of spatial models, questioning assumptions about dimensions, stable evaluation, ordered alternatives, and a common frame of reference. He distinguished position from valence issues and proposed treating relevant cognitive features as variables (Stokes, 1963, pp. 372–377). Here, voters’ understandings and encounters supply explicit inputs to the organization of electoral choice.

Behavioral work gives substance to the differences those inputs can represent. Converse (1964) reported weaker associations among issue positions in an electorate sample than among congressional candidates, instability on several repeated policy questions, and greater stability among respondents with particular interest in the issue. His account of issue publics directs attention to variation in what citizens engage with and organize politically. Zaller (1992, chap. 3) links political information acquisition to the formation of opinion statements. Framing raises a related distinction between offered and adopted meaning: Entman (1993, pp. 52–53) separates interpretations promoted by a text from those a receiver adopts, while Lakoff (1996) proposes contrasting family-morality accounts of political disagreement. These concerns motivate specifying reception as well as the content of an appeal.¹

¹Related evidence includes partisan differences in retrospective economic perceptions (Bartels, 2002), selective argument evaluation and search in two student studies of affirmative action and gun control (Taber and Lodge, 2006), and a long-run rise through 2016 in affective polarization between American party identifiers (Iyengar et al., 2019). These findings motivate attention to reception and evaluation; identifying the model’s states and parameters requires separate measurement.

The formal construction begins with an *understanding*: the representations an agent can access and distinguish, together with the meanings attached to them. Reception translates offered content into the listener’s categories. An encounter assesses conflict, filters meanings by source credibility, and selects incorporation, inaction, or rejection. Incorporation can make new distinctions available and change meanings; inaction and rejection preserve the current understanding. The main incorporation rule, expansion, retains prior meanings alongside admitted content. A revision alternative permits the replacement of conflicting prior meanings. Evaluation then combines recognition, valued cognitive change, and a charge for rejection. An evaluative anchor represents content the listener values; the appeal that best addresses her also depends on her current understanding, the source, and the feasible menu. These ingredients let a familiar speech affirm a listener without changing her understanding, and let another supply a distinction or meaning she values.

The argument makes three claims. First, listeners determine which content differences affect their responses. Credibility alone does not identify those differences: excluded content can still select rejection through conflict with the listener’s prior meanings. The opening interaction gives this mechanism a concrete form. Across the running example’s five appeals, the second voter distinguishes factual alternatives while treating their evaluative accompaniment as irrelevant to her score. The third distinguishes that accompaniment while treating the factual alternatives alike. Section 5.1 displays their responses together. The categories and meanings through which they receive the speeches explain why their political comparisons differ. For direct encounters under expansion at a common stance, the combination result separates the recognition and valued uptake gained by adding content from the losses due to blocked incorporation or a rejection charge.

Second, combining audiences changes the distinctions needed to represent competition. The second and third voters jointly distinguish every appeal on the running menu, although each alone treats some appeals alike. Theorem 5.6 gives a general sufficient-content construction at a common stance. For a specified candidate, audience, and finite election horizon, it derives speech coordinates on which agreement guarantees identical audience evaluations. The coordinates protect valued content, possible conflict with meanings participants hold or may acquire, and inputs to reports that can reach the audience before voting. The theorem permits both incorporation rules, active conflict, and feedback; it requires no convergence. It explains why a difference can become relevant to one audience through its effects on another.

Adding listeners weakly refines the exact partition of appeals by their joint evaluations. Extending communication weakly enlarges the sufficient set when its bounds and report reads are held fixed, although the exact response partition need not refine as voting is delayed. The sufficient set can contain redundant coordinates, and it need not be smaller than the available content. Its strategic use comes from preserving responses: feasible representatives preserving every voter’s score for every candidate yield the same payoffs and corresponding pure equilibria. Coalition membership determines whose distinctions enter this comparison; population weights determine how their votes count. The available menu matters as well: Sniderman and Bullock (2004) make congruent choice jointly conditional on chooser characteristics and available alternatives.

Third, these audience comparisons have strategic consequences. Theorem 6.2 classifies every pure equilibrium in the specified five-appeal immigration election by population shares and rejection cost. Increasing that cost can create or destroy equilibrium, depending on audience composition. When the voter type unmoved by the evaluative addition is largest, the type offended by it is intermediate, and no type has a majority, the factual appeal addressed to the largest type becomes the unique equilibrium choice once the charge reaches the contested factual content’s value. When the type attracted by the addition is largest and the offended type is intermediate, again without a majority, raising the charge above that value instead removes the equilibrium combining the offended type’s factual appeal with the evaluative addition. When the offended type is smallest, the other two types sustain their combined appeal as the unique equilibrium choice at every nonnegative charge. This

last result holds throughout an open region of population compositions and extends across positive additive content values.

The same election admits no common weakly single-peaked ordering of its five appeals, at any rejection charge. Its stable regions therefore show how competition can sustain an appeal without a common axis organizing voter rankings. The classification concerns the specified menu and the content distinctions and ties generated by these audiences. It makes the effect of greater aversion depend on who else values the speech and how numerous those listeners are. Separate isolation conditions explain when candidates can design components for different audiences independently, while shared majority maxima leave minority comparisons to select among the majority’s best appeals.

Communication adds further strategic effects. An opponent’s expressed objection can itself become valued content for a supporter, increasing the supporter’s evaluation of the original speech. The worked objection election shows how a sufficiently valued report makes the provocative appeal the unique equilibrium choice under its specified conditions. Continuing communication need not change every aspect of the contest together. Under expansion on a finite field, understandings eventually become constant; evaluations with the fixed initial encounter record, and hence the finite electoral game, become constant as well. Public reactions can keep cycling after that point. The objection-reporting conditions of Theorem 7.2 also permit complete settling with lasting disagreement. Revision can instead sustain changes in meanings and recurring electoral phases. These results distinguish what people retain, how they react, and the comparisons at the election date. A constant game can still lack a pure equilibrium, and no candidate-adjustment process is assumed.

The analysis connects to formal accounts of communication and adaptation. [Hinich and Munger \(1994\)](#) model ideological communication through symbols and analogies alongside political commitment. [Kollman, Miller and Page \(1992\)](#) study adaptive party search using limited polling information, and their later work relates voter-preference distributions to the difficulty of that search ([Kollman, Miller and Page, 1998](#)). [Bendor et al. \(2011\)](#) develop aspiration-based adaptation in party competition, turnout, and voter choice. The present construction follows content through understandings and encounters to derive the comparisons on which electoral incentives operate. Familiar order and stability results remain informative benchmarks: stipulated strict single-peakedness of actual responses supports the median result, and persistent pooling supplies a setting for Tarski’s fixed-point theorem.²

The representational field and coalition membership are specified inputs. The framework allows changes in access and distinctions, while the sufficient-content theorem holds the stance fixed to isolate dependencies through meanings, conflict, and reports. Its explanatory use requires restrictions on those inputs: a sufficiently rich field can represent arbitrary finite ranking profiles. Empirical applications must distinguish offered content, interpreted meanings, cognitive change, public reactions, and electoral evaluations, and specify how each is measured. The construction supplies a way to connect these tasks to the political comparisons an application seeks to explain.

Section 2 develops understandings, and Section 3 specifies encounters and evaluation. Section 4 defines the electoral game and analyzes combinations of appeals. Sections 5 and 6 establish audience relevance and its competitive consequences; Section 7 studies communication, settling, and recurrence. Sections 8 and 9 develop empirical interpretation and extensions, and Section 10 concludes.

²The median benchmark has two source-neutral candidates, uniform voting ties, an odd unit-weight electorate, and feasible peaks on the stipulated common order. Its hypothesis concerns the rankings after reception. The pooling application retains a persistent base and repeatedly pools parent states; the general communication protocol allows other report rules.

2 Understandings: representations, stances, meanings.

Voters can use shared representations while differing in what they access, distinguish, and mean by them. This section builds their *understandings* from three layers: representational resources, epistemic stances toward those resources, and the meanings attached to perceived categories. The construction makes differences in access and interpretation comparable within one state space.

Two structural results guide the argument. Understandings form a complete lattice: any collection has a greatest common understanding and a least understanding containing them all (Theorem 2.11). These operations identify semantic common ground and pooled representational capacity. Reception across categories is also well defined, even when the sender and receiver have no visible categorical meet (Proposition 2.10). Shared resources can carry a message without either person first adopting the other’s distinctions.

The immigration example makes both layers and the lattice operations concrete. The section then introduces the evaluative anchor and connects possible changes in understanding to political encounters. Which differences between appeals affect an audience’s responses will depend on the reception, evaluation, and communication processes built on these foundations.

2.1 What representations are—and aren’t.

Political knowledge can be modeled using probability distributions or partitions over a set Ω of mutually exclusive and exhaustive states of the world. The present construction begins with the representational resources agents use to conceive of political life. A voter might model Congress as a hierarchy controlled by a decisive leader, as a network of committees and agenda-setting procedures, or as a narrative of heroes and villains contesting legitimacy. Each schema can organize understanding, including when simplified or mistaken. We ask which schemas an agent accesses, which distinctions they draw, and what they mean by them.

We collect these resources in a shared *representational field*. It lets us compare understandings of the same domain; an electoral ordering, when one exists, will depend on the responses these understandings and the available messages generate.

Definition 2.1. *The representational field $\mathcal{R}$ is a nonempty set whose elements are representational resources: models, schemas, symbols, narratives, formal structures, and other devices through which domains of political life can be conceived, described, or contested.*³

Elements of $\mathcal{R}$ are *tools for making, framing, contesting, or refusing claims*. A Petri net model of congressional procedure, a conspiracy theory about legislative capture, and a spatial model of ideological competition all belong to $\mathcal{R}$, regardless of their epistemic status. No resource is privileged as the correct representation.

Throughout, $\mathcal{R}$ and the meaning space M (definition 2.5) are finite. The understanding lattice $\mathcal{U}_{\mathcal{R}}$ is therefore finite, a standing assumption used in the communication results of Section 7 and the comparisons in Section 9.

The field is *shared* as a common ambient space, with neither uniform access nor uniform interpretation. The word *immigration*, for example, can be a shared token whose policy implications, evaluative associations, and social identifications are supplied by each agent’s semantics. The field specifies representational possibilities; a feasible menu will specify which appeals candidates can offer.⁴

³The field is fixed within a given model. Field variation can be modeled by comparing different finite sets of resources; Section 9.1 supplies restriction and embedding maps for those comparisons.

⁴Sniderman and Bullock (2004, pp. 337–339) hypothesize *menu dependence*: consistency of political choice depends jointly on citizens’ orientations and the organization of the alternatives they face. Their distinction between the properties of choosers and of choices motivates attending to both voter understandings and the electoral menu.

To fix ideas, consider immigration as a representational domain. The field $\mathcal{R}$ includes, among many other elements, three representations that will recur as a running example: r_1 (border enforcement mechanisms), r_2 (paths to citizenship), and r_3 (the economic impact of immigration). These are representational resources, not policy positions. A voter may engage with all three, with a subset, or—as the epistemic layer will formalize—may be unable to distinguish some of them from one another. Different voters may assign categorically different meanings to the same representation: r_1 can carry the meaning “enforcement works” or the meaning “enforcement is cruel,” and a single voter can hold both meanings simultaneously. The next three subsections build the apparatus to make these differences precise.

2.2 Epistemic stances.

One voter might distinguish several legal statuses while another distinguishes forms of economic contribution, with neither drawing all the distinctions available to the other. An *epistemic stance* records which resources are accessible and how finely the agent distinguishes them.

Definition 2.2. Fix a marker $\star \notin \mathcal{R}$ and write $\mathcal{R}_\star = \mathcal{R} \cup \{\star\}$. An epistemic stance is a pair $e = (\mathcal{R}_e, \sim_e)$ where $\sim_e$ is an equivalence relation on $\mathcal{R}_\star$, its invisible class is $I_e = [\star]_e$, and the agent’s visible scope is $\mathcal{R}_e = \mathcal{R} \setminus I_e$. Thus visibility is determined by the marked partition: the invisible resources are $I_e \setminus \{\star\}$. The perceived representations $\mathcal{R}_e / \sim_e$ are the classes other than I_e in $\mathcal{R}_\star / \sim_e$ —the finest distinctions the agent can draw within their visible scope. The set of all epistemic stances is denoted $E_{\mathcal{R}}$.

The marker is a bookkeeping device, not a representational resource or a primitive meaning. It identifies the invisible class even when every resource is visible, in which case $I_e = \{\star\}$. Without the marker, two identical partitions could have different classes designated invisible; partition refinement would then fail to distinguish the stances.

Consider a variant of the immigration example on $\{r_1, r_2, r_3\}$. The enforcement and integration voters access all three resources and distinguish each separately. Suppose C groups enforcement mechanisms and economic impact as the harsh side of the debate, while distinguishing paths to citizenship from both. Thus $\mathcal{R}_{e_C} = \{r_1, r_2, r_3\}$ with $r_1 \sim_{e_C} r_3$: C has the same access but fewer distinctions. This coarsening is a separate illustrative assumption; the worked election lets C distinguish all three resources.

Representational richness. It is natural to compare epistemic stances by the representational capacity they confer. Say that e_2 is *representationally richer* than e_1 , written $e_1 \preceq_E e_2$, when $\sim_{e_2}$ refines $\sim_{e_1}$ as partitions of $\mathcal{R}_\star$:

$$e_1 \preceq_E e_2 \iff \sim_{e_2} \subseteq \sim_{e_1} \text{ as binary relations on } \mathcal{R}_\star,$$

i.e., every $\sim_{e_2}$ -class is contained in a $\sim_{e_1}$ -class. The marker makes access and discrimination part of the same comparison. Since the fine class containing $\star$ must lie inside the coarse class containing $\star$, $I_{e_2} \subseteq I_{e_1}$ and hence $\mathcal{R}_{e_1} \subseteq \mathcal{R}_{e_2}$. A richer stance preserves existing access and distinctions; it may split an invisible block to make resources accessible or split visible blocks to distinguish them. Gaining access while losing an existing distinction can instead produce an incomparable stance.

Richness concerns representational capacity, not epistemic accuracy, calibration, or rationality. Finer distinctions may be misleading or irrelevant to a decision. The phrase *representationally richer* describes the resources and discriminations available; it makes no claim of Blackwell dominance or improved localization of truth.

Lemma 2.3. $(E_{\mathcal{R}}, \preceq_E)$ is a partially ordered set.

Incomparability. Two epistemic stances e_1 and e_2 are *incomparable* in $\preceq_E$ when neither $e_1 \preceq_E e_2$ nor $e_2 \preceq_E e_1$: each agent can represent something the other cannot.

For example, let agent i access a nonempty set $\mathcal{R}_P$ of Petri net representations of legislative procedure, and agent j a disjoint nonempty set $\mathcal{R}_N$ of narrative representations. The former models flows of tokens through transitions; the latter uses protagonists, obstacles, and institutional drama. Since each accesses resources the other cannot, $e_i = (\mathcal{R}_P, \sim_P)$ and $e_j = (\mathcal{R}_N, \sim_N)$ are incomparable.

Applying this comparison requires specifying access and distinctions. Lakoff (1996) proposes contrasting Strict Father and Nurturant Parent models of political morality. Such a contrast can be represented through different meanings at a common stance, different epistemic stances, or both. The mapping into particular stances and semantics is the manuscript’s modeling choice; incomparability follows only when the specified partitions fail to refine one another.

Incomparable stances can also have overlapping visible scopes. A shared resource can carry communication between agents whose surrounding schemas differ. If a recipient groups together representations that a sender distinguishes, pooling their meanings can preserve the associations while losing which source category carried each one. When the recipient preserves the relevant distinctions, this particular loss need not occur. The reception rule below specifies how such overlap supports communication.

The space of epistemic stances also admits operations for extracting shared categorical ground (the meet) and for combining access and distinctions (the join). These operations apply whether or not the stances are comparable. Their semantic counterparts will define common ground between understandings, while the epistemic join bounds the acquisition available in an encounter.

The epistemic lattice. Formally, the set $E_{\mathcal{R}}$ admits the following meet and join operations. For a nonempty family $\{e_j\}_{j \in J}$ with $e_j = (\mathcal{R}_j, \sim_j)$:

1. *Meet* $\bigwedge_{j \in J} e_j$: the most representationally rich stance lying below every e_j . Its partition of $\mathcal{R}_*$ is $\sim^\wedge = \text{EqCl}(\bigcup_j \sim_j)$ —merge any elements that some e_j identifies, then close transitively. Its invisible class is $[\star]_{\sim^\wedge}$ and its visible scope is $\mathcal{R} \setminus [\star]_{\sim^\wedge}$, which is contained in, but need not equal, $\bigcap_j \mathcal{R}_{e_j}$. The meet is the representational *common ground*.
2. *Join* $\bigvee_{j \in J} e_j$: the least representationally rich stance lying above every e_j . Its partition of $\mathcal{R}_*$ is $\sim^\vee = \bigcap_j \sim_j$ —two elements are identified only when every stance already treats them as indistinguishable. The visible scope is $\bigcup_j \mathcal{R}_{e_j}$. The join is the representational *horizon*: what an agent fluent in all the schemas could access.

In the legislative example, the meet of e_i and e_j is $e_\perp$ —they share no representational ground at all. Their join is $(\mathcal{R}_P \cup \mathcal{R}_N, \sim_{P \cup N})$ —the stance of an agent at home in both Petri nets and narrative. In this disjoint-scope example, reception at the initial stances has no overlap through which to pass meanings. Acquiring access to the other’s resources can create such overlap using elements already in $\mathcal{R}$; the acquisition rule in Section 3.3 specifies how far an encounter may extend the receiver’s stance. In the immigration case, the meet of the enforcement voter’s stance and the coarsened stance e_C is e_C itself: it preserves only distinctions both agents make and is strictly below the enforcement voter’s stance.

Overlapping access alone need not supply a shared category. On $\mathcal{R} = \{a, b\}$, compare the partitions $\{a\} \mid \{b, \star\}$ and $\{a, b\} \mid \{\star\}$. The first agent sees only a ; the second sees a and b as one inseparable category. Their meet merges a with b and b with $\star$, leaving nothing visible, although both initially access a . The reception rule below nevertheless permits communication through the shared resource a .

Lemma 2.4. $(E_{\mathcal{R}}, \preceq_E)$ is a complete lattice with bottom element $e_\perp$ (the single class $\mathcal{R}_*$, with empty visible scope) and top element $e_\top = (\mathcal{R}, \text{id}_{\mathcal{R}_*})$ (full visible scope, each resource its own class). The meet of the empty family is $e_\top$; its join is $e_\perp$.

The bottom $e_{\perp}$ represents total representational blindness relative to the chosen field; the top $e_{\top}$ represents full access with maximal discrimination. For a small field such as the three-resource immigration example, full access and discrimination can be an appropriate assumption about a voter.

2.3 Semantics.

The epistemic layer specifies access and distinctions; the semantic layer specifies what perceived representations *mean*. Two agents with the same stance may read congressional procedure as collective deliberation and as elite capture, respectively. We assign a semantic mapping to each stance and order understandings by both their distinctions and their semantic content.

The meaning space.

Definition 2.5. *The meaning space M is a nonempty set of primitive meanings.*

Like $\mathcal{R}$, the space M is left unstructured for generality. In applications, its elements might include:

- *Policy implications*: “entails higher taxes,” “requires redistribution,” “commits troops”
- *Evaluative content*: “just,” “threatening,” “wasteful,” “legitimate”
- *Social associations*: “for people like me,” “an elite agenda,” “a working-class cause”
- *Factual associations*: “correlated with crime,” “associated with economic growth”
- *Affective content*: dread, resentment, solidarity, hope

A representation of the Affordable Care Act might carry associations concerning health-care access, government intervention, personal responsibility, and partisan identity. The set-valued range in the definition below allows several associations to be attached at once. Their contribution to electoral response is specified in Section 3.5.

To see the semantic layer in isolation, suppose all three immigration voters share the top epistemic stance $e_{\top}$ —full scope, each representation in its own class—so that the action is purely in what meanings they attach. Let $M = \{\mathbf{T}, \mathbf{F}, \mathbf{G}, \mathbf{B}\}$, where $\mathbf{T}$ and $\mathbf{F}$ are factual primitives (“the policy works,” “the policy fails”) and $\mathbf{G}$ and $\mathbf{B}$ are evaluative primitives (“desirable,” “undesirable”). A representation can carry any subset of M : assigning $\{\mathbf{T}, \mathbf{G}\}$ to r_1 says “enforcement works and is good”; assigning $\{\mathbf{T}, \mathbf{B}\}$ says “enforcement works but is bad.” The enforcement voter assigns $\mu_A(r_1) = \{\mathbf{T}, \mathbf{G}\}$, $\mu_A(r_2) = \{\mathbf{F}, \mathbf{B}\}$, $\mu_A(r_3) = \{\mathbf{T}, \mathbf{B}\}$: enforcement is effective and good, paths to citizenship are based on false premises and bad, and immigration’s economic impact is real but harmful. The integration voter assigns $\mu_B(r_1) = \{\mathbf{T}, \mathbf{B}\}$, $\mu_B(r_2) = \{\mathbf{T}, \mathbf{G}\}$, $\mu_B(r_3) = \{\mathbf{T}, \mathbf{G}\}$: enforcement is real but cruel, paths to citizenship are viable and good, and immigration is economically beneficial. The affect voter assigns $\mu_C(r_1) = \{\mathbf{B}\}$, $\mu_C(r_2) = \{\mathbf{G}\}$, $\mu_C(r_3) = \emptyset$: enforcement is bad and citizenship is good, with no factual meanings and no meanings attached to economic impact.

These three semantics illustrate pure semantic disagreement. Voters A and B agree on the factual status of enforcement ($\mathbf{T} \in \mu_A(r_1) \cap \mu_B(r_1)$) but disagree evaluatively ($\mathbf{G}$ versus $\mathbf{B}$). Voter C’s displayed semantics attach only evaluative meanings. Her stance nevertheless distinguishes enforcement, citizenship, and economic impact, just as A’s and B’s do. Her response also depends on her prior, credibility, and updating; Sections 5.1 and 6.1 specify these inputs and compute their electoral consequences.

Semantic mappings. We now link epistemic stances to meanings via semantics.

Definition 2.6. For an epistemic stance $e = (\mathcal{R}_e, \sim_e)$, a semantics is a function

$$\mu_e : \mathcal{R}_\star / \sim_e \longrightarrow 2^M,$$

assigning to each class of the marked partition a set of meanings, subject to $\mu_e(I_e) = \emptyset$. The perceived representations $\mathcal{R}_e / \sim_e$ of Definition 2.2 are the visible classes of this partition. Even at full scope the marker's singleton class receives empty semantics. A semantics μ_e is e -consistent when it satisfies the condition $\mu_e(I_e) = \emptyset$ (a typing condition, not a requirement of conflict-freedom). Let M_e denote the set of all e -consistent semantics. Write 0_e for the semantics assigning $\emptyset$ to every class.

Invisible resources have empty semantics because the agent cannot access them. An encounter can make such a resource visible and supply meanings for it.

Two agents sharing the same epistemic stance e but holding different semantics $\mu_e \neq \mu'_e$ exhibit *pure semantic disagreement*: they access the same representations and draw the same distinctions, but interpret those representations differently. This is a distinct kind of political disagreement from epistemic incomparability. Both forms of disagreement can coexist and interact, as the reframing discussion below illustrates.

Remark 2.7 (Relations between meanings). *Distinct meanings can occur independently: a voter may assign factual effectiveness T without evaluative approval G , or both together. A systematic implication from T to G would require an additional restriction on permitted semantics, such as closure under a specified implication rule. The conflict relation Cf in Section 3.1 instead records which meaning pairs count as incompatible during an encounter. An individual understanding or evaluative anchor may associate T and G without imposing that association on every voter. Restrictions on these associations are possible sources of structure in electoral responses; whether they support a common single-peaked order also depends on reception, updating, evaluation, and the feasible menu.*

2.4 Understandings.

An *understanding* combines a stance with its semantics. The common-stance immigration voters differ semantically; the coarsened C variant also differs epistemically. Packaging the layers together lets us compare states across both kinds of difference.

Definition 2.8. An understanding is a pair $u = (e, \mu_e)$ consisting of an epistemic stance $e \in E_{\mathcal{R}}$ and an e -consistent semantics $\mu_e \in M_e$. The space of understandings is

$$\mathcal{U}_{\mathcal{R}} = \bigsqcup_{e \in E_{\mathcal{R}}} M_e = \{(e, \mu_e) \mid e \in E_{\mathcal{R}}, \mu_e \in M_e\},$$

where $\bigsqcup$ denotes disjoint union: each epistemic stance defines a fiber of compatible semantics, and $\mathcal{U}_{\mathcal{R}}$ assembles all fibers.

Semantic transport. The fiber structure requires us to distinguish the pooling of associations during reception from the extraction of content common to finer distinctions. For a fixed stance e , write $\mu \preceq_M \nu$ for pointwise inclusion on its visible classes. Suppose $e_1 \preceq_E e_2$. Every fine class D lies in a unique coarse class C_D , which may be invisible. The following maps compare semantics at these levels:

$$\begin{aligned} \text{Ref}_{e_1 \rightarrow e_2}(\mu)(D) &= \mu(C_D), \\ \text{Crs}_{e_2 \rightarrow e_1}(\nu)(C) &= \bigcup_{\substack{D \in \mathcal{R}_{e_2}/\sim_{e_2} \\ D \subseteq C}} \nu(D), \\ \text{All}_{e_2 \rightarrow e_1}(\nu)(C) &= \bigcap_{\substack{D \in \mathcal{R}_{e_2}/\sim_{e_2} \\ D \subseteq C}} \nu(D), \end{aligned}$$

where the last two formulas apply to visible coarse classes C ; both maps assign $\emptyset$ to I_{e_1} . Every visible coarse class contains at least one fine class. Refinement also respects invisible semantics: the fine invisible class has an invisible coarse parent. Newly visible classes inherit $\mu(I_{e_1}) = \emptyset$, so refinement alone creates no meanings at newly accessible resources.

Union-pooling Crs collects all associations attached to any finer subclass and will drive reception in the encounter model. Intersection transport All retains only associations attached to every finer subclass; it supplies the common content needed for the understanding meet.⁵

Proposition 2.9. *For $e_1 \preceq_E e_2$, the maps Ref, Crs, and All above are monotone maps between the indicated semantic fibers, and*

$$\text{Ref}_{e_1 \rightarrow e_2}(\mu) \preceq_M \nu \iff \mu \preceq_M \text{All}_{e_2 \rightarrow e_1}(\nu).$$

Thus $\text{Ref} \dashv \text{All}$ is a Galois connection. If $\mathcal{R}_{e_1} = \mathcal{R}_{e_2}$, there is also the adjunction $\text{Crs} \dashv \text{Ref}$; this second claim does not in general hold when visible scope expands. Refinement has identity maps and composes: for $e_1 \preceq_E e_2 \preceq_E e_3$, $\text{Ref}_{e_2 \rightarrow e_3} \circ \text{Ref}_{e_1 \rightarrow e_2} = \text{Ref}_{e_1 \rightarrow e_3}$.

The first equivalence says that a coarse meaning can be retained below a finer understanding exactly when every finer subclass carries it. The restriction on the second adjunction matters: coarsening may erase meanings at resources outside the coarse scope, while refinement assigns empty semantics to those resources when they become visible.

Reception across incomparable stances. Neither agent’s vocabulary need refine the other’s. For arbitrary stances e_i and e_j , define the *overlap reception map* from the sender’s semantics to the receiver’s vocabulary by

$$\text{Rec}_{e_j \rightarrow e_i}(\nu)(C) = \bigcup_{\substack{D \in \mathcal{R}_{e_j}/\sim_{e_j} \\ D \cap C \neq \emptyset}} \nu(D), \quad C \in \mathcal{R}_{e_i}/\sim_{e_i},$$

⁵These downward transports answer different questions. If a source attaches “good” to a and “bad” to b , union-pooling into the category $\{a, b\}$ retains both associations, representing the ambivalence that coarse reception can produce. Intersection transport retains neither: there is no association shared across the two subclasses. More generally, union asks whether a meaning occurs anywhere in the bundle, while intersection asks whether it occurs throughout the bundle. The latter may also be useful for cautious summarization or a communication rule restricted to shared associations. Neither operation assesses the truth of a meaning. The choice between them is substantive for reception, although intersection has a necessary formal role in the richness order adopted here. Both erase content that becomes invisible; pooling such content into the invisible class would violate semantic consistency.

and set $\text{Rec}_{e_j \rightarrow e_i}(\nu)(I_{e_i}) = \emptyset$. A receiver category collects associations from each sender category with which it shares a resource. The common ambient field supplies this correspondence; a sender's association with a bundle is available at its overlapping parts. Reception expresses the message in the receiver's categories; it does not itself incorporate meanings into her prior or give her the sender's distinctions.

Proposition 2.10. *For arbitrary e_i, e_j , $\text{Rec}_{e_j \rightarrow e_i} : M_{e_j} \rightarrow M_{e_i}$ is monotone and, writing $k = e_i \vee e_j$, satisfies*

$$\text{Rec}_{e_j \rightarrow e_i} = \text{Crs}_{k \rightarrow e_i} \circ \text{Ref}_{e_j \rightarrow k}.$$

If $e_i \preceq_E e_j$, it reduces to $\text{Crs}_{e_j \rightarrow e_i}$; if $e_j \preceq_E e_i$, it reduces to $\text{Ref}_{e_j \rightarrow e_i}$.

For example, a sender may attach “good” to the fused category $\{a, b\}$ while a receiver sees only a . Their epistemic meet is empty, but overlap reception delivers “good” at the receiver's a -class, leaving b invisible. Partial reception can therefore occur without shared categorical ground.⁶

The order on understandings. Using the refinement operator to compare meanings across epistemic levels, we extend the order on epistemic stances to the full space of understandings:

$$(e_1, \mu_{e_1}) \preceq_{\mathcal{U}} (e_2, \mu_{e_2}) \iff e_1 \preceq_E e_2 \text{ and } \text{Ref}_{e_1 \rightarrow e_2}(\mu_{e_1}) \preceq_M \mu_{e_2},$$

where $\preceq_M$ is the pointwise inclusion order on M_{e_2} : $\nu \preceq_M \nu'$ iff $\nu(D) \subseteq \nu'(D)$ for all $D \in \mathcal{R}_{e_2}/\sim_{e_2}$. Unpacking the refinement: the condition requires $\mu_{e_1}(C) \subseteq \mu_{e_2}(D)$ for every fine class D at e_2 's level that is contained in the coarse class C at e_1 's level. An understanding (e_2, μ_{e_2}) lies above (e_1, μ_{e_1}) when it is epistemically richer *and* each fine distinction D in e_2 's partition carries at least all the meanings e_1 attributes to the encompassing coarse class C . In the immigration example, $u_A \preceq_{\mathcal{U}} u_B$ would require $\mu_A(r_i) \subseteq \mu_B(r_i)$ for all i (since both voters share $e_{\top}$). This fails at r_1 : $\mu_A(r_1) = \{\mathbf{T}, \mathbf{G}\}$ is not contained in $\mu_B(r_1) = \{\mathbf{T}, \mathbf{B}\}$, because $\mathbf{G} \notin \mu_B(r_1)$. The same check in reverse fails at r_2 . The two voters are incomparable: neither understanding contains the other in the richness order. Whether their electoral responses nevertheless admit a common single-peaked order is a separate question.

The order $\preceq_{\mathcal{U}}$ again measures representational richness. Inherited meanings must persist at every finer distinction; accuracy, rationality, and political appropriateness are separate properties.

Theorem 2.11. *($\mathcal{U}_{\mathcal{R}}, \preceq_{\mathcal{U}}$) is a complete lattice. For any nonempty family $\{(e_j, \mu_{e_j})\}_{j \in J} \subseteq \mathcal{U}_{\mathcal{R}}$, writing $e^{\wedge} = \bigwedge_j e_j$ and $e^{\vee} = \bigvee_j e_j$, the meet and join are*

$$\bigwedge_{j \in J} (e_j, \mu_{e_j}) = (e^{\wedge}, \mu_{\wedge}), \quad \bigvee_{j \in J} (e_j, \mu_{e_j}) = (e^{\vee}, \mu_{\vee}),$$

where

$$\mu_{\wedge}(C) = \bigcap_{j \in J} \text{All}_{e_j \rightarrow e^{\wedge}}(\mu_{e_j})(C), \quad \mu_{\vee}(D) = \bigcup_{j \in J} \text{Ref}_{e_j \rightarrow e^{\vee}}(\mu_{e_j})(D),$$

for $C \in \mathcal{R}_{e^{\wedge}}/\sim_{e^{\wedge}}$ and $D \in \mathcal{R}_{e^{\vee}}/\sim_{e^{\vee}}$ respectively. Both semantics assign $\emptyset$ to their invisible class. The global bottom is $u_{\perp} = (e_{\perp}, 0_{e_{\perp}})$, and the global top has stance $e_{\top}$ and meaning set M at every visible singleton. The empty-family join and meet are these bottom and top elements, respectively.

⁶An alternative route through $h = e_i \wedge e_j$ is $\text{Ref}_{h \rightarrow e_i} \circ \text{Crs}_{e_j \rightarrow h}$. This rule first expresses the message in common coarse categories and transfers nothing when $h = e_{\perp}$. It agrees with overlap reception when the stances are comparable, but can differ otherwise. It may also blur more associations: if the sender distinguishes $\{a, b\}$ from $\{c\}$ and the receiver distinguishes $\{a\}$ from $\{b, c\}$, their meet fuses all three resources and pools the sender's associations across both receiver classes. We use overlap reception to carry content through shared resources in the receiver's current categories. Neither route transfers content between disjoint visible scopes; doing so would require a bridge resource, expanded access, or an additional translation relation.

The lattice operations have direct political interpretations. The meet $u_i \wedge u_j$ is their *semantic common ground*: the greatest understanding below both in the understanding order. The join $u_i \vee u_j$ is their *pooled representational toolkit*: the least understanding above both, combining their distinctions and meanings. Evaluation uses meets with an evaluative anchor to value received content and cognitive change (Definition 3.7). On incorporation under expansion, full credibility and maximal acquisition make the encounter’s output the join of prior and broadcast (Proposition 3.5).

The immigration voters make the lattice operations concrete. Consider the meet of voters A and B (at the shared stance $e_\top$, the meet reduces to pointwise intersection of meaning sets):

$$u_A \wedge u_B : \quad r_1 \mapsto \{\mathbf{T}\}, \quad r_2 \mapsto \emptyset, \quad r_3 \mapsto \{\mathbf{T}\}.$$

Their common semantics is $(\{\mathbf{T}\}, \emptyset, \{\mathbf{T}\})$: the two voters share the factual judgment that enforcement has consequences and that immigration has economic impact, but they share *nothing* evaluative and have zero semantic overlap on paths to citizenship. This is the formal content of the familiar observation that “they agree on the facts but not on the values”—the meet strips away everything contested and leaves only what survives intersection.

Now consider the meet of voters A and C:

$$u_A \wedge u_C : \quad r_1 \mapsto \emptyset, \quad r_2 \mapsto \emptyset, \quad r_3 \mapsto \emptyset.$$

Write 0_e for empty semantics at every class and $u_{e,0} = (e, 0_e)$ for the bottom of the fixed semantic fiber. The result is $u_{e_\top,0}$: the voters share no meanings, but both still see and distinguish all three representations. It is strictly above the global empty-scope bottom $u_\perp$. Empty common semantics does not preclude reception: a message can introduce an association absent from the receiver’s prior.

Voters A and C are also incomparable, with empty common semantics at their shared stance. By contrast, every meaning C holds is present in B’s understanding, so $u_C \preceq u_B$.

The join $u_A \vee u_B = (\{\mathbf{T}, \mathbf{G}, \mathbf{B}\}, \{\mathbf{T}, \mathbf{F}, \mathbf{G}, \mathbf{B}\}, \{\mathbf{T}, \mathbf{G}, \mathbf{B}\})$ pools both meaning systems, producing an understanding that is internally conflicted—at r_2 it simultaneously carries $\mathbf{T}$ and $\mathbf{F}$, $\mathbf{G}$ and $\mathbf{B}$ —and is the least understanding containing both voters’ semantic systems. It need not be maximal in the full understanding lattice. At r_2 the pooled understanding registers both *viable and good* and *false and bad*. The join records these tensions without resolving them.

These computations distinguish semantic richness from electoral orderability. Incomparable understandings cannot be ranked by inclusion, but voters’ electoral responses might still be single-peaked on a common ordering of feasible messages. Section 5 derives audience-, candidate-, and horizon-dependent comparisons of appeals from reception, priors, credibility, updating, evaluation, and communication (Theorem 5.6). For the running election, Proposition 6.3 establishes the absence of a common weakly single-peaked order from the resulting rankings.

Figure 1 summarizes the lattice structure of the three immigration voters.

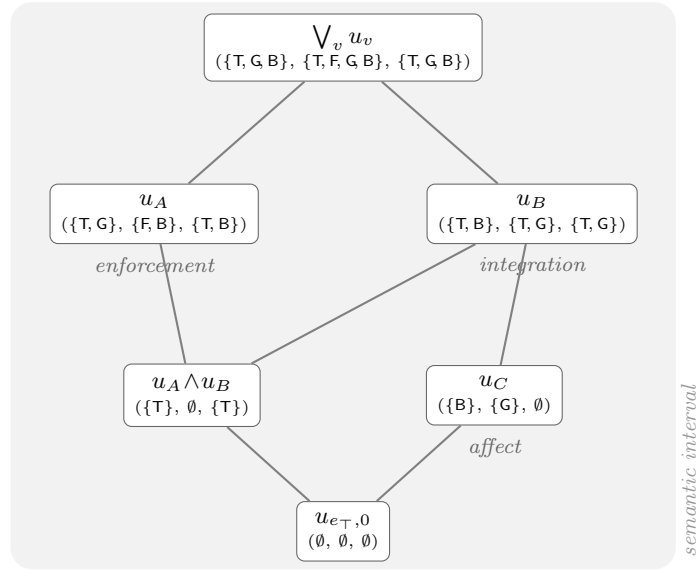


Figure 1: Order relations among selected understandings at the common stance e_T . Nodes show semantic content $(\mu(r_1), \mu(r_2), \mu(r_3))$; edges are covering relations among the displayed nodes, not all covers in the full lattice. Among these understandings, u_A and u_B are incomparable (neither lies above the other); u_C lies below u_B . The three-way meet $\bigwedge_v u_v = u_{e_T, 0}$ has empty meanings but full epistemic scope. The displayed nodes lie in the interval between that meet and the join; the diagram alone makes no claim about feasible majority-preference paths.

Epistemic regression. The state space can represent a loss of access or distinctions. For example, an agent who coarsens an epistemic partition while retaining the same visible scope moves downward in $\preceq_E$. To move downward in the understanding order as well, the resulting coarse meanings must be included in every corresponding prior fine class; intersection transport supplies the greatest semantics satisfying that condition. If coarsening instead pools new associations, the old and new understandings can be incomparable. Similarly, losing distinctions while making previously invisible resources accessible does not give a downward epistemic move: an increase in visible scope precludes it. The order does not assign a common numerical weight to gains in access and losses of discrimination.

The state space’s ability to describe these changes must be distinguished from the transitions generated by a particular update rule. On incorporation, the acquisition rule in Section 3.3 preserves or refines the stance under either named semantic rule; expansion additionally preserves inherited meanings. Inaction and rejection leave the whole understanding unchanged. Actual loss of epistemic distinctions would require a further transition rule.

2.5 The evaluative anchor.

An agent’s *evaluative anchor* $u_v^* \in \mathcal{U}_{\mathcal{R}}$ represents dispositions through which political content can be recognized as valuable. It need not coincide with her current understanding or be consciously articulated in it: a listener can value an association she does not yet hold. No comparability between the two understandings is required. The anchor need not be a feasible or optimal speech. The best available appeal depends on the listener’s prior and the whole encounter as well as on what she values.

Section 3 uses meets with this anchor to value received content and cognitive change (Definition 3.7); Definition 4.1 includes it among the electoral primitives. In the running election,

each displayed understanding serves as both its listener’s prior and her anchor: $u_v^0 = u_v^* = u_v$ for $v \in \{A, B, C\}$. This equality is an assumption of that example.

C’s anchor contains evaluative meanings but no factual meanings. At the common stance, a factual-only receipt has the same meet with her anchor as the empty receipt, so its normalized recognition is zero for every admissible valuation. This calculation alone does not determine a general encounter’s score: changes in valued prior content or acquired distinctions must also be evaluated.

2.6 From understandings to political encounters.

Entman (1993, pp. 52–53) describes framing through selection and salience, directed toward interpretations of problems, causes, evaluations, or remedies. He distinguishes the communicator’s framing intention, the frame in the text, and the receiver’s thinking; a frame’s presence in a text does not guarantee its adoption. This distinction motivates separating the broadcast, the understanding produced by an encounter, and the encounter’s evaluation. Adopting an interpretation need not make a listener favor its speaker.

The state space represents several forms of reframing. A familiar category can acquire a moral association alongside its existing meanings. A listener can also acquire access or distinguish cases previously grouped together, and semantic change can accompany either. Section 3 specifies which changes an encounter generates.⁷ The feasible broadcast map $\phi : X \rightarrow \mathcal{U}_{\mathcal{R}}$ in Section 4.1 determines what candidates can offer. Credibility and the update operator determine the resulting understanding (Definitions 3.2 and 3.3); evaluation then supplies the listener’s score.

Consider a separate immigration illustration on a field containing high-skill workers, low-skill workers, undocumented entry, and a statistical display organized by immigration status. Its meaning space includes economic contribution, humanitarian obligation, procedural status, and threat. The listener initially distinguishes the first three resources, assigns them existing meanings, and cannot access the display. The broadcast expresses only the display and undocumented entry as distinct categories, attaching threat to each. Suppose the listener’s filter admits threat at both source classes. On incorporation with maximal acquisition and expansion, the display becomes visible carrying threat, and undocumented entry gains threat alongside its prior meanings. The distinct skill categories receive no content and retain their meanings. An association on the display alone would not spread to those separate categories.

With an empty filter, incorporation under the same acquisition rule still makes the display visible, but with empty semantics; expansion retains the transported prior meanings. Inaction and rejection instead preserve the entire prior, including its stance. Of the two named incorporation rules, revision can replace a conflicting prior meaning while expansion retains it; thresholds determine which regime an encounter enters. A simpler reframe adds an admitted association to an already visible category at an unchanged stance.⁸

The lattice identifies common and pooled understandings. Political relevance asks which differences between appeals affect a listener, and how those dependencies combine across a coalition. Section 3 specifies encounters; Section 5 derives response equivalence and, at a common stance, sufficient content from encounters and communication. The members’ meet records common understanding; the coalition’s sufficient content preserves their responses. These audience comparisons need not inherit the lattice structure.

⁷A separate salience variable could represent changes in the relative prominence of associations while access, distinctions, and attached meaning sets remain fixed. A pure prominence change would require an additional state variable or an explicitly specified effect on reception or evaluation.

⁸If the listener also groups both skill categories as legal entry, retaining undocumented entry as illegal, while acquiring the display, gained access and lost discrimination make the new stance incomparable with the prior. The state space represents this change, but generating it requires a coarsening rule beyond the acquisition interval in Section 3.3.

3 Encounters and their evaluation.

A campaign appeal can supply valued content to one listener, contribute nothing to another, and provoke a third person’s rejection. This section constructs the encounter and its evaluation, separating raw conflict, admitted content, cognitive change, and approval. The distinction allows electoral comparisons to depend on content a listener rejects as well as content she accepts.

Three results identify the main routes. Under expansion, widening credibility weakly increases direct approval while leaving the conflict regime unchanged. Raising the rejection threshold can instead increase approval without changing understanding or recognition: switching rejection to inaction removes exactly the rejection charge. Finally, richer priors and broadcasts weakly increase conflict exposure when their added distinctions preserve inherited meanings (Proposition 3.16). Greater richness can therefore expose more grounds for rejecting an appeal. The acquisition example also shows how a richer cognitive output can accompany lower recognition and approval when the listener’s categories change.

Write $u_v^0 = (e, \mu)$ for listener v ’s prior understanding and $b = (f, \nu)$ for the broadcast understanding offered by a source j . The broadcast specifies the distinctions and meanings expressed in this encounter. It need not describe everything the speaker understands. All understandings belong to the finite field introduced in Section 2. Parameters may differ across listeners; comparisons below hold them fixed except where a stated parameter is varied.

3.1 Conflict structure.

Conflict concerns the relation between meanings attached to the same perceived subject. The listener’s categories therefore determine where conflict can be encountered. A source may distinguish two subjects that the listener groups together, bringing associations from both into one received category. Overlap reception supplies the relevant translation before the listener decides what to retain.

Definition 3.1 (Conflict). *A conflict relation is a symmetric, irreflexive relation $\text{Cf} \subseteq M \times M$. For a set of meanings $A \subseteq M$, write*

$$O(A) = \{m \in M : \text{some } m' \in A \text{ satisfies } (m, m') \in \text{Cf}\}.$$

A meaning set is conflict-free when it contains no related pair. A semantics is coherent when its assignment at every visible class is conflict-free.

Coherence is a property an understanding may have. The semantic space also contains conflicted assignments, including the pooled understandings constructed in Section 2. An application specifies which meanings conflict: factual negations, opposed evaluations, or other incompatible associations can enter the relation. This specification separates the presence of a conflict from the listener’s response to it.⁹

The raw message received at the prior stance is $\text{Rec}_{f \rightarrow e}(\nu)$. Define its conflict neighbors at a visible receiver class C and the resulting conflict set by

$$F_{vb}(C) = O(\text{Rec}_{f \rightarrow e}(\nu)(C)), \tag{1}$$

$$\text{Conf}(u_v^0, b) = \{C \in \mathcal{R}_e / \sim_e : \mu(C) \cap F_{vb}(C) \neq \emptyset\}, \tag{2}$$

$$d(u_v^0, b) = |\text{Conf}(u_v^0, b)|. \tag{3}$$

⁹Festinger (1957, chap. 1) relates cognitive inconsistency to pressure for change. Harmon-Jones and Mills (2019, pp. 3–4) distinguish inconsistent cognitions from experienced discomfort and discuss several routes of adjustment, including consonant additions and changes in importance. Here the conflict relation specifies incompatible content; the response rule specifies what the encounter does to an understanding.

Thus d counts affected receiver classes. Several conflicting meanings at one class contribute one unit; a conflict at another class contributes another. The count is computed from the unfiltered message. A listener can detect that a claim conflicts with her understanding while declining to admit that claim. An empty conflict set means that the prior and received message supply no conflicting pair at any visible class; either may still contain internal tensions.

For the immigration example, take T/F and G/B as the conflicting factual and evaluative pairs. The enforcement voter’s prior and the integration broadcast are

$$\mu_A = (\{T, G\}; \{F, B\}; \{T, B\}), \quad \nu_L = (\{T, B\}; \{T, G\}; \{T, G\}).$$

The three entries concern enforcement, citizenship, and economic impact. Reception at their common stance preserves these entries. The prior meanings exposed to conflict are respectively $\{G\}$, $\{F, B\}$, and $\{B\}$, so $d(u_A^0, b_L) = 3$. Four prior meanings are implicated across three categories. The response depends on that class count and on A’s thresholds.

3.2 Source credibility.

A listener may admit factual claims from a source while excluding its evaluative associations, or admit a source’s social associations while excluding its factual claims. Credibility specifies this receptivity at the source’s categories. It governs the content available for incorporation and recognition, separately from the raw conflict assessment.

Definition 3.2 (Credibility). *For listener v and source j at stance f , a credibility filter is a map*

$$\alpha_{vj}^f : \mathcal{R}_f / \sim_f \longrightarrow 2^M.$$

The filtered source semantics is

$$\nu^\alpha(D) = \alpha_{vj}^f(D) \cap \nu(D) \quad \text{on visible source classes,} \quad \nu^\alpha(I_f) = \emptyset.$$

When a source can use different stances, credibility is specified by a family of these maps.

Filtering precedes translation into the listener’s categories. At a receiving stance k , the admitted receipt is $\text{Rec}_{f \rightarrow k}(\nu^\alpha)$. A meaning can pass the filter at one source category and be blocked at another; overlap reception then pools the admitted assignments that reach each receiver category. The filter is full when every visible source class admits M and empty when every class admits nothing. An empty filter still permits the raw message to register a conflict.

The running example gives A and B factual-only filters, $\{T, F\}$, and C an evaluative filter, $\{G, B\}$, at each source class. Consider the evaluative appeal

$$g = (\{B\}; \{G\}; \{G\}).$$

A and B admit none of it; C admits all of it. Yet A’s prior conflicts with g at all three classes, whereas B’s and C’s priors conflict with it nowhere. Equal admission sets therefore need not produce equal responses. The raw conflict test preserves a route by which excluded content can matter.

The prior, credibility filter, and evaluative anchor describe different features of a listener. A meaning already held need not be admitted from this source, and an admitted meaning need not be valued. This separation follows the broader distinction between receiving political messages, accepting their considerations, and forming a response (Zaller, 1992, chap. 3). Source cues can help generate credibility filters, but source labels alone do not identify them. Observed partisan differences in perceptions and affect likewise motivate several possible locations for political attachment within the model: prior associations, source receptivity, and evaluation (Bartels, 2002; Iyengar et al., 2019). An application must distinguish these locations through observations of reception and change.

3.3 Acquiring distinctions.

An appeal can make a subject available to thought or distinguish subjects the listener previously treated together. Fix a deterministic *epistemic acquisition rule*

$$\Gamma : E_{\mathcal{R}} \times E_{\mathcal{R}} \longrightarrow E_{\mathcal{R}}, \quad e \preceq_E \Gamma(e, f) \preceq_E e \vee f.$$

Minimal acquisition chooses e ; maximal acquisition chooses $e \vee f$. Intermediate choices preserve the listener's distinctions and acquire some of those offered by the broadcast. The rule depends only on the two stances and is held fixed independently of the semantic contents, credibility, and thresholds. Its interval restriction alone does not require monotonicity in its arguments.

Acquisition is invoked on incorporation. The conflict count has already been assessed at the prior stance; admitted meanings are subsequently received at the stance selected by Γ . This order permits a distinction and its associated content to become available during the same encounter. On inaction and rejection, the listener keeps the prior stance.

To see what acquisition contributes, let $\mathcal{R} = \{r_p\}$. Initially the listener has the single marked block $\{r_p, \star\}$, so r_p is invisible and the semantics is empty. The source separates $\{r_p\}$ from $\{\star\}$ and associates a meaning m with r_p . There is no visible receiver class at which conflict can arise. Maximal acquisition makes $\{r_p\}$ visible. Refining the prior alone leaves it without a meaning; Example 3.4 completes the encounter after the incorporation rule is defined. The resource belongs to the specified field throughout. What changes is the listener's access to it.

3.4 The signed response.

Two thresholds distinguish incorporating a message, leaving it without cognitive effect, and rejecting it. Social judgment research provides a motivation for distinguishing these responses. [Hovland, Harvey and Sherif \(1957, pp. 246–250\)](#) examine acceptable and objectionable positions and the evaluation of communications; among influenced participants, an expanded range of acceptable positions could coexist with retention of the selected own stand. The present construction represents retained and acquired content separately and assigns rejection its own evaluative consequence.¹⁰

Let $0 \leq \tau_{1v} \leq \tau_{2v} \leq \infty$. The encounter's *regime* is

$$\zeta_v(u_v^0, b) = \begin{cases} + & d(u_v^0, b) \leq \tau_{1v}, \\ 0 & \tau_{1v} < d(u_v^0, b) \leq \tau_{2v}, \\ - & d(u_v^0, b) > \tau_{2v}. \end{cases} \quad (4)$$

The labels mean incorporation, inaction, and rejection, ordered $- < 0 < +$. Equal thresholds leave no inaction band; infinite τ_{2v} rules out rejection. The lower threshold determines the greatest conflict count that permits incorporation. The upper threshold determines when conflict elicits rejection rather than inaction.

For a potential incorporation, put $e^+ = \Gamma(e, f)$ and define

$$P^+ = \text{Ref}_{e \rightarrow e^+}(\mu), \quad A^+ = \text{Rec}_{f \rightarrow e^+}(\nu^\alpha).$$

A semantic incorporation rule Σ^+ maps this transported prior and admitted receipt to a semantics at e^+ . Its output is empty on the invisible class, and $\Sigma^+(P, 0) = P$. The default is pointwise *expansion*,

$$\Sigma_{\text{exp}}^+(P, A)(C) = P(C) \cup A(C). \quad (5)$$

¹⁰ [Jager and Amblard \(2005, p. 296\)](#) provide a two-threshold model with approach, no movement, and movement away on a scalar attitude dimension. Here the discrepancy measure counts conflicting receiver classes, and rejection preserves the understanding while affecting evaluation. The thresholds and the semantic incorporation rule are distinct model parameters.

It retains inherited meanings and adds those admitted from the broadcast. A named alternative is *revision*,

$$\Sigma_{\text{rev}}^+(P, A)(C) = (P(C) \setminus O(A(C))) \cup A(C),$$

which gives the admitted meanings priority over conflicting prior assignments. The results below identify their incorporation rule; Section 7 also studies revision.

Definition 3.3 (Signed encounter). *For the specified conflict relation, credibility, acquisition rule, thresholds, and incorporation rule, the cognitive output is*

$$T(u_v^0, b) = \begin{cases} (e^+, \Sigma^+(P^+, A^+)) & \zeta_v = +, \\ u_v^0 & \zeta_v \in \{0, -\}. \end{cases}$$

The encounter records the pair $(T(u_v^0, b), \zeta_v)$. Write $\bar{e}$ for the stance of its cognitive output.

The regime distinguishes responses that can yield the same understanding. Inaction and rejection preserve the prior, and incorporation can also leave it unchanged. Under expansion, newly visible classes inherit empty semantics and can receive meanings through A^+ ; previously visible classes retain their inherited meanings.¹¹

Example 3.4 (Completing articulation). *Return to the initially invisible resource r_p . Its conflict count is zero, so the encounter selects incorporation. With maximal acquisition and full credibility, the transported prior at $\{r_p\}$ is empty and the admitted receipt is $\{m\}$. Both expansion and revision assign $\{m\}$, since the empty prior leaves nothing to remove. The appeal has supplied both a distinction and an association; its accuracy and its evaluative significance require further information.*

Proposition 3.5 (Incorporation as a join). *Let $b^\alpha = (f, \nu^\alpha)$. Under expansion and maximal acquisition, an incorporating encounter has output*

$$T(u_v^0, b) = u_v^0 \vee b^\alpha.$$

Under full credibility this is $u_v^0 \vee b$.

Lemma 3.6 (Expansion preserves the prior). *Under expansion, $u_v^0 \preceq_{\mathcal{H}} T(u_v^0, b)$ in every regime.*

Proof. On incorporation, refinement preserves the prior meanings and union includes them at every refined class. The other regimes return the prior itself. $\square$

For A's encounter with the integration broadcast, thresholds $(2, 2)$ select rejection: $d = 3$, and A retains $(\{T, G\}; \{F, B\}; \{T, B\})$. At thresholds $(3, 3)$, the same encounter incorporates. Its expansion output depends on credibility:

$$\begin{aligned} \text{full credibility:} & \quad (\{T, G, B\}; \{T, F, G, B\}; \{T, G, B\}), \\ \text{factual-only credibility:} & \quad (\{T, G\}; \{T, F, B\}; \{T, B\}). \end{aligned}$$

The second output adds the factual association T at citizenship while retaining F there. The first also admits the opposed evaluative associations. The regime determines whether incorporation occurs; the filter determines its available content.

¹¹A different timing would first incorporate at e and then refine, giving semantics $\text{Ref}_{e \rightarrow e^+}(\mu \cup \text{Rec}_{f \rightarrow e}(\nu^\alpha))$. It leaves newly visible classes empty and can carry a pooled association into every child of a class split during the encounter. For example, pooling $\{g\}$ and $\{h\}$ at an initially combined category and then splitting gives both meanings to both children; acquisition followed by reception can give one to each. The chosen timing lets the newly acquired distinction govern reception immediately. Other timing or exposure-order rules can introduce different forms of path dependence.

3.5 Direct evaluation.

An encounter's cognitive output does not exhaust its electoral significance. A familiar claim can affirm what the listener already holds, while a new association can make an appeal more valuable by adding to her understanding. Rejection can carry a cost even when she retains every prior meaning. To distinguish these possibilities, fix the listener's evaluative anchor u_v^* and a common content valuation. The anchor describes what she can value in an understanding, separately from what she currently holds and from which meanings she admits from this source.

Definition 3.7 (Anchored value). *A content valuation is a map $\mathcal{W} : \mathcal{U}_{\mathcal{R}} \rightarrow \mathbb{R}$ strictly increasing in the understanding order. Define*

$$F_v(u) = \mathcal{W}(u \wedge u_v^*).$$

The meet selects the greatest understanding below both its arguments: their semantic common ground. An encounter has greater anchored value when it supplies richer common ground with what the listener values. Content outside the anchor can still affect reception and conflict, but this valuation does not attach a separate penalty to every unvalued meaning. The anchor itself need not coincide with any feasible speech. The most attractive available appeal will depend on the prior and the entire encounter as well as on the anchor.

Write $u_v^1 = T(u_v^0, b)$ and let $\bar{e}$ be its stance. The admitted message used for recognition is the understanding

$$r_v^\alpha(b) = (\bar{e}, \text{Rec}_{f \rightarrow \bar{e}}(\nu^\alpha)).$$

This definition applies in every regime. Recognition on incorporation uses the acquired stance; recognition on inaction or rejection uses the unchanged prior stance. In each case, the reference is $u_{\bar{e},0} = (\bar{e}, 0)$, the empty semantics at that same stance.

Definition 3.8 (Encounter evaluation). *The initial encounter supplies recognition and a regime,*

$$R_v^{\text{dir}}(b) = F_v(r_v^\alpha(b)) - F_v(u_{\bar{e},0}), \quad \zeta_v^{\text{dir}}(b) = \zeta_v(u_v^0, b).$$

For fixed $\lambda_v \geq 0$ and $\delta_v \geq 0$, evaluate an understanding u relative to this encounter by

$$\mathcal{E}_v(u; b) = R_v^{\text{dir}}(b) + \lambda_v [F_v(u) - F_v(u_v^0)] - \delta_v \mathbf{1}\{\zeta_v^{\text{dir}}(b) = -\}. \quad (6)$$

The direct score is $s_v(b) = \mathcal{E}_v(u_v^1; b)$. Write $G_v(b) = F_v(u_v^1) - F_v(u_v^0)$ for the direct valued change.

Recognition credits admitted, anchored content whether familiar or new. Subtracting the empty-message value at the same stance means that a distinction without an attached meaning earns no recognition by itself. Its contribution to valued understanding can enter through G_v . The coefficient λ_v weights that change relative to recognition, whose coefficient is normalized to one. The charge δ_v represents an adverse evaluation of the rejected appeal. It is constant for the listener; rejection selects whether it applies.

The three terms make the link from an encounter to evaluation explicit. They are a behavioral specification alongside the cognitive rule. They do not follow from the lattice order alone. The recognition term and the rejection indicator in (6) belong to the initial direct encounter and remain fixed when a different endpoint u is inserted. This keeps one evaluation rule available for later applications while specifying exactly which information it uses.

Proposition 3.9 (Recognition, change, and rejection). *Recognition satisfies $R_v^{\text{dir}}(b) \geq 0$, with equality for an empty admitted receipt. Inaction and rejection give $G_v(b) = 0$, hence scores $R_v^{\text{dir}}(b)$ and $R_v^{\text{dir}}(b) - \delta_v$, respectively. Under expansion, $G_v(b) \geq 0$ also on incorporation, whose score is therefore nonnegative. If $u_v^0 \succeq_{\mathcal{U}} u_v^*$, expansion gives $G_v(b) = 0$ in every regime. A global-bottom broadcast has score zero under either named rule, or any incorporation rule satisfying $\Sigma^+(P, 0) = P$.*

Rejection need not give a negative total score: recognizable content can offset some or all of its charge. Conversely, the revision instance can produce a negative valued change on incorporation by displacing an anchored prior meaning. At one class with prior and anchor $\{p\}$, a fully admitted, conflicting $\{q\}$ replaces p under revision when thresholds $(1, 1)$ permit incorporation; with unit values the gain is -1 . The regime records how the encounter is processed; the score records its evaluation. Neither alone specifies what the listener publicly expresses.

The valuation fixes cardinal units for the tradeoff. Replacing $\mathcal{W}$ by $a\mathcal{W} + c$, with $a > 0$, scales recognition and valued change by a . Scaling δ_v by the same factor then scales every score by a and preserves rankings. Keeping the numerical rejection charge fixed while rescaling content value would change that tradeoff.

Example 3.10 (Affirmation and learning). *At one visible class, let the prior be $\{p\}$ and the anchor $\{p, q\}$. Use unit content values, full credibility, $\lambda_v = 1$, and no conflicts. All three messages below incorporate under expansion:*

message	$\emptyset$	$\{p\}$	$\{q\}$
recognition	0	1	1
valued change	0	0	1
score	0	1	2

The familiar p affirms without changing the prior. The message q receives the same recognition and also adds valued content. Changing the prior to $\{q\}$ reverses the singleton messages' scores while leaving the anchor fixed.

For the immigration encounters, let each prior also serve as its listener's anchor and use unit content values. Expansion can add meanings to these priors, but it cannot increase their common ground with an anchor they already contain. Every direct valued change is therefore zero. On the evaluative appeal g , the three listeners differ as follows:

Listener	Regime	Recognition	Valued change	Score
A	—	0	0	$-\delta_A$
B	+	0	0	0
C	+	2	0	2

These are the outcomes at thresholds $(2, 2)$. B's incorporation receives no meanings and changes nothing. C admits all of g and adds G at economic impact, where her prior and anchor are empty. Her recognition comes from the other two classes; her cognitive change adds no anchored value. A rejects and keeps her prior, with a score determined entirely by the rejection charge.

Example 3.11 (Recognition on rejection). *A's integration encounter offers the factual content $\{T\}$ at each of the three classes alongside g . Her factual filter admits those three assignments, and she recognizes T at enforcement and economic impact. The raw conflict count remains three, so thresholds $(2, 2)$ select rejection. She retains all prior meanings, has recognition 2 and valued change 0, and scores the appeal at $2 - \delta_A$. At $\delta_A = 3$, that score is -1 . It exceeds her score for g alone by the two recognized factual meanings.*

Example 3.12 (The value of a thinner appeal). *The same evaluative channel can make a thinner appeal more attractive. At one class, take prior and anchor $\{p\}$, full credibility, conflict between p and q , thresholds $(0, 0)$, and unit values. The message $\{p\}$ incorporates with score 1. The richer message $\{p, q\}$ is rejected, leaves the prior intact, and scores $1 - \delta_v$. Its extra content reduces approval when $\delta_v > 0$; at zero charge the messages tie. The retained understanding and recognition are identical in the two encounters.*

3.6 Parameter dependence.

Credibility and the two thresholds affect different parts of the encounter. Credibility changes the filtered source semantics while leaving the raw conflict count fixed. The incorporation threshold determines whether acquisition and semantic change occur. The rejection threshold distinguishes two responses with identical cognitive outputs but different evaluative charges. These separations give direct comparisons when the other inputs and parameters are held fixed.

Lemma 3.13 (Credibility in a direct encounter). *Fix the prior, raw broadcast, thresholds, acquisition rule, anchor, valuation, λ_v , and δ_v , and use expansion. If $\alpha_{vj}^f(D) \subseteq \tilde{\alpha}_{vj}^f(D)$ at every visible source class, the regime and output stance are unchanged, the cognitive output weakly increases in the understanding order, and the direct score weakly increases.*

The enlarged filter admits at least as much content at the same output stance. This can increase recognition even when the encounter is rejected; on incorporation it can also increase valued change. The comparison fixes the prior. Different histories of admission can leave listeners with different priors, so the lemma does not rank entire trajectories generated by two filters.

Lemma 3.14 (The two thresholds). *Fix the prior, broadcast, filter, acquisition rule, and evaluation parameters. Under expansion, increasing τ_{1v} at fixed τ_{2v} weakly increases the cognitive output. If the output stance remains fixed, as under minimal acquisition or at a common source and receiver stance, it also weakly increases the direct score. For any fixed incorporation rule, increasing τ_{2v} at fixed τ_{1v} leaves cognition and recognition unchanged. An encounter switched from rejection to inaction gains exactly δ_v in score; other direct scores are unchanged. Both comparisons respect $0 \leq \tau_{1v} \leq \tau_{2v}$.*

For example, change A’s thresholds from (2, 2) to (2, 3) while holding the speech g fixed. Its conflict count is still three, but the regime becomes inaction. A keeps the same prior, admits no content, and recognizes nothing; her score changes from $-\delta_A$ to zero. This is a different route to no cognitive change from B’s empty incorporation. Changing A’s thresholds instead to (3, 3) permits incorporation. Thresholds leave the source-class filtered message fixed, though acquisition on incorporation can change its reception into the listener’s categories.

Remark 3.15 (Acquisition and recognition). *A higher incorporation threshold can increase understanding while reducing the direct score when reception changes stance. Let a listener group two resources x_1, x_2 into one visible class with prior meaning $\{p\}$ and anchor meaning $\{r\}$. The source distinguishes the resources, offers $\{q\}$ at x_1 and $\{r\}$ at x_2 , and faces a filter that excludes q and admits r . Only p and q conflict. Use maximal acquisition and $\tau_{2v} = 1$. The raw conflict count at the prior stance is one. With a strictly increasing valuation that counts visible classes plus meaning assignments over visible resources, the two encounters give*

τ_{1v}	regime	R_v^{dir}	G_v	s_v
0	0	2	0	2
1	+	0	0	0

At the coarse stance, received r applies to the grouped category and overlaps the anchor. After acquisition it applies only to x_2 . Taking the meet with the coarse anchor retains meanings shared across both children, leaving no r in that common ground. The expanded understanding is richer, but recognition is lower. The calculation is given with the proofs.

Prior richness has a further effect: it can expose more categories to conflict. The result uses the understanding order’s inheritance of meanings under refinement.

Proposition 3.16 (Richness and conflict exposure). *Fix the field, conflict relation, and overlap reception. If $u \preceq_{\mathcal{U}} u'$ and $b \preceq_{\mathcal{U}} b'$, then*

$$d(u, b) \leq d(u', b').$$

At common thresholds, the corresponding regimes satisfy $\zeta(u', b') \leq \zeta(u, b)$ in the order $- < 0 < +$. When receiver stances coincide and the broadcast is fixed, nested prior semantics give literal inclusion of their conflict sets.

An old conflict has a witness: a prior meaning, an opposed source meaning, and a resource shared by their categories. Refinement carries these meanings into the children containing that resource, preserving a conflict there. Distinct old receiver classes supply distinct children. The general comparison is therefore a comparison of counts; the classes themselves can differ across stances. At a fixed stance, adding prior meanings simply leaves every existing conflict in place and may expose another class.

The counting unit gives this result its interpretation. Splitting a class can increase the number of conflicts counted even while the inherited meanings stay the same. Equal numerical thresholds consequently refer to equal tolerated counts, which need not represent equal psychological tolerance across different categorizations. More precise categorization accompanied by selective revision of prior associations is a different change from refinement that preserves those associations in every child.

Nor does greater prior richness order the cognitive outputs of separate encounters. At one class, with p/q conflict, full credibility, and thresholds zero, an empty prior incorporates $\{q\}$ while prior $\{p\}$ rejects it and remains $\{p\}$. The priors are ordered; the two outputs are incomparable. The exposure proposition characterizes the encounter's conflict assessment, not a general advantage in accuracy or approval.

An appeal can supply valued content, change an understanding, and provoke rejection in different combinations across its listeners. Section 4 places their evaluations in an electoral contest. Section 5 asks which differences between appeals can affect an audience's responses and how those differences depend on who belongs to the audience.

4 Appeals and electoral competition.

An appeal can combine content valued by different audiences while introducing conflicts with some listeners' priors. This section identifies the resulting gains and losses and conditions supporting a common appeal in equilibrium.

First, against fixed rivals, improving every voter's score weakly raises a candidate's share. The gain is strict where a voter's strict score increase crosses, reaches, or leaves a tie with the best rival. Under direct, common-stance expansion, adding content has four effects: increased recognition, additional valued uptake, uptake blocked at a threshold, and a rejection charge. Only the last two can reduce approval; an incorporated combination weakly dominates its components.

A feasible appeal that everyone incorporates and whose admitted content contains every rival's admitted content maximizes every voter's score. Under source neutrality, all candidates choosing it is a symmetric equilibrium. The worked election shows how introducing conflict can instead shift the unique equilibrium from a combined appeal to one addressing the larger audience. The closing subsection carries the evaluator into communication after the speech, using the election-date understanding with the initial recognition and rejection record.

4.1 The electoral setup.

Definition 4.1 (Semantic election). *A semantic election has finite sets of candidates $\mathcal{C}$, with $|\mathcal{C}| \geq 2$, voters $V \neq \emptyset$, and feasible platforms $X \neq \emptyset$. A broadcast map $\phi : X \rightarrow \mathcal{U}_{\mathcal{R}}$ specifies the understanding expressed by each platform. The conflict relation Cf , acquisition rule Γ , incorporation rule Σ^+ , and strictly increasing valuation $\mathcal{W}$ are fixed as in Section 3; expansion is the default incorporation rule. Each voter has a prior u_v^0 , anchor u_v^* , weight $\omega_v > 0$, and parameters*

$$\lambda_v \geq 0, \quad 0 \leq \tau_{1v} \leq \tau_{2v} \leq \infty, \quad \delta_v \geq 0.$$

For each candidate, α_{vc} is the family of credibility filters indexed by the candidate's possible broadcast stances. Write $\Omega = \sum_{v \in V} \omega_v$.

Weights can represent the population mass of a voter type; unit weights give an electorate of individual voters. The menu has no prescribed policy order or distance, and distinct platforms may express the same understanding. A candidate chooses what to broadcast from this feasible menu.

Each voter's own prior is reused for every candidate's encounter. Instantiating Section 3's cognitive map and evaluator with the candidate's parameters gives

$$u_v^{c,1}(x) = T_v^c(u_v^0, \phi(x)), \quad s_v^c(x) = \mathcal{E}_v^c(u_v^{c,1}(x); \phi(x)). \quad (7)$$

The superscript on $\mathcal{E}$ records which source's parameters enter the same evaluation formula, including initial recognition. Identical cognitive outputs therefore need not imply equal evaluations of two sources. Separate encounters specify the base comparison; sequential exposure to rival candidates would require specifying their order and how earlier encounters affect later ones.

Call the election *source-neutral* when every primitive and initial condition entering evaluation is independent of candidate identity. In the specification above, credibility is the only candidate-varying primitive, so $\alpha_{vc} = \alpha_v$ suffices. We then write $s_v(x)$. This condition permits differences among voters in priors, anchors, filters, and thresholds.

The *best available appeals* to voter v from candidate c form the nonempty set

$$\mathcal{A}_v^c(u_v^0) = \arg \max_{x \in X} s_v^c(x).$$

This set depends on the feasible menu and the entire encounter. Even at a fixed anchor, changing the prior or the source can change which appeal best addresses the voter.

4.2 Candidate strategy.

Definition 4.2 (Voting and expected share). *At a campaign profile $\mathbf{x} = (x_c)_{c \in \mathcal{C}}$, voter v chooses uniformly among her highest-scoring candidates,*

$$K_v(\mathbf{x}) = \arg \max_{c \in \mathcal{C}} s_v^c(x_c).$$

Candidate c maximizes expected vote share

$$\mathcal{V}_c(\mathbf{x}) = \frac{1}{\Omega} \sum_{v \in V} \omega_v \frac{\mathbf{1}\{c \in K_v(\mathbf{x})\}}{|K_v(\mathbf{x})|}. \quad (8)$$

Definition 4.3 (Symmetric equilibrium). *A profile in which every candidate chooses x^* is a symmetric equilibrium if no candidate can increase expected share by choosing another feasible platform while all rivals retain x^* .*

This definition also applies with source-dependent responses, when identical platforms can earn different shares. Under source neutrality, define the strict-support weight

$$N(x, y) = \sum_{v \in V} \omega_v \mathbf{1}\{s_v(x) > s_v(y)\}.$$

We say that x *strictly majority-defeats* y when $N(x, y) > \Omega/2$, and call x a *strict Condorcet winner* when it does so against every other platform. Here strict majority means an absolute majority of the electorate. With indifferent voters, it is stronger than having more strict supporters than strict opponents.

Proposition 4.4 (Scores and votes). *Fix candidate c 's rivals and suppose $s_v^c(y) \geq s_v^c(x)$ for every voter. Changing from x to y weakly increases c 's share. The increase is strict exactly when some voter satisfies*

$$s_v^c(x) < s_v^c(y), \quad s_v^c(x) \leq m_v \leq s_v^c(y), \quad m_v = \max_{d \neq c} s_v^d(x_d).$$

In a source-neutral two-candidate election, a deviation from (x, x) to y changes the deviator's share by

$$\frac{N(y, x) - N(x, y)}{2\Omega}. \tag{9}$$

Consequently, (x^, x^*) is an equilibrium exactly when $N(y, x^*) \leq N(x^*, y)$ for every $y \in X$. A strict Condorcet winner is sufficient.*

A score increase matters electorally where it changes a voter's allocation. At a symmetric two-candidate profile, each strict supporter gained contributes half her weight and each strict opponent lost costs half hers. An absolute majority is consequently unnecessary: with unit weights, a deviation supported by two voters, opposed by one, and leaving seven indifferent is profitable.

With $m \geq 2$ source-neutral candidates initially at x , the corresponding change is

$$\frac{(m-1)N(y, x) - N(x, y)}{m\Omega}. \tag{10}$$

A new strict supporter moves from an m -way tie to the deviator alone, whereas a strict opponent costs only the deviator's previous $1/m$ share. Additional candidates can therefore make a shared platform harder to sustain. The condition $(m-1)N(y, x) \leq N(x, y)$ for every y characterizes these symmetric equilibria; the existence and structure of other pure-equilibrium profiles require further analysis of the audience rankings and feasible appeals.

4.3 Combining content.

To isolate the effect of adding meanings, throughout this subsection fix a common stance for receivers and broadcasts, use expansion on incorporation and identity otherwise, and hold priors, anchors, credibility filters, thresholds, $\lambda_v \geq 0$, and $\delta_v \geq 0$ fixed. For broadcasts at this stance, $b \preceq c$ means pointwise inclusion of their meanings. Their union is pointwise union. This order induces a preorder on platforms through ϕ ; a union is an available strategy only when some feasible platform broadcasts it.

Write $P_v = u_v^0$ and let $A_v(b)$ be the admitted understanding at the common stance. Put $R_v(b) = R_v^{\text{dir}}(b)$ and $\zeta_v(b) = \zeta_v^{\text{dir}}(b)$, and define potential valued uptake by

$$D_v(b) = F_v(P_v \cup A_v(b)) - F_v(P_v) \geq 0.$$

The union here is within the common stance. Potential uptake contributes to the score only when the encounter incorporates:

$$s_v(b) = R_v(b) + \lambda_v \mathbf{1}\{\zeta_v(b) = +\} D_v(b) - \delta_v \mathbf{1}\{\zeta_v(b) = -\}. \quad (11)$$

These local broadcast scores equal $s_v^c(x)$ when $b = \phi(x)$ and the source parameters are those of c .

Proposition 4.5 (The effects of adding content). *Under the direct, common-stance expansion specification above, if $b \preceq c$, then*

$$\begin{aligned} s_v(c) - s_v(b) = & [R_v(c) - R_v(b)] \\ & + \lambda_v \mathbf{1}\{\zeta_v(c) = +\} [D_v(c) - D_v(b)] \\ & - \lambda_v [\mathbf{1}\{\zeta_v(b) = +\} - \mathbf{1}\{\zeta_v(c) = +\}] D_v(b) \\ & - \delta_v [\mathbf{1}\{\zeta_v(c) = -\} - \mathbf{1}\{\zeta_v(b) = -\}]. \end{aligned} \quad (12)$$

Every bracketed quantity is nonnegative.

The four terms identify increased recognition, additional valued uptake, uptake blocked by leaving incorporation, and a charge introduced by entering rejection. Both possible losses occur at thresholds. Within an unchanged regime, adding content weakly raises the score; an incorporated combination weakly dominates each component. More content can also reduce the set of listeners who incorporate it. These conclusions use monotonicity of anchored value, without requiring additive values for meanings. The decomposition also bounds a combination's loss relative to its components by their blocked valued uptake and any new rejection charge; the bound is given with the proof.

For an audience $Q \subseteq V$ and a fixed source's parameters, define the *incorporation domain*

$$\mathcal{I}_Q = \{x \in X : \zeta_v(\phi(x)) = + \text{ for every } v \in Q\}.$$

Proposition 4.6 (Incorporation and comparison). *Under the subsection's direct, common-stance expansion specification, $\mathcal{I}_Q$ is downward closed in the feasible content preorder. If $x, y \in \mathcal{I}_Q$ and $A_v(\phi(x)) \preceq A_v(\phi(y))$ for every $v \in Q$, every member weakly prefers y .*

Incorporation permits disagreement. The immigration example's factual appeals are

$$f_A = (\{\mathbf{T}\}; \{\mathbf{F}\}; \{\mathbf{T}\}), \quad f_B = (\{\mathbf{T}\}; \{\mathbf{T}\}; \{\mathbf{T}\}).$$

With thresholds $(2, 2)$, both voters incorporate both appeals. A scores them 3 and 2, while B scores them 2 and 3. These incomparable appeals divide the audience despite their common incorporation. Nor must a best appeal incorporate: recognition can make a rejected speech optimal.¹²

Proposition 4.7 (Admitted domination). *Consider a source-neutral two-candidate election satisfying the subsection's direct, common-stance expansion hypotheses. If a feasible x^* is incorporated by every voter and*

$$A_v(\phi(y)) \preceq A_v(\phi(x^*)) \quad \text{for every } v \in V \text{ and } y \in X,$$

then x^ maximizes every voter's score and (x^*, x^*) is a symmetric equilibrium.*

¹²At one visible class, take prior $\{p\}$, anchor $\{q\}$, full credibility, only p/q conflict, thresholds $(0, 0)$, unit values, $\lambda = 1$, and $\delta = 1/2$. The menu $\emptyset, \{p\}, \{q\}$ receives scores 0, 0, 1/2; the rejected $\{q\}$ is optimal.

The argument allows rivals that fail to incorporate. It also supplies a symmetric equilibrium with any finite number of candidates at least two: no deviation improves any voter's score. Thus this sufficient condition survives the changed incentives in (10).

A greatest feasible speech guarantees admitted domination, while incorporation remains a separate condition. At one class with full credibility, prior $\{r\}$, anchor $\{p\}$, only q/r conflict, thresholds $(0, 0)$, unit values, and $\lambda = \delta = 1$, the menu $\{p\}, \{p, q\}$ receives scores 2 and 0. The greatest speech is internally coherent but rejected because of the prior. A merely maximal speech need not be optimal.¹³

Example 4.8 (Combining appeals to different audiences). *Take one visible class, two source-neutral candidates, and voter types 1 and 2 with weights 2 and 1, respectively. Type 1 has prior $\{r\}$, anchor $\{r, p\}$, and filter $\{p\}$; type 2 has an empty prior, anchor $\{q\}$, and filter $\{q\}$. Both have thresholds $(0, 0)$ and $\lambda = 1$. Use unit content values and the menu $\{p\}, \{q\}, \{p, q\}$. Their direct scores are*

<i>Conflict relation</i>	<i>Type</i>	$\{p\}$	$\{q\}$	$\{p, q\}$
<i>Empty</i>	1	2	0	2
	2	0	2	2
<i>Only r/q</i>	1	2	$-\delta$	$1 - \delta$
	2	0	2	2

Without conflict, both candidates choosing $\{p, q\}$ is the unique pure equilibrium. With r/q conflict, both choosing $\{p\}$ is the unique pure equilibrium for every $\delta \geq 0$.

Adding q to $\{p\}$ benefits type 2 by two points. With conflict, it costs type 1 one point of blocked uptake and the rejection charge: this listener still recognizes p , but rejects the combined appeal and consequently does not acquire the admitted meaning p . The combination therefore trades the support of the larger audience for that of the smaller one. It remains feasible and internally coherent in both specifications.

Changing the thresholds to $(0, 1)$ in the conflict specification makes type 1's scores 2, 0, 1. The combination now produces inaction, losing the uptake benefit without a rejection charge. Both candidates choosing $\{p\}$ remains the unique pure equilibrium. At this common stance, repeating any fixed speech leaves these scores unchanged, as discussed below.

4.4 The horizon score.

Listeners can encounter reports and reactions after the candidate speaks. Fix an integer horizon $H \geq 1$, common to the candidates, and a separately attributed process for each candidate's platform. Each process starts from the fixed vector of voter priors, begins with the direct encounter, and thereafter depends only on that candidate's platform and its declared communication protocol. Section 5 specifies expression, reporting, attribution, and delivery.

Definition 4.9 (Horizon score). *If $u_v^{c,H}(x)$ is voter v 's understanding at voting, define*

$$S_v^c(x; H) = \mathcal{E}_v^c(u_v^{c,H}(x); \phi(x)), \quad x \in X. \quad (13)$$

At a profile $\mathbf{x}$, voters compare $S_v^c(x_c; H)$ using Definition 4.2.

Only the endpoint argument of the evaluator varies with H . Initial recognition and the initial rejection indicator remain fixed, and the anchor is fixed throughout the trajectory. Later encounters

¹³At one visible class, with an empty prior, full credibility, anchor $\{p, q\}$, no conflict, $\lambda = 1$, and values 1 for p and 2 for q , the menu $\{p\}, \{q\}$ has two maximal speeches, scoring 2 and 4.

have their own regimes; their effect on this score enters through the endpoint understanding. At $H = 1$ the score is the direct score. Evaluation at a finite horizon requires no convergence.¹⁴

For example, at one common visible class, a fully receptive supporter with empty prior semantics, anchor $\{p, h\}$, unit values, $\lambda = 1$, and no conflicts scores the speech $\{p\}$ at 2: one point of recognition and one of valued uptake. If a report supplying h is incorporated next, the endpoint score becomes 3. Recognition of the initial speech remains 1. The communication protocol determines which reactions supply such a valued report.

Source neutrality at a horizon requires candidate symmetry of every operative rule and initial condition, including expression, reporting, attribution, delivery, and horizon. Together with the separate-process requirement, this gives $S_v^c(x; H) = S_v(x; H)$, so the vote identities apply at that horizon. A joint process that reads rival platforms would require a profile-dependent score and its own strategic analysis.

For repetition of a fixed speech without reports at a common source and receiver stance, expansion settles the understanding after one encounter. The stance cannot change, and repeating an already incorporated receipt adds nothing; an initially inactive or rejecting encounter repeats from the same prior. The fixed initial record therefore gives the same score at every horizon. Section 7 gives the broader one-encounter result under maximal acquisition and distinguishes settled understandings from later signs. Partial acquisition can instead reveal distinctions across several encounters.

The game now makes audience response consequential: a content difference matters electorally when it changes evaluations enough to affect allocations. The next section asks which content differences suffice to preserve an audience’s evaluations.

5 Audience and coalition relevance.

The differences between appeals that an electoral analysis must retain depend on whose responses matter. One audience distinguishes factual alternatives while ignoring their evaluative accompaniment; another responds to that accompaniment while treating the factual alternatives alike. Together they can distinguish appeals that neither separates alone.

This section derives a set of speech coordinates sufficient to preserve an audience’s evaluations. The main result, Theorem 5.6, includes content that supplies valued meanings, affects a present or future conflict, or changes a public report reaching the audience before voting. It allows incorporation, inaction, rejection, and feedback, and requires no convergence. With the construction’s other inputs fixed, adding listeners or extending the communication horizon weakly enlarges the sufficient set. Two electoral consequences follow: an edit can affect votes only among listeners whose retained coordinates it meets, and feasible representatives preserving every voter’s scores give a game on representative appeals with the same payoffs and corresponding pure equilibria. The construction supplies sufficient coordinates; exact equality of evaluations may identify still more appeals.

Throughout the content constructions, priors, anchors, speeches, and reports share one stance e . A *coordinate* pairs a visible class with a meaning, in the finite set $\mathcal{Z}_e = (\mathcal{R}_e / \sim_e) \times M$. Identify a semantics with its subset of $\mathcal{Z}_e$; unions, intersections, and the conflict-neighbor operation O act classwise. We suppress the common stance in these operations. An *audience* $Q \subseteq V$ is a nonempty set of voters whose responses we examine; *coalition* names that group’s electoral interpretation. The construction takes this membership as given.

¹⁴Updating can depend on encounter order even when evaluation uses the endpoint. Evaluation could additionally weight the intervening history by replacing its endpoint change with $\sum_{t=1}^H w_{v,t} [F_v(u_v^{c,t}(x)) - F_v(u_v^0)]$, where the weights are fixed across candidates, nonnegative, and sum to one. Putting all weight at H recovers (13).

5.1 The audience table.

Return to the immigration example’s three classes, in the order enforcement, citizenship, and economic impact. Keep the priors and credibility filters of Sections 2 and 3, with each prior also serving as its listener’s anchor. A and B admit factual meanings; C admits evaluative meanings. Use expansion, unit content values, $\lambda_v = 1$, thresholds (2, 2), and source neutrality. Alongside Section 4’s factual appeals f_A and f_B , take the evaluative appeal

$$g = (\{B\}; \{G\}; \{G\}).$$

The five feasible appeals are $f_A, f_B, g, f_A + g, f_B + g$, where addition denotes classwise union. We identify their platform names with their broadcasts in this example. At rejection charge $\delta = 3$, their direct scores are

Audience member	f_A	f_B	g	$f_A + g$	$f_B + g$
A	3	2	−3	0	−1
B	2	3	0	2	3
C	0	0	2	2	2

A assigns every appeal a different score. B identifies f_A with $f_A + g$ and f_B with $f_B + g$, leaving g as a third response class. C identifies the two factual appeals and, separately, the three appeals containing g . The joint responses of B and C distinguish all five. B separates the factual alternatives that C treats alike, while C separates the evaluative additions that B ignores.

The same evaluative content has three roles. A admits none of g , but it conflicts with her prior at every class and selects rejection. B also admits none of it, and it conflicts with none of his prior meanings. C admits it all and recognizes valued content at enforcement and citizenship. She also acquires G at economic impact, where her anchor is empty; that addition contributes no valued gain. The different response spaces therefore reflect differences in conflict and valuation as well as credibility. Within this menu, the incorporation domain is exactly $\mathcal{I}_{\{A,B\}} = \{f_A, f_B\}$.

5.2 Direct relevance.

First describe the distinctions in evaluations themselves. For a fixed candidate c and horizon H , define *response equivalence* by

$$x \sim_{Q,H}^c y \iff S_v^c(x; H) = S_v^c(y; H) \text{ for every } v \in Q. \quad (14)$$

The unanimous comparison $x \preceq_{Q,H}^c y$ requires the corresponding weak inequalities. It is a preorder on platforms and a generally incomplete partial order on response classes. Enlarging Q restricts both relations on $X \times X$: the equality partition refines, and a previously unanimous comparison may disappear. Either relation can remain unchanged. Under horizon source neutrality we suppress c , as in the running example.

The number of response classes does not by itself describe this comparison. A and the joint audience B–C both distinguish all five appeals, but A prefers f_A to f_B , whereas B–C weakly ranks f_B above f_A . For B–C, g and f_A are incomparable: B prefers the factual appeal and C the evaluative one. Audience composition thus affects both which appeals are identified and how the remaining classes compare.

These response classes describe the computed scores. A sufficient content set instead identifies coordinates on which agreement guarantees equal scores, using the primitives of the encounter. For direct evaluation, two routes suffice: admitted content within the anchor, and raw content that can conflict with the prior.

Proposition 5.1 (Direct sufficient content). *Fix a candidate, a common stance, and all encounter primitives. Use either expansion or revision on incorporation, and identity on inaction and rejection. For voter v , put*

$$\Lambda_v^{c,\text{dir}} = (\alpha_{vc} \cap \mu_v^*) \cup O(\mu_v^0). \quad (15)$$

Replacing the broadcast semantics b by $b \cap \Lambda_v^{c,\text{dir}}$ preserves the initial regime, recognition, anchored endpoint, valued change, and direct score. Their audience-wide counterparts are preserved by the union of these sets over $v \in Q$. Under fixed repetition without reports, the anchored endpoint and the score evaluated with the initial record are preserved at every horizon.

Credibility limits the first route but not the second. Removing unadmitted content is harmless to recognition, yet may change the regime and thereby change what happens to the rest of the appeal. Revision adds no new route to the direct set: a meaning capable of removing a prior association is already a conflict neighbor of that prior. The proof preserves the relevant intersections with the anchor, so it requires no additive valuation. Each named rule has a fixed cognitive output after one common-stance encounter with a repeated speech. Later regimes may nevertheless differ between an original and clipped speech. The repetition clause concerns the endpoint and its fixed-record evaluation.

Let $\mathcal{B}_X = \bigcup_{x \in X} \phi(x)$ collect the coordinates available in the immigration menu. Intersecting the direct sets with this pool gives

$$\begin{aligned} \Lambda_A^{\text{dir}} \cap \mathcal{B}_X &= (\{\mathsf{T}, \mathsf{B}\}; \{\mathsf{F}, \mathsf{T}, \mathsf{G}\}; \{\mathsf{T}, \mathsf{G}\}), \\ \Lambda_B^{\text{dir}} \cap \mathcal{B}_X &= (\{\mathsf{T}\}; \{\mathsf{F}, \mathsf{T}\}; \{\mathsf{T}\}), \\ \Lambda_C^{\text{dir}} \cap \mathcal{B}_X &= (\{\mathsf{B}\}; \{\mathsf{G}\}; \emptyset), \\ \Lambda_{\{B,C\}}^{\text{dir}} \cap \mathcal{B}_X &= (\{\mathsf{T}, \mathsf{B}\}; \{\mathsf{F}, \mathsf{T}, \mathsf{G}\}; \{\mathsf{T}\}). \end{aligned}$$

A retains all three coordinates of g through conflict while admitting none. B retains none; C retains its first two through valued admission. Economic-impact G is absent from C's sufficient set despite being admitted. The joint set retains the differences needed to separate the five joint responses.

Three audience objects now have distinct jobs. The incorporation domain $\mathcal{I}_Q$ selects feasible appeals that everyone in Q incorporates. A sufficient coordinate set specifies a projection of content. The exact response partition identifies appeals with equal evaluations. Agreement on retained coordinates implies equal evaluations, so the projection's partition refines the response partition. Neither object is the incorporation domain.

Varying the rejection charge makes the distinction concrete. A's scores on the five appeals are

$$(3, 2, -\delta, 3 - \delta, 2 - \delta).$$

Her sufficient coordinates stay fixed, while her response partition has three classes at $\delta = 0$, four at $\delta = 1$, and five at every other nonnegative charge. Thus the projection and response partitions coincide at the displayed $\delta = 3$, and at every positive charge other than 1. B's, C's, and their joint partitions are unchanged for all $\delta \geq 0$. The sufficient sets track possible routes to evaluation even where those routes produce equal scores.

5.3 Public expression.

An audience can also value a reaction to an appeal. An objection, endorsement, or visible refusal supplies content that the original speaker did not express directly. To determine which speech coordinates can matter through these reactions, specify what participants express, what their reports mean, and who receives them.

Definition 5.2 (Observation structure). *Fix a candidate's process on a finite directed graph of participants containing the voters and any nonvoting intermediaries. An edge $j \rightarrow i$ delivers a report from j to i one round after its production. Participant j has a fixed semantic read mask J_j . The edge's deterministic report rule r_{ji} reads only $P_j^t \cap J_j$ and the regime ζ_j^t , where P_j^t is j 's semantics after round t . From fixed priors P_j^0 , initialize $\zeta_j^0 = \diamond$ and require $r_{ji}(A, \diamond) = \emptyset$ for every A . The raw receipt in round $t + 1$ is*

$$b_i^{t+1} = b \cup \bigcup_{j:j \rightarrow i} r_{ji}(P_j^t \cap J_j, \zeta_j^t). \quad (16)$$

Apply the signed encounter to (P_i^t, b_i^{t+1}) , filtering the pooled receipt once with i 's fixed credibility mask toward the candidate. All participants update simultaneously.

The external speech is thus delivered every round, and round 1 is the direct encounter. Reports are events attributed to that speech; each candidate's process starts separately from the fixed priors and reads only its own declared variables. The rule has no additional access to platform labels, raw speech syntax, rival processes, or undeclared history. The same candidate can be received differently by different participants. Source neutrality requires symmetry of the full protocol across candidates.

A read mask limits which parts of a participant's understanding her report can consult. It can describe a selective summary or a publicly expressed judgment about particular subjects. Even an empty mask permits reports depending on the regime alone, as in the following example. The report rule can respond differently to incorporation, inaction, and rejection; it need not preserve an order on meanings or evaluations. Feedback edges allow such responses to return to participants who helped produce them.

For the common-stance semantics, write $F_v(P) = F_v((e, P))$. Section 4's horizon evaluator becomes

$$S_v^c(x; H) = R_v^{c, \text{dir}}(\phi(x)) + \lambda_v[F_v(P_v^H) - F_v(P_v^0)] - \delta_v \mathbf{1}\{\zeta_v^1 = -\}. \quad (17)$$

The recognition term and rejection indicator come from the initial encounter; later encounters enter through the endpoint. The observed event need not inherit the value of the judgment it reports.

Example 5.3 (An objection becomes part of an appeal). *Use one visible class, unit values, $\lambda = 1$, thresholds $(0, 0)$, and only p/r conflict. An opponent holds and values r , and her credibility filter is empty. A supporter starts empty, values p, h , and admits precisely those meanings. The speech supplies p . The opponent's report to the supporter is h when her regime is $-$, and empty otherwise. It reads no private semantic coordinates. The meaning h is an observed objection attributed to the candidate's speech. The first two rounds give*

	P_{opponent}	$P_{\text{supporter}}$	S_{opponent}	$S_{\text{supporter}}$
1	$\{r\}$	$\{p\}$	$-\delta$	2
2	$\{r\}$	$\{p, h\}$	$-\delta$	3

The opponent retains her meaning and rejects the appeal. Her expressed objection contributes a meaning valued by the supporter. The supporter need not endorse the objection's judgment.

Expression of the objection and its positive value to the supporter are explicit primitives. Private rejection alone supplies no report. Declared read masks make it possible to trace which changes in a speech can affect expression. Unrestricted reporting gives no guarantee of a nontrivial reduction; the sufficient set may include the whole speech domain.

5.4 Relevance through communication.

Two examples identify what the direct construction must acquire when reports are possible. Both use one visible class, unit values, $\lambda = 1$, zero thresholds, and expansion.

Example 5.4 (A future conflict). *A target starts empty, values h , and admits p, q, h ; only p/q conflict. A relay with empty prior semantics and filter reports p, h after each encounter. Compare the external speeches $\{q\}$ and $\emptyset$. Both initially score zero, but the target acquires q only from the first. At round 2 she rejects the receipt $\{q, p, h\}$ and retains q ; following the empty speech, she incorporates $\{p, h\}$. The horizon-2 scores are therefore 0 and 1. The direct mask $\{h\}$ removes q and changes the later score. That initially unvalued acquisition matters because it can conflict with a later receipt.*

Example 5.5 (Relevance through another audience). *A supporter starts empty and admits and values p, h . An opponent holds r , admits only q , and reports h to the supporter on rejection. Only r/q conflict. The external speeches $\{p\}$ and $\{p, q\}$ give the supporter the same direct score, 2. Only the second makes the opponent reject and object. At horizon 2 the supporter's scores are 2 and 3, although she never admits q . The route from q to her evaluation passes through the opponent's response.*

We can cover both routes without calculating the audience's rankings first. Fix a candidate and suppress its index locally. Choose a fixed set of possible external coordinates containing every feasible speech. For participant i , let $\mathcal{B}_i$ contain that set and the full permitted output ranges of incoming report rules. These bounds cover both an original speech and its projections, including reports produced by either process. They are held fixed across the comparisons below. Define

$$\mathcal{S}_i = P_i^0 \cup (\alpha_i \cap \mathcal{B}_i), \quad (18)$$

$$\mathcal{D}_i = \mu_i^* \cup J_i \cup [\mathcal{S}_i \cap O(\mathcal{B}_i)], \quad (19)$$

$$\Lambda_i = \mathcal{B}_i \cap [(\alpha_i \cap \mathcal{D}_i) \cup O(\mathcal{S}_i)]. \quad (20)$$

An intermediary whose evaluation is unused can have an empty anchor here. The first set bounds meanings a participant can hold. The second retains held meanings that are valued, read by reports, or capable of conflict with a possible receipt. The third retains raw meanings that can populate those states or conflict with any possible held meaning. Keeping both sides of possible conflicts preserves the gate controlling the entire encounter.

Let $d(i, Q)$ be the shortest directed observation distance from i into Q , with $d(i, Q) = 0$ on Q and infinity when there is no path. For an integer $H \geq 1$, put

$$\Lambda_{Q,H}^c = \bigcup_{d(i,Q) \leq H-1} \Lambda_i, \quad \pi_\Lambda(b) = b \cap \Lambda_{Q,H}^c. \quad (21)$$

An encounter at distance d can affect the audience no earlier than round $d + 1$. The construction therefore includes only participants whose own encounters can matter by voting.

Theorem 5.6 (Sufficient content at the election date). *Fix a candidate, the common stance, the observation structure, all initial states and encounter primitives, and the bounds used in (18)–(21). Use either expansion or revision on incorporation and identity otherwise. Replacing the external speech b by $\pi_\Lambda(b)$ preserves every audience member's horizon- H score. Consequently,*

$$\phi(x) \cap \Lambda_{Q,H}^c = \phi(y) \cap \Lambda_{Q,H}^c \implies x \sim_{Q,H}^c y. \quad (22)$$

Feedback and active regimes are allowed. The process need not converge.

The proof follows the rounds that can affect the audience by H . It preserves the possible conflict witnesses, additions and any revision deletions within retained state, and the declared inputs to reports. Anchor intersections and the initial evaluative record consequently agree. An included intermediary’s retained state and regime need only agree through the rounds that can reach the audience in time. Taking the union over all ancestors instead gives a single set preserving audience scores at every horizon. The small direct set of Proposition 5.1 is not generally the horizon-1 specialization of this construction: the latter also protects possible states and declared reads. The two sets do coincide in the immigration example after intersection with its feasible coordinates.

The earlier examples show what these formulas retain. In Example 5.4, take the target’s raw bound to be $\{p, q, h\}$. Her possible-state bound is also $\{p, q, h\}$: all three meanings can be admitted. The anchor retains h , and potential conflict retains p, q in state and receipt. The construction consequently keeps q even though it has no direct anchored value. In Example 5.5, empty read masks give local retained receipts $\{p, h\}$ for the supporter and $\{q\}$ for the opponent, using the feasible external coordinates and report ranges as bounds. The supporter’s horizon-1 set excludes q ; her horizon-2 set includes it through the opponent. What she needs to distinguish about the speech depends on whose reactions can reach her.

The common stance makes these coordinate comparisons stable across encounters. Adoption and revision can change the assigned meanings, and reports can alter subsequent regimes, while the classes to which those meanings attach remain fixed. A construction covering changing stances would also have to preserve the effects of acquisition and transport on those classes. The present result isolates the routes through meanings, conflict, and expression within the specified categories.

At fixed bounds and read masks, $\Lambda_{Q,H}^c$ weakly enlarges when Q or H grows. It can stay unchanged, and its retained coordinates can remain redundant on the feasible menu. Projection is idempotent because it intersects with a fixed set. The exact response partition refines when listeners are added, but need not refine when voting is delayed.¹⁵

Corollary 5.7 (An influence bound). *Suppose candidate c changes from x to y , with $\phi(x) \triangle \phi(y) \subseteq E$. Define*

$$I_H^c(E) = \{v \in V : E \cap \Lambda_{\{v\},H}^c \neq \emptyset\}.$$

Only these voters can change their evaluation of c . Using horizon- H scores, against fixed rival platforms and their separately attributed processes,

$$|\mathcal{V}_c(y, \mathbf{x}_{-c}) - \mathcal{V}_c(x, \mathbf{x}_{-c})| \leq \frac{\sum_{v \in I_H^c(E)} \omega_v}{\Omega}. \quad (23)$$

The bound allows any finite number of candidates and source-dependent responses.

Retention identifies possible influence; an affected score need not change a vote. In the immigration example, economic-impact G lies outside B ’s and C ’s sets and inside A ’s. Adding just that coordinate to f_B leaves B ’s and C ’s scores unchanged. At thresholds $(1, 1)$ it changes A ’s score from 2 to $2 - \delta$ by introducing rejection. This is a broadcast-level comparison: the modified broadcast is absent from the five-platform menu.

5.5 Feasible reduction.

A representation of speech content becomes a restriction of candidate strategy when its representatives are feasible. To preserve the whole election, retain the evaluations of every voter for every candidate.

¹⁵With one class, unit values, $\lambda = 1$, no conflicts, and full credibility, let a voter hold p_2 and value p_1, p_2 . The speeches $\{p_1\}$ and $\{p_2\}$ score 2 and 1 at $H = 1$. A relay with empty prior semantics and filter reports p_1 after each encounter, giving both speeches score 2 at $H = 2$. With empty read masks and raw bounds $\{p_1, p_2\}$, the sufficient set is $\{p_1, p_2\}$ at both horizons.

One sufficient content set is the union of $\Lambda_{V,H}^c$ over candidates. Alternatively, use the joint score map directly. In either case the finite menu permits a feasible representative from each realized class.

This step connects an audience’s representation to candidate incentives. A candidate may still have several ways to express content that yields the same score vector. Selecting one feasible expression for each class removes those duplications while preserving every voter’s comparison with every rival. If all score vectors are distinct, every platform remains. The reduction’s guarantee concerns the comparisons and incentives preserved, whether the resulting menu is much smaller or unchanged.

Corollary 5.8 (Feasible strategic reduction). *Let $\rho : X \rightarrow X$ choose and fix feasible representatives satisfying*

$$S_v^c(\rho(x); H) = S_v^c(x; H) \quad \text{for every } v \in V, c \in \mathcal{C}, x \in X.$$

Replacing each platform by its representative preserves the profile’s payoff vector. Every full-game deviation has an equivalent deviation in the reduced menu $\rho(X)$. A reduced profile is a pure equilibrium of the reduced game exactly when it is an equilibrium of the full game, and projecting any full-game pure equilibrium gives a reduced equilibrium. These conclusions permit source-dependent responses and any finite number of candidates.

Under horizon source neutrality, the map also preserves both strict-support weights $N(x, y)$ and $N(y, x)$. It therefore preserves the strict absolute-majority relation and the supporters-minus-opponents margin. Every strict majority step projects to a strict step; its endpoints cannot collapse to one representative. Reachability between retained endpoints is preserved, although an original path can leave the representative menu.

Literal clipping is one possible choice of representative when the menu is closed under it. For a simple failure of closure, take one voter with empty prior, anchor $\{p\}$, full credibility, no conflicts, unit values, and $\lambda = 1$. The feasible broadcasts are $\emptyset$ and $\{p, q\}$. Retaining p preserves the scores 0 and 2, but the description $\{p\}$ is infeasible. Restricting the menu to literal clipping fixed points would leave only $\emptyset$, whose symmetric profile admits a profitable deviation to $\{p, q\}$. Choosing the original $\{p, q\}$ as its class’s representative preserves that strategy and comparison.

How much structure does the underlying field impose on these comparisons? The following representation result locates the explanatory role of restrictions on audiences and appeals.

Proposition 5.9 (Representation of rankings). *For fixed finite nonempty voter and platform sets V, X , there is one sufficiently large finite representational field, with a common one-class stance and fixed pairwise disjoint nonempty voter anchor blocks, in which every profile of weak rankings on X is represented by varying only the broadcast map. The construction uses either named incorporation rule, empty prior semantics, full credibility, no conflicts, $\lambda_v = 1$, and valuation differences given by cardinality at the common stance. Each direct score equals twice the platform’s rank in the voter’s weak ranking, with ranks increasing from zero.*

The field and anchors can thus remain fixed while broadcasts represent arbitrary finite ranking profiles. The conditions on feasible content, credibility, conflict, and public expression supply the substantive restrictions on competition. In McKelvey’s Euclidean distance-preference model, absence of a total median yields global majority reachability (McKelvey, 1976, Theorem 2); his broader analysis identifies restrictive conditions on the frontiers of reachable sets (McKelvey, 1979). Here the reduction identifies an audience-dependent representation on which the same electoral comparisons can be studied. Conditions bounding reachability must constrain comparisons or permissible proposal sequences within that retained game.

The relevant distinctions also depend on what is being compared. At fixed primitives, changing positive population weights changes margins while leaving the all-type response partition unchanged.

Adding an audience member can refine that partition; adding time can enlarge a sufficient coordinate set without separating any additional evaluations. These are changes in electoral relevance at a fixed common stance. Perceptual distinctions, equality of responses, coalition membership, and population weight each have a separate role. The next section uses these audience responses to study which appeals electoral competition sustains.

6 Electoral competition among appeals.

Which appeals can electoral competition sustain when their content matters differently to different audiences? The immigration example gives a complete answer on its five-appeal menu. Theorem 6.2 classifies every pure equilibrium by population shares and the cost of rejection. Increasing that cost creates equilibrium for one region of population compositions and destroys it for another; a third region sustains the same unique equilibrium appeal at every charge. These equilibria occur even though no common single-peaked ordering of the appeals exists.

Two further results explain how audience structure enters competition. Under the isolation conditions of Proposition 6.4, candidates can optimize separate components of a speech independently, even with feedback inside each audience. When a strict majority shares a set of best appeals, Proposition 6.6 shows how that majority restricts the contest and the minority selects within the remaining set. Its application to public objection gives the communication mechanism an electoral consequence: sufficiently valued opposition to a speech can make that speech the unique equilibrium appeal.

The final subsection identifies conditions supporting familiar conclusions about common ground and median positions. The results concern scores at the election date. The communication processes producing those scores can be studied without first establishing that they settle.

6.1 The running election.

Section 5 derived different politically relevant distinctions for the three immigration audiences. A responds to factual recognition and to conflict with evaluative content she does not admit. B distinguishes factual alternatives while ignoring their evaluative accompaniment. C responds to that accompaniment and treats the factual alternatives alike. Their population shares determine how these responses combine in the electoral contest. Changing positive shares changes vote margins while leaving the three types' response classes, and their joint response partition, unchanged.

Take two source-neutral candidates and positive type shares a, b, c , summing to one. Retain the common stance, priors, and filters of the running example, with anchors equal to priors, expansion, thresholds $(2, 2)$, unit content values, and $\lambda_v = 1$. The feasible menu is exactly

$$X = \{f_A, f_B, g, f_A + g, f_B + g\},$$

where addition is classwise union. These are all nonempty coherent unions of the three specified appeals; the incompatible factual accounts are never offered together. Only A rejects any of them, so write $\delta = \delta_A$. The scores become

Type	f_A	f_B	g	$f_A + g$	$f_B + g$
A	3	2	$-\delta$	$3 - \delta$	$2 - \delta$
B	2	3	0	2	3
C	0	0	2	2	2

At thresholds $(3, 3)$, nothing is rejected: A's row is $(3, 2, 0, 3, 2)$, and at equal shares the equilibrium appeals are $f_A + g$ and $f_B + g$. Tolerance thus changes which combinations are sustainable with

anchors and menu fixed. Varying the charge instead changes the electoral consequence of A's rejection while holding the conflict assessment fixed. At zero charge, adding g to either factual appeal costs A nothing and attracts C, so each factual appeal loses to its combined version. Positive charges can make that same addition electorally costly.

For the displayed (2, 2) election, the scores also apply at every horizon under fixed repetition without reports. Section 4's evaluator keeps the initial recognition and rejection record fixed, and the first cognitive output persists. Public reports can change endpoint gains; the objection election below uses that route.

The two-candidate game permits a short reduction of the equilibrium calculation. For any source-neutral scores at a fixed horizon, let

$$\mathcal{X}^* = \{x \in X : N(y, x) \leq N(x, y) \text{ for every } y \in X\},$$

where $N(y, x)$ is the weight strictly preferring y to x . Thus $\mathcal{X}^*$ consists of appeals unbeaten by any feasible deviation.

Lemma 6.1 (The pure-equilibrium set). *In a two-candidate source-neutral election with uniform voting ties, the pure equilibria are exactly $\mathcal{X}^* \times \mathcal{X}^*$. Each candidate earns one half at every such profile.*

Copying the rival guarantees half the vote. At equilibrium both candidates must therefore earn half, and each chosen appeal must prevent a profitable deviation by the other candidate. The lemma includes asymmetric profiles: different appeals can be equilibrium choices when they tie in the electoral comparison.

Theorem 6.2 (Population composition and rejection cost). *In the five-appeal election specified above, membership of $\mathcal{X}^*(\delta; a, b, c)$ is given by the following conditions. All shares are positive.*

Appeal	$\delta = 0$	$0 < \delta < 1$	$\delta = 1$	$\delta > 1$
f_A	never	$a \geq \frac{1}{2}$	$a \geq \frac{1}{2}$	$a \geq \frac{1}{2}$
f_B	never	$b \geq \frac{1}{2}, a \geq c$	$b \geq a \geq c$	$b \geq a \geq c$
g	never	never	never	never
$f_A + g$	$a \geq b$	$c \geq a \geq b$	$c \geq a \geq b$	$c \geq \frac{1}{2}, a \geq b$
$f_B + g$	$b \geq a$	$a \leq \min(b, c)$	$a \leq \min(b, c)$	$a \leq \min(b, c)$

The full pure-equilibrium set is the square of the set so identified.

Three open regions give the comparative static its political content:

$$\begin{aligned}
b > a > c, \quad b < \frac{1}{2} : \mathcal{X}^* &= \emptyset \quad (0 < \delta < 1), \quad \mathcal{X}^* = \{f_B\} \quad (\delta \geq 1); \\
c > a > b, \quad c < \frac{1}{2} : \mathcal{X}^* &= \{f_A + g\} \quad (0 \leq \delta \leq 1), \quad \mathcal{X}^* = \emptyset \quad (\delta > 1); \\
a < \min(b, c) : \mathcal{X}^* &= \{f_B + g\} \quad (\delta \geq 0).
\end{aligned}$$

On the first region, A stops preferring $f_A + g$ to f_B at $\delta = 1$. B's greater weight than C then blocks that deviation; above the boundary, A strictly joins B in supporting f_B . On the second region the same comparison turns against the combination that C helped sustain, leaving every appeal vulnerable. On the third, B and C together sustain $f_B + g$ at every charge. The effect of greater aversion to a speech therefore depends on who else values its content and how those audiences compare in size.

Zero cost always permits equilibrium. The boundary ties also matter: when $\delta = 0$ and $a = b$, the two combined appeals give four pure profiles; at equal shares and $\delta = 1$, the equilibrium appeals are $f_B, f_A + g, f_B + g$, giving nine. The table describes the specified positive-share electorate and menu, including its structural ties.

With positive additive coordinate values, the same classification places the positive-charge boundary at $\kappa = R_A^{\text{dir}}(f_A) - R_A^{\text{dir}}(f_B) > 0$, the value of A's contested factual coordinate. Writing $r = R_A^{\text{dir}}(f_B)$, A's row is

$$(r + \kappa, r, -\delta, r + \kappa - \delta, r - \delta).$$

B's and C's rankings are unchanged. Only A's comparison of f_B with $f_A + g$ crosses at $\delta = \kappa$. The equilibrium regions thus extend across positive content values as the charge is measured against that coordinate's value.

Do the stable regions conceal a common single-peaked order?

Proposition 6.3 (No common single-peaked axis). *For every $\delta \geq 0$, no ordering of the five appeals makes all three types' rankings weakly single-peaked: every proposed ordering gives some type a strictly lower score at an intermediate appeal than at two appeals on either side. This holds for all positive additive coordinate values.*

For positive charge, three appeals suffice. A ranks $f_B + g$ uniquely worst among $f_B, f_A + g, f_B + g$; B ranks $f_A + g$ worst; C ranks f_B worst. Whichever appeal lies in the middle of a proposed order creates a strict valley for one type. At zero charge the full menu supplies the obstruction. The example consequently supports pure equilibrium across regions of compositions without a common single-peaked axis. Its claim concerns the orderings of this finite menu; higher-dimensional representations are a separate question. The stability is nevertheless more than a single balanced population. Every composition satisfying $a < \min(b, c)$ sustains $f_B + g$, throughout the charge range, and the positive-value extension preserves this conclusion. This persistence is conditional on the audiences and feasible appeals specified here. Enlarging the menu or changing which content the types identify can introduce comparisons outside the classification.

6.2 Isolated audiences.

A candidate can sometimes combine appeals to separate audiences without sacrificing approval in either. The decisive condition is independence of their responses. Disjoint credibility filters help, but raw content can still trigger conflict, and a report can carry one audience's reaction to another. The following result states sufficient conditions under which the components can be designed independently.

Use Section 5's common-stance coordinate space $\mathcal{Z}_e$ and observation protocol, including its separate candidate processes, delayed reports, and declared state and regime reads. Partition all participants, including intermediaries, into blocks $I_1, \dots, I_k$.

Proposition 6.4 (Independent design under isolation). *Let $E_1, \dots, E_k \subseteq \mathcal{Z}_e$ be pairwise disjoint coordinate domains. For each candidate c , assume:*

1. *If $i \in I_j$, then $P_i^0 \subseteq E_j$ and $\alpha_i^c \subseteq E_j$. Across distinct blocks, $E_j \cap O(E_\ell) = \emptyset$.*
2. *Every observation parent of a member of I_j belongs to I_j , and its report carries only coordinates in E_j .*
3. *Each component menu X_j is a nonempty finite collection of subsets of E_j . The feasible menu is $X = \prod_j X_j$, with broadcast $\bigcup_j z_j$ for platform $(z_1, \dots, z_k)$.*

Fix the priors, anchors, parameters, and finite horizon H , and use either named incorporation rule. Each participant's trajectory, regimes, and reports depend only on its block's component. Against fixed rivals, a candidate's payoff is a sum of block contributions, each depending on that component alone, and its best-response set is the product of the componentwise best-response sets. These conclusions permit source dependence and any finite number of candidates.

Feedback inside a block can affect what its members think and how they vote. The proposition requires those effects to remain functions of the same component. It imposes no additive content valuation: electoral payoff separates because it sums voters' allocations across blocks. Anchors need not themselves be confined to the domains. The result gives a precise sense in which one campaign can address several audiences at once. A change in one component may produce an elaborate response within its audience, yet the candidate can choose that component without reconsidering the others. A public reaction crossing that boundary can make one component consequential for another audience: the audience addressed by a component can differ from the audience whose votes it ultimately affects.

Corollary 6.5 (A jointly optimal polysemic appeal). *Under Proposition 6.4 and horizon source neutrality, suppose each block has a feasible component maximizing every voter in that block's score. The product appeal gives a symmetric equilibrium for any finite candidate set $\mathcal{C}$ with $|\mathcal{C}| \geq 2$. It is polysemic when voters in at least two distinct blocks receive nonempty components, $\alpha_v \cap z_j^* \neq \emptyset$.*

Disjoint domains make these nonempty receipts different, although they arise from one broadcast. Merely offering two nonempty components does not ensure this: both could be filtered out. The equilibrium conclusion uses common maxima; the reception condition identifies when that equilibrium also expresses different meanings to different audiences.

Three changes can defeat independence. Section 5's cross-audience report example lets a meaning in one component cause a report valued in another, changing the latter's score from 2 to 3. A cross-block conflict can have an effect even without admission: with unit content values, prior and anchor $\{\mathbf{p}\}$, filter $\{\mathbf{p}\}$, zero thresholds, and $\mathbf{p}/\mathbf{r}$ conflict, $\{\mathbf{p}\}$ scores 1 while $\{\mathbf{p}, \mathbf{r}\}$ scores $1 - \delta$. Finally, feasible combinations may be restricted. Two unit-weight, empty-prior, conflict-free voters with anchors and filters $\{\mathbf{p}\}$ and $\{\mathbf{q}\}$ can both be satisfied by $\{\mathbf{p}, \mathbf{q}\}$. Removing that speech while retaining the empty and singleton broadcasts prevents joint satisfaction. Against an empty rival, either singleton is a best reply, although the product of the separate component maxima is unavailable. Equilibrium can still exist; product feasibility is what licenses independent assembly.

6.3 Majorities and minorities.

Common interests need not determine a unique appeal. A majority may regard several feasible speeches as equally good while a minority distinguishes them. The next result separates the majority's power to restrict the contest from the comparisons selecting within that restriction.

Proposition 6.6 (Majority response). *Consider two source-neutral candidates at a fixed finite horizon, with uniform voting ties. Let $Q \subseteq V$ have weight above $\Omega/2$, and suppose its members have the same nonempty maximizing set $\mathcal{X}_Q$. Let $N_{V \setminus Q}(y, x)$ be the weight of minority voters in $V \setminus Q$ strictly preferring y to x . Every symmetric-equilibrium appeal belongs to $\mathcal{X}_Q$. For $x \in \mathcal{X}_Q$, the symmetric profile is an equilibrium exactly when*

$$N_{V \setminus Q}(y, x) \leq N_{V \setminus Q}(x, y) \quad \text{for every } y \in \mathcal{X}_Q.$$

The full pure-equilibrium set is the square of the set satisfying this condition.

Outside $\mathcal{X}_Q$, every majority voter strictly prefers a deviation into it. Inside it, that group is indifferent and the minority determines whether a deviation gains votes. The majority need not share an anchor, a prior, or reasons for liking those speeches. Agreement on the entire set of best available appeals is enough. Its size gives the minority room to select among appeals that satisfy the majority equally well. In the immigration election at type weights $(1, 3, 1)$ and $\delta = 3$, B is a strict majority with maximizing set $\{f_B, f_B + g\}$. The other two types balance one another's comparisons, leaving both appeals as equilibrium choices. If instead the minority comparison leaves no member of the majority's maximizing set unbeaten, there is no pure equilibrium. A cycle can coexist with equilibrium when another member of that set defeats every alternative in the cycle.¹⁶

Public reactions can change the majority's maximizing set itself. The next election makes the consequence of observed opposition explicit.

Example 6.7 (When objection makes provocation win). *Two supporters and one opponent have unit weights and share one visible class. Use expansion, zero thresholds, $\lambda = 1$, and source-neutral encounters. Supporters start empty and admit and value $\{c_1, c_2, c_3, p, h\}$. Each meaning has unit value except h , whose value is $\eta > 0$. The opponent's prior and anchor are $\{r\}$, and she admits nothing. Only p and r conflict.*

The menu offers a calm appeal $\{c_1, c_2, c_3\}$ and a provocative appeal $\{p\}$, excluding their combination. Each candidate's speech repeats; the opponent's rejection produces a report $\{h\}$ to both supporters one round later, with no other reports. At $H = 2$, the scores are

Voter	Calm	Provocative
Each supporter	6	$2 + \eta$
Opponent	0	$-\delta_{\text{opp}}$

Take $\delta_{\text{opp}} > 0$. The unique pure equilibrium is calm-calm for $\eta \leq 4$ and provocative-provocative for $\eta > 4$.¹⁷

Calm gives a supporter three units of recognition and three of gain. Provocation initially gives one of each; the reported objection adds η to the endpoint gain. When this exceeds the four-unit gap, the supporters' strict majority sustains provocation. At equality they are indifferent and the opponent selects calm. Before the report arrives, at $H = 1$, calm is uniquely sustained for every $\eta > 0$. The opponent keeps her own meaning throughout: her adverse response changes the supporters' evaluation through the content of the report.

6.4 Benchmarks.

The preceding elections use the full response of the listener. A simpler comparison evaluates a broadcast directly against each voter's anchor,

$$F_v(b) = \mathcal{W}(b \wedge u_v^*).$$

This is the anchored value already used inside the encounter evaluator. When actual rankings agree with these values, the order on understandings supports conclusions about richer broadcasts and shared content. The following bridge identifies two such cases.

¹⁶The minority-selection criterion is specific to two candidates. With majority weight 3, minority weight 2, and three candidates all at the majority's unique best appeal, a deviation attracting the minority earns $2/5 > 1/3$. The unanimous-maximizer corollary above permits any candidate count.

¹⁷At $\eta = 4$, $\delta_{\text{opp}} = 0$, every voter ties the two appeals, so all four pure profiles are equilibria. The other two ranges of η have the same unique equilibria at zero opponent charge.

Proposition 6.8 (Direct-broadcast benchmark). *Fix a common stance e for priors, anchors, and broadcasts, full credibility, and incorporation at every feasible initial encounter. Write $u_{e,0} = (e, 0)$ for the understanding with empty semantics at that stance. If $\lambda_v = 0$, the score is $F_v(b) - F_v(u_{e,0})$ for any allowed incorporation rule, at every horizon even with later reports. If instead the prior has empty semantics and incorporation uses either named rule, the direct score is*

$$(1 + \lambda_v) [F_v(b) - F_v(u_{e,0})].$$

More generally the latter formula holds when $\Sigma^+(\emptyset, A) = A$. Under each stated scope, rankings and ties agree with F_v .

The zero-gain-weight case persists because recognition and rejection are recorded at the initial encounter. With positive gain weight, later reports can change the second formula's endpoint term. The empty-prior condition on the rule matters: an allowed rule that keeps the prior would give recognition without the additional gain.

Recall Section 4's strict-majority convention: a *strict Condorcet winner* x satisfies $N(x, y) > \Omega/2$ for every other feasible platform y .

Proposition 6.9 (Feasible maximality). *In a source-neutral election, if actual rankings and ties agree with the anchored benchmark values, a strict Condorcet winner's broadcast is maximal in $\phi(X)$ under the understanding order.*

Any strictly richer feasible broadcast weakly improves every anchored value. Additional content can leave all values unchanged, so maximality is distinct from being a greatest feasible broadcast or from strictly increasing everyone's score. For actual signed responses, Section 3's thinner-appeal example gives a different possibility: with positive rejection charge, every voter can prefer $\{p\}$ to the richer $\{p, q\}$, scoring them at 1 and $1 - \delta$.

A common axis supplies a further restriction on the rankings themselves. The median conclusion is the familiar majority comparison for single-peaked preferences (Black, 1948); here the hypothesis concerns scores after reception. Downs's spatial account of party competition likewise assumes a shared ordering of alternatives and single-peaked voter preferences (Downs, 1957, pp. 115–116).

Proposition 6.10 (Median benchmark). *Consider two source-neutral candidates, uniform voting ties, and an odd electorate with unit weights. Suppose an injective ordering of the feasible menu makes every voter's actual scores strictly single-peaked, with a feasible peak. The median peak is a strict Condorcet winner and its symmetric profile is an equilibrium.*

Strict single-peakedness requires scores to rise toward each voter's peak and fall away on either side. A policy-focused reception regime is compatible with this hypothesis, but policy content, full credibility, and incorporation alone do not supply it. The theorem applies to any fixed election-date scores satisfying the ranking condition. It provides an explicit point of contact with spatial competition. An empirically defensible common order can support a median description even when scores arise through interpretation and response. The running election shows another possibility: stable competition among appeals whose derived rankings admit no such order.

The majority-response result also clarifies the role of a shared anchor.

Corollary 6.11 (A majority anchor). *In a two-candidate source-neutral election with uniform ties and agreement with benchmark rankings and ties, suppose a group of weight above $\Omega/2$ shares anchor u_A^* . If some feasible broadcast contains u_A^* , every symmetric-equilibrium broadcast b contains it. For a minority sharing anchor u_B^* , every broadcast b containing u_A^* satisfies*

$$0 \leq \mathcal{W}(u_B^*) - F_B(b) \leq \mathcal{W}(u_B^*) - \mathcal{W}(u_A^* \wedge u_B^*).$$

The deficit is positive exactly when $u_B^ \not\leq b$. A feasible broadcast containing all anchors gives every voter maximal benchmark value and zero deficit.*

The deficit measures unsatisfied anchored value. The same bound applies voter by voter to heterogeneous minority anchors. Different, even incomparable, anchors permit such a shortfall, while feasible joint satisfaction removes it.

Common ground also gives a natural interval in the understanding lattice. Let $L = \bigwedge_v u_v^*$ and $U = \bigvee_v u_v^*$.

Proposition 6.12 (Clipping and completing anchored content). *For any broadcast b , clipping $\pi_U(b) = b \wedge U$ preserves every $F_v(b)$, and π_U is idempotent. Completion $P(b) = (b \vee L) \wedge U$ lies in $[L, U]$ and weakly increases every anchored value. If $L \not\leq b$, every increase is strict. These conclusions hold on the full understanding lattice, without distributivity or a common stance.*

Thus upper clipping removes content outside the pooled anchors without changing benchmark evaluations. Completion additionally supplies the common anchor. If $L \not\leq b$, a feasible completion excludes the symmetric profile at b under source neutrality and agreement with benchmark rankings and ties, for any finite candidate set $\mathcal{C}$ with $|\mathcal{C}| \geq 2$. A feasible join U maximizes every anchored value.¹⁸

Feasibility also matters for majority paths. With one visible class, take anchors $\{\mathbf{q}, \mathbf{a}_1, \mathbf{a}_2\}$, $\{\mathbf{q}, \mathbf{b}_1, \mathbf{b}_2\}$, and $\{\mathbf{q}, \mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}$. Offer only

$$x = \{\mathbf{q}, \mathbf{a}_1, \mathbf{a}_2, \mathbf{c}_1\}, \quad y = \{\mathbf{q}, \mathbf{a}_1, \mathbf{b}_1, \mathbf{b}_2\}, \quad z = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}.$$

Empty priors, full credibility, no conflicts, either named rule, unit cardinality differences, and $\lambda = 1$ give voter score rows $(6, 4, 0)$, $(2, 6, 4)$, and $(4, 2, 6)$, with columns x, y, z . With unit voter weights, x defeats y , y defeats z , and z defeats x , each two to one. Both x, y contain the common meaning $\mathbf{q}$; z omits it. The improvement path $x \rightarrow z \rightarrow y$ must therefore leave the anchor interval. Section 5's feasible representatives preserve comparisons by preserving scores; interval membership alone supplies no such guarantee.

The classification and the benchmarks identify different sources of structure. Audience responses and population composition sustain some appeals and leave others vulnerable. Isolation licenses independent assembly; common majority maxima let minority comparisons select; a common single-peaked order supports a median outcome. These conclusions make the restrictions on competition explicit. The next section studies how continued communication changes the understandings and reports on which the election-date evaluations depend.

7 Communication.

What keeps changing if communication continues before the vote? Section 5 identified the routes through which an appeal and the reactions to it can matter to an audience. Allowing more rounds gives those dependencies more time to operate. Under expansion, however, their effects on evaluations eventually stop changing: Proposition 7.1 shows that understandings, scores, and the finite electoral game all become constant. Public reactions can continue cycling after that point.

Settling can also occur in the presence of feedback and lasting disagreement. Under the reporting condition in Theorem 7.2, objections accumulate until the communication process becomes constant, with some participants continuing to reject and others continuing to incorporate. The opponent's objection valued by a supporter is one such outcome. The final two subsections explain what can recur. A ring of reactions cycles while meanings and evaluations remain fixed; under revision, a similar dependence between participants can keep meanings changing and produce recurring electoral phases. The incorporation rule thus helps determine which consequences of communication persist.

¹⁸The ranking bridge matters. At one visible class with full credibility, unit cardinality differences, prior $\{\mathbf{r}\}$, anchor $\{\mathbf{p}\}$, $\mathbf{p}/\mathbf{r}$ conflict, zero thresholds, $\lambda = 1$, $\delta = 2$, and expansion, the empty broadcast scores 0 while its anchor completion $\{\mathbf{p}\}$ scores -1 .

7.1 The augmented state and accumulation.

Unless stated otherwise, use Section 5’s common-stance observation protocol. The participant set I includes voters and any intermediaries, and $\mathcal{Z}_e = (\mathcal{R}_e/\sim_e) \times M$ is the set of visible semantic coordinates. Participant i ’s state after round t is the pair (P_i^t, ζ_i^t) : her held meanings and the regime of her latest encounter. The initial marker $\zeta_i^0 = \diamond$ produces no reports, so round 1 is the direct encounter. Every later receipt pools the same external speech with reports from the preceding round. The fixed report rules read only their declared parent variables. The profile of pairs is therefore a complete autonomous state.

Call the process *settled* when that augmented profile becomes constant. Cognitive settling means only that the understandings become constant. Its electoral consequence follows from the inherited score:

$$S_v^c(x; H) = R_v^{c, \text{dir}}(\phi(x)) + \lambda_v[F_v(P_v^H) - F_v(P_v^0)] - \delta_v \mathbf{1}\{\zeta_v^1 = -\}.$$

The initial recognition and rejection record stay fixed. Only the endpoint varies with the horizon.

Proposition 7.1 (Accumulation and electoral stability). *Under expansion on a finite representation and meaning field, along any specified sequence of encounters:*

1. *Each understanding is nondecreasing in richness and eventually constant. Participant i undergoes at most $n(m+1) - \ell(u_i^0)$ strict changes, where $n = |\mathcal{R}|$, $m = |M|$, and $\ell(u)$ counts visible classes plus their meaning assignments. At a common stance the bound is $|\mathcal{Z}_e| - |P_i^0|$.*
2. *With the fixed initial record and valuation parameters, each horizon score is nondecreasing in H and eventually constant.*
3. *For finitely many candidates, voters, and feasible platforms, following the same separately attributed processes as H varies, the payoff table and pure-equilibrium set eventually become constant.*

The cognitive conclusion allows changing stances and arbitrary history-dependent inputs. The eventual equilibrium set may be empty.

The proof counts what can still be acquired. Each strict increase adds a distinction or a meaning assignment, and the finite field permits only finitely many such additions. Under the autonomous protocol, every recurrent orbit consequently has constant understandings and at most $3^{|I|}$ regime profiles; the proof also gives a bound on when recurrence begins. An input chosen using an additional history or clock can postpone the last acquisition arbitrarily, so the number of cognitive changes alone does not bound their dates.

Individual scores increasing need not make a candidate’s vote share increase: rivals’ scores may also rise. Nor does a settled understanding imply that reactions have settled. With one class, empty prior meanings, full credibility, zero thresholds, and p/q conflict, repeating $\{\mathbf{p}, \mathbf{q}\}$ gives

$$\emptyset \xrightarrow{+} \{\mathbf{p}, \mathbf{q}\} \xrightarrow{-} \{\mathbf{p}, \mathbf{q}\} \xrightarrow{-} \dots$$

The first encounter adds the meanings; the next rejects what now conflicts with them. For any fixed raw receipt, either named incorporation rule fixes the common-stance understanding after one encounter and the pair after at most two. Proposition A.1 supplies the statement and its acquisition extensions. This also explains the running election’s invariance under repetition without reports.

Scores can become constant before cognition does, if subsequent additions carry no anchored value. Once all voter understandings are fixed, the baseline game is fixed, while regimes and the reports they produce can still change. These are distinct observations of the same process.

7.2 Settling with standing disagreement.

Continue Section 5's objection mechanism. The opponent keeps r , rejects the speech $\{p\}$, and reports an objection. The supporter admits p and the valued report h . From round 2 their states are

$$(P_{\text{supporter}}, \zeta_{\text{supporter}}) = (\{p, h\}, +), \quad (P_{\text{opponent}}, \zeta_{\text{opponent}}) = (\{r\}, -).$$

The objection continues to be expressed, and the supporter already holds its meaning. The scores remain 3 and $-\delta$. Section 6's objection election likewise retains its horizon-2 scores at every later horizon. The opponent's retained judgment and the supporter's favorable response to its expression are compatible with a settled communication process.

A general reporting condition supports this outcome even when observations feed back. Order the ordinary regimes by $- < 0 < +$. A sign-only report is *antitone* if lowering the parent's regime can only enlarge the report. A fixed objection token supplied on rejection and absent otherwise satisfies this condition. More generally, reports can add different content as the response moves from incorporation to inaction to rejection.

Theorem 7.2 (Objection feedback settles). *At the common stance, use expansion and the cold-start observation protocol. Suppose every report reads only its parent's regime and is antitone in the regime order. On any observation graph,*

$$P_i^{t+1} \supseteq P_i^t, \quad \zeta_i^{t+1} \leq \zeta_i^t \quad (t \geq 1).$$

The augmented profile is constant from round

$$1 + \sum_{i \in I} (|\mathcal{Z}_e| - |P_i^0|) + 2|I|$$

at the latest. The terminal profile may contain incorporation, inaction, and rejection.

The mechanism is cumulative exposure. Held meanings grow; a lower regime enlarges reports; and each receipt continues to contain the external speech. Conflict exposure therefore cannot fall along this cold-start trajectory. There are only finitely many meanings to add and two downward regime steps per participant. Feedback can propagate objections through the network, but under these conditions it eventually exhausts the changes it can induce. The result requires no agreement among participants.¹⁹

Initialization matters. Two participants holding p , with full credibility, zero thresholds, repeated speech $\{p\}$, and p/q conflict can each receive q exactly when the other rejects. The cold response $(+, +)$ stays fixed. Starting the same ordinary transition at $(-, +)$ produces perpetual alternation. Reports that lose content when a regime falls also require a different argument.

A familiar pooling benchmark provides another sufficient condition. With expansion, unlimited tolerance, full semantic read masks, and reports transmitting held meanings on a complete graph, put $A_i = \alpha_i \cap b$. The direct encounter gives $P_i^1 = P_i^0 \cup A_i$. Subsequent rounds apply

$$F_b(P)_i = P_i \cup A_i \cup \left(\alpha_i \cap \bigcup_{j \neq i} P_j \right).$$

This map is a closure operator: its output is the least fixed extension of its input, reached by the cold process at round 2. Its fixed points form a complete lattice, with least profile A and greatest

¹⁹Section 3's joint exposure result extends the argument to changing stances when the actual composed receipts are antitone in regimes in the full richness order and contain the base broadcast in that order. The state must then include the full understanding. The proof gives the corresponding rank bound.

profile containing every visible meaning assignment. Tarski (1955) supplies the lattice conclusion; the pooling calculation supplies the one-round closure (Proposition A.2 and corollary A.3).

Two further conditions explain how observation structure and memory support settling. On a graph without directed cycles, participants with no observation parents face fixed inputs; once they settle, their children’s inputs become fixed. Both incorporation rules settle by round $2d$, where d is the largest number of participants on a directed path (Proposition A.4).

Reports can also draw on retained meanings and a finite cumulative record of expressed event types. Let B count the initially unused semantic coordinates and event types across participants. Under expansion and autonomous event updates, the cold-start bounds on any graph are $B + 1$ rounds for retained state and $B + 2$ for regimes (Proposition A.5). An event recorded as having occurred can remain available after the corresponding current status has changed. In the ring below, reporting whether a predecessor has ever rejected yields settling, whereas reporting her current regime permits recurrence.

7.3 Recurrence.

The reporting condition governs the conclusion. An objection may elicit another objection, while elsewhere favorable reception may provoke an adverse response. Mixed dependencies can sustain changing reactions to a speech whose meaning nobody changes.

Example 7.3 (A ring of reactions). *Arrange $k \geq 2$ participants in a directed ring. Each holds $\mathbf{p}$ at one visible class, has full credibility and zero thresholds, and receives the speech $\{\mathbf{p}\}$ every round. Only $\mathbf{p}/\mathbf{q}$ conflict. Encode rejection by $x_i^t = 1$ and incorporation by $x_i^t = 0$. The report to i supplies $\mathbf{q}$ when the predecessor’s rejection bit differs from a fixed edge label $\varepsilon_i \in \{0, 1\}$. An edge labeled 0 reports on rejection; one labeled 1 reports on non-rejection.*

Every conflicting receipt is rejected, so all meanings remain $\{\mathbf{p}\}$. The direct response is $x^1 = (0, \dots, 0)$, and thereafter

$$x_i^{t+1} = x_{i-1}^t \oplus \varepsilon_i \quad (t \geq 1).$$

Three participants with $\varepsilon = (1, 1, 0)$ follow

$$000 \longrightarrow 110 \longrightarrow 101 \longrightarrow 000.$$

The reactions have period three. With anchor $\{\mathbf{p}\}$ and unit values, every baseline score is 1 at every horizon.

This is the standard parity structure of a synchronously updated Boolean circuit (Demongeot, Noual and Sené, 2012). Even parity gives two fixed profiles and periods dividing k ; odd parity gives no fixed profile and periods dividing $2k$, with none dividing k . The appendix gives the short proof. The cold start determines which orbit is reached: on three participants, all copying is fixed, all inversion alternates, and one inversion gives period six. Reaction data can therefore display sustained movement while the associated election-date evaluations are already constant.

Endorsement reporting has a useful conditional bound. When understandings are fixed throughout and reports are monotone in the ordinary regime order, the induced regime map is antitone. The direct response bounds its range from above; the even and odd subsequences then become constant, giving eventual period at most two (Proposition A.6). This holds under either incorporation rule. An acquisition transient can instead place the eventual regime map on a six-cycle (Example A.7). The initial semantic state helps select the subsequent reaction dynamics.

The finite-state bound $3^{|I|}$ is attainable by per-edge reporting with self-observation and a sufficiently rich representation field. Proposition A.8 gives the construction. Such flexibility makes restrictions on reporting substantive: the observation graph alone does not determine whether reactions settle.

7.4 Revision.

Expansion preserves what an incorporating encounter previously added. Revision permits an admitted meaning to replace conflicting held meanings. Fixed-input and acyclic settling still hold, as does the endorsement result when understandings are fixed throughout. The accumulation argument depends on expansion, however, and revision can transmit recurring reactions into recurring semantic change.

Example 7.4 (Reactions that change retained meanings). *Use one visible class, an empty speech delivered every round, and three participants X, Y, R . Participant X holds x , admits nothing, and receives O_Y exactly when Y rejected in the preceding round. Participant Y holds y , admits nothing, and receives E_X exactly when X incorporated. Both have zero thresholds. The driver conflicts are x/O_Y and y/E_X . Their regimes follow*

$$(+, +) \longrightarrow (+, -) \longrightarrow (-, -) \longrightarrow (-, +) \longrightarrow (+, +).$$

The receiver R starts empty, admits p, q , and has thresholds $(1, 1)$. She receives p when X 's preceding regime is $+$ and q when it is $-$. These meanings conflict, with no further conflict pairs. With revision, her semantics from round 1 are

$$\emptyset, \{p\}, \{p\}, \{q\}, \{q\}, \{p\}, \dots$$

Every receipt is coherent. The full state and its cognitive projection have period four from round 2.

The receiver always incorporates: one visible class puts her conflict count within tolerance. Each change of report replaces the previous meaning. With expansion she instead holds both meanings from round 4, while the two drivers continue their cycle. The reporting mechanism and delivery remain the same.

This distinction can change electoral outcomes. The five-voter election in Example A.9 has two source-neutral candidates choosing between appeals x and y . From horizon 3, its three receiver voters score x at 1 or 0 by phase against y 's $1/2$. Their majority converts that reversal into an electoral cycle. Under revision its unique pure equilibrium is (x, x) at horizons $H \geq 3$ with $H \bmod 4 \in \{2, 3\}$, and (y, y) otherwise. Under expansion it is (y, y) at every horizon. The comparison holds the communication processes fixed and varies when the electorate votes. Together with the settling results, the example links the timing of competition to what audiences retain and what their reports continue to make relevant.

8 Empirical interpretation.

The framework gives an empirical study several comparisons to make. At a common stance, messages agreeing on an audience's sufficient content yield the same evaluations, even through communication. In direct encounters under expansion, combining content can reduce approval through blocked valued uptake or a new rejection charge. Observing another audience can change approval through the valued content its reaction leaves with the listener. These implications identify what a design should vary and what it must hold fixed. Their interpretation depends on measuring reception, retained meanings, evaluation, and expression separately.

8.1 Response components as measurement tasks.

Comprehending an objectionable claim, admitting it from a source, and incorporating it into an understanding are distinct responses. At a common stance, write $A_v(b) = \alpha_v \cap b$ for admitted content. The table describes the expansion instance; changing stances additionally requires identifying how content is transported. Potential valued uptake $D_v(b)$ is defined in Section 4.

Component	Formal object	Measurement task	Insufficient evidence
Admission	$A_v(b)$	Elicit content the listener admits from this source.	Comprehension alone: excluded content can still trigger conflict.
Uptake	$A_v(b) \setminus P_v^0$, added on incorporation	Compare retained assignments before and after exposure.	No change: incorporation can add nothing.
Blocked uptake	The component's $\lambda_v D_v(b)$, lost when the combination leaves incorporation	Estimate valued uptake forgone when a combination leaves incorporation.	An unincorporated meaning alone: it may already be held or unvalued.
Derogation	Charge δ_v on rejection	Separate the charge from recognition and valued change.	Equal cognition or total evaluation alone does not quantify the charge.
Expression	Report r_{ji}	Record content expressed and delivered to another listener.	Private rejection alone; report wording alone does not identify its source regime.

Voter A illustrates the separation. The appeal $f_B + g$ passes the same factual content through her filter at thresholds $(2, 2)$ and $(2, 3)$. She retains the same understanding, but scores the appeal at $2 - \delta_A$ on rejection and 2 on inaction. Lemma 3.14 identifies this conditional contrast. Using it to estimate a charge requires an independent regime classification, a cardinal evaluation scale, and a comparison preserving the other encounter primitives. At zero charge even the evaluations coincide.

The distinctions have precedents. Zaller (1992, chap. 3) separates receiving and accepting messages from the accessibility of considerations governing response; that accessibility can vary rather than accumulate permanently. Hovland, Harvey and Sherif (1957, pp. 246–248) distinguish acceptable, objectionable, and unclassified positions. An unclassified position need not express psychological indifference. These connections motivate separate measurements without supplying a measurement rule for each primitive.

8.2 Relevance as a testable invariance.

Theorem 5.6 gives the central comparison. For a fixed candidate and horizon, messages agreeing on $\Lambda_{Q,H}^c$ receive equal scores from every member of Q . Against unchanged rivals, those listeners also allocate their votes identically. Total share is preserved when the edit misses every voter's sufficient set; otherwise Corollary 5.7 bounds the share change by the affected weight.

A matched-message design first elicits categories, prior meanings, admitted content, and values. It specifies conflict, report rules, declared reads, and bounds covering possible receipts, including reports from intermediaries. Both treatments must lie within those bounds. Initial states, source parameters, the common stance, horizon, and communication rules are held fixed; messages vary outside or inside the resulting sufficient set. The set is specified independently of the evaluations being tested. This attention to supplied alternatives has a precedent in the account of chooser characteristics and menus in Sniderman and Bullock (2004).

Adding economic-impact G to f_B gives a concrete comparison. That coordinate lies outside B's and C's direct sufficient sets, leaving their scores unchanged. It lies inside A's set but leaves her score unchanged at thresholds $(2, 2)$; at $(1, 1)$, her score changes from 2 to $2 - \delta_A$, a loss when the charge is positive. The edited speech can be an experimental treatment without entering the five-appeal menu whose equilibria were classified.

The test is one-sided. Agreement on sufficient content implies equal scores; disagreement permits either equality or change. A systematic difference under the predicted invariance challenges the specified model together with its measurement mapping. A statistical test therefore needs an account of evaluation error. Modeled updating of held meanings remains allowed. Teaching new distinctions

leaves the common-stance theorem’s scope, while later reports require the horizon construction rather than simply reusing a direct mask. Fixed report rules can produce different realized reports.

8.3 Combined appeals and observed reactions.

The combination decomposition (Proposition 4.5) suggests comparing components with their union at a common stance, using direct encounters, expansion, and fixed source and listener primitives.²⁰ Its four contributions are increased recognition, added valued uptake, blocked component uptake, and a new rejection charge. They guide estimation; potential uptake on a blocked branch requires counterfactual valuation. Matched conditions should preserve the initial prior rather than expose the same listener sequentially without accounting for learning.

For the type-1 listener in Example 4.8, $\{p\}$ scores 2. Adding excluded q blocks incorporation, giving 1 at thresholds $(0, 1)$ and $1 - \delta$ at $(0, 0)$. The first loss comes from blocked valued uptake; the second comparison isolates the charge under the stipulated controls. Within an unchanged regime, adding content weakly raises the score under this specification. Independent design requires the conditions of Proposition 6.4: disjoint domains containing priors, full filters, and report outputs, product feasibility, and restrictions on conflict and observation.

A second design holds the initial speech encounter fixed and varies access to an opponent’s later objection. The election-date evaluator gives

$$\Delta S_v = \lambda_v [F_v(u_v^H \mid \text{report}) - F_v(u_v^H \mid \text{withholding})].$$

For $\lambda_v > 0$, the effect is positive exactly when the report condition leaves greater anchored value than withholding. At $\lambda_v = 0$ it vanishes. Example 5.3 raises the supporter’s score from 2 to 3. Measure attribution, admission, and retained valued content alongside approval; the approval contrast alone does not identify those mediators. In Example 6.7, $\eta = 4$ is the supporters’ indifference threshold from $H = 2$, with unit calm-content values and $\lambda = 1$; the positive opponent charge selects calm at equality. An application must estimate that comparison for its content values and feasible menu. Voting should be measured against a fixed rival: a score increase matters when it reaches, leaves, or crosses the rival comparison.

Timing also distinguishes a retained record that someone objected from a report that someone currently objects. Section 7’s finite-record result gives settling under expansion with a finite event alphabet, autonomous record updates, and reporting determined by the cumulative retained state; current-status reports can recur. A one-encounter measure can miss the supporter’s later uptake even though evaluations eventually settle under expansion (Proposition 7.1). Source-trust ratings alone do not identify content filters. Partisan differences in factual perceptions (Bartels, 2002) motivate eliciting reception directly.

8.4 Majority priorities.

Proposition 6.6 begins with measured maximizing sets over a specified menu and horizon. For two source-neutral candidates with uniform ties, a group above half the electorate’s weight sharing the entire nonempty maximizing set restricts equilibrium appeals to that set; minority comparisons select its unbeaten members. If none is unbeaten, no pure equilibrium exists. At $\delta = 3$, weights $(1, 3, 1)$ leave both f_B and $f_B + g$ available. Weights $(2, 4, 1)$ select the former, and $(1, 4, 2)$ the latter. B remains a strict majority in each comparison.

Anchor comparisons supply a separate benchmark when actual rankings and ties agree with anchored values. A broadcast containing the majority anchor bounds the minority’s deficit; feasible

²⁰Revision gives a different comparison. At one common class, take prior and anchor $\{p\}$, full credibility, only p/q conflict, thresholds $(1, 1)$, and unit values. The empty receipt scores 0; adding q replaces p and scores $-\lambda_v$, while both encounters incorporate.

joint containment makes it zero (Corollary 6.11). Establishing such containment requires measuring access, distinctions, and valued meanings. Factor-analytic orthogonality alone establishes neither lattice incomparability nor a common maximizing set.

8.5 Scope and identification.

Thresholds are cutoffs applied to counts of conflicting receiver classes. Response proportions alone do not identify them: identical distance cutoffs can yield different proportions on the same ordered menu when respondents occupy different positions. The receiver's categories also determine the counting unit. Refinement preserving inherited associations can increase the conflict count; dividing by the number of visible classes does not remove this dependence. Splitting a nonconflicting class can change the fraction from $1/2$ to $1/3$; splitting the conflicting class instead can give $2/3$. Categories and discrepancy rules must accompany any threshold estimate.

Measured psychological characteristics likewise require a bridge to the primitives. Taber and Lodge (2006) measure prior attitudes and sophistication while randomizing study conditions, and distinguish attitude position from strength. Neither characteristic directly measures semantic content or a conflict cutoff. Proposition 5.9 makes the role of restrictions explicit: on a sufficiently rich finite field, varying broadcasts alone realizes every finite weak-ranking profile. An application obtains empirical content by specifying and measuring its messages, reception, and evaluation. Its object is the joint pattern of understanding, response, expression, and choice, connected to observations through an explicit measurement model.

9 Extensions.

Access conditions, feasible proposals, and source effects offer further applications of the framework. Common restriction preserves an ordering of understandings, while different access can make ordered understandings incomparable (Propositions 9.2 and 9.3). A feasible join of legislators' anchors maximizes their anchored direct values (Proposition 9.4). The final remarks explain how source effects and incorporation choices alter the arguments supporting electoral symmetry and semantic relevance.

9.1 Field variation and differential access.

Keep the meaning universe M and marker $\star$ fixed. An access window $S \subseteq \mathcal{R}$ specifies which resources remain available in a comparison.

Definition 9.1 (Field restriction and fragmentation). *For $u = (e, \mu)$, the restriction $\rho_S(u) \in \mathcal{U}_S$ has the nonempty intersections of the marked e -classes with $S \cup \{\star\}$. Each visible $C \cap S$ inherits $\mu(C)$; the marked class has empty semantics. Invisible extension $\iota_S : \mathcal{U}_S \rightarrow \mathcal{U}_{\mathcal{R}}$ adds $\mathcal{R} \setminus S$ to the marked class, leaving visible classes and their meanings unchanged. The ambient view is $\mathbf{V}_S = \iota_S \circ \rho_S$. A fragmentation profile $(S_v)_v$ assigns a window to each voter. The empty window is allowed.*

Unavailable resources thus differ from visible resources with empty meanings. Let t_S distinguish every resource in S , attach all of M there, and leave the rest invisible.

Proposition 9.2 (Common access preserves order). *Restriction is monotone. Invisible extension is an order embedding preserving binary meets and joins, with image*

$$\iota_S(\mathcal{U}_S) = \{(e, \mu) : \mathcal{R}_e \subseteq S\} = \downarrow t_S.$$

Moreover,

$$\rho_S \iota_S = \text{id}, \quad \mathbf{V}_S \mathbf{V}_T = \mathbf{V}_{S \cap T}, \quad \mathcal{R}_{e_{\mathbf{V}_S(u)}} = S \cap \mathcal{R}_{e_u}.$$

Each view is monotone and idempotent, fixing exactly the understandings supported on its window.

A common window can therefore collapse a strict comparison to equality. Cutting a visible class makes a view incomparable with its original; retaining or removing whole classes leaves it below the original.²¹

Proposition 9.3 (Differential access and incomparability). *Let $u_i = (e_i, \mu_i)$, $u_1 \preceq_{\mathcal{W}} u_2$, $w_i = \bigvee_{S_i}(u_i)$, and $A_i = S_i \cap \mathcal{R}_{e_i}$. The views are incomparable if either*

1. *both $A_1 \setminus A_2$ and $A_2 \setminus A_1$ are nonempty; or*
2. *$S_1 = \mathcal{R}$, $\mathcal{R}_{e_1} \setminus S_2 \neq \emptyset$, and $\rho_{S_2}(u_1) \prec_{\mathcal{W}_{S_2}} \rho_{S_2}(u_2)$.*

The first condition gives each view exclusive access: windows $\{a, c\}$ and $\{b, c\}$ suffice for comparable full-scope understandings. The second combines lost access with a retained advantage. On distinguished a, b , let u_1 hold $\{p\}$ at both and u_2 additionally hold q at a . Keeping u_1 whole and restricting u_2 to a produces incomparable views. These conditions are sufficient, not necessary.

In the immigration example, give A window $\{r_1, r_3\}$ and B window $\{r_2, r_3\}$. Their restricted anchors share only $\{T\}$ at economic impact in their meet; the other resources become invisible. The comparison identifies surviving common content between already incomparable anchors.

Among active online news readers, [Flaxman, Goel and Rao \(2016\)](#) associate social and search channels with both greater segregation between users and greater exposure to opposing perspectives within users. Connecting such evidence to these maps requires measuring access, categories, and meanings; the association does not identify windows or a causal access effect.

9.2 The legislative benchmark.

Suppose legislators evaluate proposals by $D_i[b] = \mathcal{W}(b \wedge u_i^*)$, and write $L = \bigwedge_i u_i^*$, $U = \bigvee_i u_i^*$.

Proposition 9.4 (A common maximum). *If U is feasible, it maximizes every legislator's direct value and belongs to the strict-majority core.*

Proof. Every proposal satisfies $D_i[b] \leq \mathcal{W}(u_i^*) = D_i[U]$. No legislator strictly prefers an alternative. $\square$

For the immigration anchors, $U = (\{T, G, B\}; \{T, F, G, B\}; \{T, G, B\})$. L has empty semantics at the full stance. Shared anchor content can be imposed as an admissibility condition; joint satisfaction does not require it. Trade-offs require costs, restrictions, or different payoffs. The join can contain conflict, so applying the benchmark to encounter evaluations requires Section 6's ranking-and-tie conditions. A bargaining account would additionally specify proposal rules and continuation incentives.

9.3 Source effects.

Articulation can govern which offered distinctions a receiver acquires. For a finite family $\mathcal{D}(e_j)$ of stances $d \preceq_E e_j$, let $\theta_i(d)$ be acquisition thresholds and χ_{ij} the source's articulatory strength for receiver i . One possible rule is

$$\Gamma_{ij}(e_i, e_j; \chi_{ij}) = e_i \vee \bigvee \{d \in \mathcal{D}(e_j) : \chi_{ij} \geq \theta_i(d)\}.$$

The empty join is $e_{\perp}$. Outputs lie between e_i and $e_i \vee e_j$; generating e_j and passing every threshold recovers maximal acquisition. With the family and thresholds fixed, increasing articulatory strength

²¹The conservative map $u \wedge t_S$ always gives the greatest understanding below u supported on S . It removes whole classes meeting the discarded region. Restriction itself need not preserve meets.

refines the acquired stance. It can prevent pooling: a source attaching $\mathbf{p}$ only to a reaches an empty-prior listener grouping a, b , with full credibility. Without acquiring the offered distinction she receives $\mathbf{p}$ on the grouped class; acquiring it gives $\mathbf{p}$ at a alone. The resulting understandings are incomparable. With parameters fixed, Section 7’s stance-only fixed-input bounds apply; common-stance results have no acquisition to explain. Acquiring distinctions remains distinct from admitting meanings or tolerating conflict.

A further extension adds a fixed term ξ_{vc} to $S_v^c(x; H)$. It preserves each candidate’s response equivalence while changing comparisons across candidates.²² Source differences can remove equal shares at identical platforms and the common-set equilibrium characterization. Fixed-rival score comparisons and score-preserving reductions retain their scopes.

9.4 Incorporation choices.

At a visible class, expansion and revision retain admitted content, stay within $P \cup A$, are idempotent at fixed receipt A , and yield A from an empty prior. Both satisfy $\Sigma^+(P, \emptyset) = P$; invisible semantics remain empty. Under Section 7’s hypotheses, expansion’s extensivity supports accumulation and fixed-receipt idempotence supports fixed-input and acyclic settling. Extensivity also supports objection settling with antitone sign-only reports, and finite-record settling with autonomous cumulative records and retained-state reporting.

Containment supports isolation because its induction preserves whole inputs. Relevance additionally requires retained inputs to determine retained outputs; the combination calculus uses expansion’s union-based gain. The illustrative rule $P \cup (A \setminus O(P))$ preserves the prior while withholding conflicting additions, separating extensivity from complete admitted-content retention. Each rule’s dependence on its inputs determines which arguments extend.

10 Conclusion.

The distinctions organizing political competition depend on who is listening. An appeal’s content meets an audience’s categories, assigned meanings, and dispositions toward the source. Some differences matter because listeners value them, others because they encounter resistance, and still others because the reactions they elicit become meaningful to someone else. The immigration example makes this dependence visible: at the displayed rejection charge of 3, five appeals yield five response classes for A, three for B, two for C, and five for B and C together. Content excluded by a listener’s credibility filter can still matter through conflict with what she holds.

The sufficient-content construction turns these possibilities into a statement about political relevance. At a common stance, Theorem 5.6 derives coordinates on which agreement guarantees identical audience evaluations at the election date under either incorporation rule, allowing active conflict and feedback. Its inputs identify what listeners value, what may conflict with their retained meanings, and what can reach them through reports. For a specified coalition, the construction combines these dependencies. Adding listeners weakly refines the exact response partition; extending communication weakly enlarges the sufficient set at fixed bounds and read masks, although exact response partitions need not refine. The ambient field and coalition membership are specified, and the sufficient content is not shown minimal. The result locates the politically consequential comparisons within that field.

Those comparisons have strategic consequences. The direct common-stance expansion calculus identifies how adding content can gain recognition and valued uptake while costing blocked uptake

²²With one voter, normalized base scores $s(x) = 0, s(y) = 2$, and bonuses $(1, 0)$, candidate 1’s shares at $(x, x), (x, y), (y, x), (y, y)$ are 1, 0, 1, 1. Pure equilibria are $\{y\} \times \{x, y\}$. Empty prior, anchor/filter $\{\mathbf{p}\}$, no conflicts, unit values, $\lambda = 1$, and empty/ $\{\mathbf{p}\}$ appeals realize the base-score gap under either named rule.

or a rejection charge. In the running election, Theorem 6.2 classifies pure equilibria by audience shares and rejection cost on the specified menu with its structural ties. One appeal is uniquely stable throughout an open region of compositions for every nonnegative rejection cost. Elsewhere, raising that cost creates a pure equilibrium in one region and removes one in another. No common weakly single-peaked order supports the preferences in this election. With the stipulated single-peaked ordering of actual responses, the median benchmark recovers a familiar spatial equilibrium. Communication supplies another route: an opponent's expressed objection can add anchored value for a supporter. Under expansion on a finite field, understandings and evaluations with the fixed initial record eventually become constant, even while public regimes cycle. The resulting pure-equilibrium set can be empty. Revision can sustain semantic and electoral phases; the electoral comparison depends on when the electorate votes. No candidate-adjustment process is assumed.

Further work can ask which sufficient sets are minimal, how acquisition of new categories changes them, and which electoral conclusions survive broader menus or altered ties. The communication results invite comparisons between current reactions and retained records, including recurrent bounds without self-observation. Empirical applications require measurement models for these distinctions and restrictions on the encounters and reports through which they matter. On a sufficiently rich field, Proposition 5.9 permits arbitrary finite ranking profiles. Explanation therefore comes from specifying how messages, reception, and evaluation organize the comparisons a political audience actually makes.

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A Supporting results and proofs.

A.1 Understandings.

Proof of Lemma 2.3. The stances are partitions of the fixed marked set $\mathcal{R}_\star$. Reflexivity and transitivity follow from inclusion of equivalence relations, with the inclusion reversed in the richness order. If $e_1 \preceq_E e_2$ and $e_2 \preceq_E e_1$, their equivalence relations on $\mathcal{R}_\star$ are equal. Their classes containing $\star$ are consequently equal, as are their visible scopes, so $e_1 = e_2$. Moreover, $e_1 \preceq_E e_2$ implies $[\star]_{e_2} \subseteq [\star]_{e_1}$ and hence $\mathcal{R}_{e_1} \subseteq \mathcal{R}_{e_2}$. $\square$

Proof of Lemma 2.4. For a nonempty family $\{e_j\}_{j \in J}$, form equivalence relations on $\mathcal{R}_\star$ by

$$\sim^\wedge = \text{EqCl} \left(\bigcup_j \sim_j \right), \quad \sim^\vee = \bigcap_j \sim_j.$$

Each determines a stance by taking the complement in $\mathcal{R}$ of its marker's class as visible scope. The first relation contains every $\sim_j$, so its stance is a lower bound. Every lower bound has an equivalence relation containing all $\sim_j$ and hence their equivalence closure; it is therefore below this stance. The second relation is contained in every $\sim_j$, so its stance is an upper bound. The relation of any other upper bound is contained in every $\sim_j$, hence in their intersection; the displayed upper bound is therefore least.

The marker's class in the join is $\bigcap_j I_{e_j}$, giving visible scope $\bigcup_j \mathcal{R}_{e_j}$. The marker's class in the meet contains every I_{e_j} and may contain more through transitive closure, so its visible scope is only required to be contained in $\bigcap_j \mathcal{R}_{e_j}$.

The universal relation on $\mathcal{R}_\star$ gives $e_\perp$, and the diagonal relation gives $e_\top$. These lie below and above every stance respectively and supply the join and meet of the empty family. Thus every family has both bounds. $\square$

Proof of Proposition 2.9. For $e_1 \preceq_E e_2$, the fine invisible class is contained in the coarse invisible class, so refinement assigns it $\emptyset$. The other two maps set the coarse invisible semantics to $\emptyset$ explicitly. All three maps therefore take values in the stated fibers; pointwise inclusion is preserved by copying, union, and intersection.

Fix a visible coarse class C . The condition $\mu(C) \subseteq \nu(D)$ for every fine $D \subseteq C$ is equivalent to $\mu(C) \subseteq \bigcap_{D \subseteq C} \nu(D)$. Across all visible coarse classes, these are exactly the comparisons $\text{Ref}(\mu) \preceq_M \nu$ and $\mu \preceq_M \text{All}(\nu)$. Fine classes with an invisible coarse parent add only the automatic condition $\emptyset \subseteq \nu(D)$ to the first comparison. This proves $\text{Ref} \dashv \text{All}$ even when scope expands.

When visible scopes agree, every visible fine class has a visible coarse parent. In this case $\bigcup_{D \subseteq C} \nu(D) \subseteq \mu(C)$ for every visible C is equivalent to $\nu(D) \subseteq \mu(C_D)$ for every visible D , proving $\text{Crs} \dashv \text{Ref}$. For contrast, take a coarse stance seeing only a and a finer stance distinguishing and seeing a, b . Let the coarse semantics be empty and the fine semantics attach a nonempty meaning set only to b . Coarsening erases that content, so $\text{Crs}(\nu) \preceq_M \mu$, but $\nu \not\preceq_M \text{Ref}(\mu)$.

Finally, a fine class and its intermediate parent have the same ancestor at any coarser level. Copying along either route therefore gives the same semantics, including when that ancestor is invisible. This proves refinement composition; refinement at an unchanged stance is the identity. $\square$

Proof of Proposition 2.10. The reception formula is a union of sender meanings at each visible receiver class and assigns $\emptyset$ at the invisible class. It is therefore a well-defined monotone map into M_{e_i} . The classes at $k = e_i \vee e_j$ are precisely the nonempty intersections of classes from the two marked partitions. For a visible receiver class C , the k -classes inside C are its nonempty intersections with sender classes. Refinement from e_j assigns each such intersection the semantics of its sender

parent; an invisible sender parent contributes $\emptyset$. Coarsening into C consequently unions exactly the meanings of visible sender classes meeting C , proving the identity.

If $e_i \preceq_E e_j$, a sender class meeting a visible receiver class lies wholly inside it, giving the coarsening formula. If $e_j \preceq_E e_i$, a visible receiver class has one sender parent; it receives that parent's meanings, or $\emptyset$ if the parent is invisible, giving refinement. $\square$

Proof of Theorem 2.11. First, the understanding relation is a partial order. Reflexivity follows from identity refinement. For transitivity use refinement composition and monotonicity from Proposition 2.9. Mutual comparison forces equal epistemic stances by Lemma 2.3, and then equal semantics by antisymmetry of pointwise inclusion.

Let $\{(e_j, \mu_j)\}_{j \in J}$ be nonempty, and let $h = \bigwedge_j e_j$ and $k = \bigvee_j e_j$. Use the semantics in the theorem statement, with their invisible values set to $\emptyset$. They lie in M_h and M_k by the transport proposition.

Meet. At a visible h -class C , the proposed semantics is

$$\mu_\wedge(C) = \bigcap_j \bigcap_{\substack{D \in \mathcal{R}_{e_j} / \sim_{e_j} \\ D \subseteq C}} \mu_j(D).$$

It is included in every term, so its refinement to each e_j is below μ_j on classes with a visible h -parent. Classes with an invisible h -parent inherit $\emptyset$. Together with $h \preceq_E e_j$, this proves the lower-bound property.

For greatestness, let (e', μ') be any lower bound. Then $e' \preceq_E h$. A visible h -class C has a unique e' -parent C' . If that parent is invisible, its refinement into C is empty. Otherwise, for every j and fine $D \subseteq C$, the lower-bound assumption gives $\mu'(C') \subseteq \mu_j(D)$, since $D \subseteq C \subseteq C'$. Intersecting these inclusions gives $\text{Ref}_{e' \rightarrow h}(\mu')(C) \subseteq \mu_\wedge(C)$. Thus (e', μ') lies below the proposed meet.

Join. At stance k , the proposed semantics is the pointwise union of $\text{Ref}_{e_j \rightarrow k}(\mu_j)$. Each member is therefore below it. If (e', μ') is any upper bound, then $k \preceq_E e'$. For each visible e' -class D , its parent at k has an ancestor at every e_j . The upper-bound assumption includes that ancestor's semantics in $\mu'(D)$; invisible ancestors contribute empty semantics. The union over j is consequently included as well, establishing $\text{Ref}_{k \rightarrow e'}(\mu_\vee) \preceq_M \mu'$. The proposed join is least.

Extrema. The empty semantics at $e_\perp$ refines to empty semantics at every stance, so $u_\perp$ is below every understanding. At $e_\top$, assigning M to each resource singleton and $\emptyset$ to $\{\star\}$ gives an upper bound for every understanding. These supply the empty-family join and meet, completing the proof. $\square$

A.2 Encounters and their evaluation.

Proof of Proposition 3.5. Write $u_v^0 = (e, \mu)$, $b = (f, \nu)$, and $b^\alpha = (f, \nu^\alpha)$, where $\nu^\alpha(D) = \alpha^f(D) \cap \nu(D)$ on visible source classes. Suppose the regime is $+$ and put $k = e \vee f$. Maximal acquisition selects k . Since $f \preceq_E k$, overlap reception from f to k is refinement, as established in Section 2. The expansion instance therefore gives the semantics

$$\text{Ref}_{e \rightarrow k}(\mu) \cup \text{Ref}_{f \rightarrow k}(\nu^\alpha),$$

with pointwise union and empty semantics on the invisible class. By the join formula in Section 2, this is precisely the semantics of $u_v^0 \vee b^\alpha$ at k . Thus $T(u_v^0, b) = u_v^0 \vee b^\alpha$. Full credibility makes $\nu^\alpha = \nu$, yielding the unfiltered join. The regime hypothesis is necessary for identifying this incorporation calculation with the selected cognitive output. $\square$

Proof of Proposition 3.9. Let $\bar{e}$ be the actual output stance and let $r = (\bar{e}, \text{Rec}_{f \rightarrow \bar{e}}(\nu^\alpha))$ be the admitted receipt as an understanding. At this stance $u_{\bar{e},0} \preceq_{\mathcal{U}} r$. Meet is order-preserving in

either argument, and the strictly increasing content valuation is in particular weakly increasing. Consequently $F_v(u_{\bar{e},0}) \leq F_v(r)$, proving $R_v \geq 0$. An empty receipt makes the two arguments equal, so recognition is zero. This argument makes no assumption about the incorporation instance or the selected regime.

On either identity branch, $T(u_v^0, b) = u_v^0$, hence $G_v = 0$. The score formula then gives $s_v = R_v$ on 0 and $s_v = R_v - \delta_v$ on $-$. Under expansion, the incorporating output contains the refined prior, and its stance refines the prior stance. Thus $u_v^0 \preceq_{\mathcal{R}} T(u_v^0, b)$. Monotonicity of F_v gives $G_v \geq 0$ on $+$; together with $\lambda_v \geq 0$ and $R_v \geq 0$, this gives a nonnegative incorporating score.

Still under expansion, suppose $u_v^* \preceq_{\mathcal{R}} u_v^0$. Every regime leaves the output weakly above the prior, so

$$T(u_v^0, b) \wedge u_v^* = u_v^* = u_v^0 \wedge u_v^*.$$

Applying the valuation shows $G_v = 0$ in every regime. Neither recognition nor the rejection indicator is determined by this equality.

Finally, a bottom broadcast has an all-invisible stance $e_{\perp}$ and empty semantics. Its raw receipt is empty, so $d = 0$ and the nonnegative thresholds select $+$. The acquisition bounds force $\Gamma(e, e_{\perp}) = e$, since $e \vee e_{\perp} = e$. There is no admitted content. Both named instances return the transported prior: expansion unions with the empty set, while revision has no conflict neighbours to remove when its admitted input is empty. Thus the output is the prior, recognition and gain vanish, and there is no rejection charge. The same conclusion holds for any other incorporation instance satisfying $\Sigma^+(P, \emptyset) = P$. $\square$

Proof of Lemma 3.13. Compare $\alpha^f(D) \subseteq \alpha'^f(D)$ at every visible class of the fixed raw source $b = (f, \nu)$. All other encounter parameters, the prior, and the evaluative objects remain fixed. Filtering gives $\nu^\alpha(D) \subseteq \nu^{\alpha'}(D)$. The raw conflict test is independent of either filter, so the regimes agree. On $+$ both encounters select the same stance $\Gamma(e, f)$; on 0 or $-$ both retain e . Write $\bar{e}$ for this common output stance.

Overlap reception preserves pointwise inclusion, hence the admitted receipt under α' is weakly above that under α in the common semantic fiber. Under expansion, union with the same refined prior preserves this comparison on $+$. On the identity branches the cognitive outputs are equal. Thus in every regime the larger filter gives a weakly richer cognitive output.

For recognition, the two receipts share the subtraction term $F_v(u_{\bar{e},0})$. Monotonicity of F_v therefore gives $R_v^{\alpha'} \geq R_v^\alpha$. The cognitive comparison likewise gives $G_v^{\alpha'} \geq G_v^\alpha$. The rejection charge is unchanged because the regimes coincide. With $\lambda_v \geq 0$, substitution in the direct score yields $s_v^{\alpha'} \geq s_v^\alpha$. Each comparison concerns this fixed encounter; no comparison between the priors reached after different earlier encounters is used. $\square$

Proof of Lemma 3.14. Fix the prior, raw source, acquisition rule, filter, and evaluative parameters, and write $d = d(u_v^0, b)$. First compare $\tau_1 \leq \tau'_1 \leq \tau_2$ under expansion. If $d > \tau_2$, both regimes are $-$ and both outputs equal the prior. Otherwise both regimes are the same except possibly when $\tau_1 < d \leq \tau'_1$. In that case the regime changes from 0 to $+$, and the output changes from the prior to its weakly richer expansion output. This proves cognitive monotonicity; strict change is not required.

If the actual output stance is unchanged across the comparison, reception uses the same stance and the same filtered message. The filtered receipt and recognition are therefore unchanged. At the possible 0-to- $+$ crossing, the gain changes from zero to a nonnegative value, while neither regime incurs the rejection charge. The direct score consequently weakly increases. Minimal acquisition is sufficient to keep the output stance fixed, as is a common source and receiver stance by the acquisition bounds. Without an unchanged output stance, the recognition comparison does not follow: both the receipt and its reference understanding can change.

Next compare $\tau_2 \leq \tau'_2$ with τ_1 fixed and legal. If $d \leq \tau_1$, both encounters incorporate using identical inputs and the same incorporation instance. If $d > \tau_1$, both retain the prior. The output stance, filtered receipt, recognition, and cognitive output are therefore identical for this comparison, for any fixed incorporation instance. Only when $\tau_2 < d \leq \tau'_2$ does the regime change, from $-$ to 0 . The score then rises by exactly δ_v ; otherwise it is unchanged. $\square$

Calculation for the acquisition example. For the two-resource example accompanying Lemma 3.14, write e_c for the stance with visible class $\{x_1, x_2\}$ and e_f for its split into singletons. The prior at e_c carries $\{p\}$, the anchor at e_c carries $\{r\}$, and the source at e_f carries $\{q\}$ at x_1 and $\{r\}$ at x_2 . The only conflict pair, up to symmetry, is (p, q) , so raw reception gives $d = 1$. The filter excludes q and admits r . Use maximal acquisition, expansion, $\tau_2 = 1$, and the valuation

$$\mathcal{W}(e, \mu) = |\mathcal{R}_e / \sim_e| + \sum_{x \in \mathcal{R}_e} |\mu([x]_e)|.$$

This valuation is strictly increasing in the understanding order: strict epistemic refinement increases the number of visible classes without decreasing the resourcewise semantic count, while a strict semantic increase at a fixed stance increases that count.

At $\tau_1 = 0$ the regime is 0 , and the admitted meanings pool to $\{r\}$ at e_c . The receipt's anchored value is 3 and the empty receipt's anchored value is 1 , so $R_v = 2$. The output is the prior and $G_v = 0$. At $\tau_1 = 1$ the regime is $+$ and acquisition selects e_f . The receipt is $(\emptyset, \{r\})$ and the output is $(\{p\}, \{p, r\})$. To meet either understanding with the coarse anchor, the semantics common to both fine classes must be retained. The receipt and the output therefore each have empty meet semantics with that anchor, as do the prior and the empty receipt at e_f . All four anchored values are 1 . Thus $R_v = G_v = 0$ after acquisition, and the direct score falls from 2 to 0 although the output becomes richer.

Proof of Proposition 3.16. Write $u = (e, \mu)$, $u' = (e', \mu')$, $b = (f, \nu)$, and $b' = (f', \nu')$, with $u \preceq_{\mathcal{U}} u'$ and $b \preceq_{\mathcal{U}} b'$. For every $C \in \text{Conf}(u, b)$ choose a prior meaning $p \in \mu(C)$ and a conflicting received meaning q . By overlap reception there is a visible source class D with $q \in \nu(D)$ and $C \cap D \neq \emptyset$. Choose a resource $x \in C \cap D$, and let C' and D' be its classes at e' and f' respectively. These are visible classes contained in C and D .

The understanding comparisons preserve the refinement of every old association, so $p \in \mu'(C')$ and $q \in \nu'(D')$. Because $x \in C' \cap D'$, raw reception of b' at e' supplies q to C' . The original conflict pair (p, q) consequently witnesses $C' \in \text{Conf}(u', b')$. Distinct old receiver classes have disjoint refined children. The selected witness child for each old conflicting class therefore defines an injection

$$\text{Conf}(u, b) \hookrightarrow \text{Conf}(u', b').$$

Taking cardinalities proves $d(u, b) \leq d(u', b')$. This is a count comparison across partitions, not inclusion of their class sets.

For the fixed-stance special case, hold the source fixed and suppose $\mu(C) \subseteq \mu'(C)$ at each receiver class. The raw receipt and its conflict neighbours are then unchanged. Every prior witness for conflict remains a witness at the same class, giving literal conflict-set inclusion in this special case. Finally, with fixed thresholds the map from d to ζ is weakly decreasing under the order $- < 0 < +$. The count inequality therefore implies the asserted regime comparison. No filter, acquisition, coherence, or semantic-update assumption enters this raw-exposure argument. $\square$

A.3 Appeals and the electoral game.

Proof of Proposition 4.4. Fix a candidate c and the rivals' platforms. For each voter v , let $m_v = \max_{j \neq c} s_v^j(x_j)$ be her highest rival score and let $k_v \geq 1$ be the number of rivals attaining it. If c

receives score t from this voter, its vote probability is

$$h_v(t) = \begin{cases} 0, & t < m_v, \\ 1/(k_v + 1), & t = m_v, \\ 1, & t > m_v. \end{cases}$$

This function is weakly increasing. For $a \leq b$, it increases strictly between a and b exactly when $a < b$ and $a \leq m_v \leq b$. Apply this observation with $a = s_v^c(x)$ and $b = s_v^c(y)$. The candidate's share change is the sum of these probability changes with positive coefficients ω_v/Ω . Under the assumed weak score increase for every voter, all summands are nonnegative, and the sum is positive exactly when at least one voter satisfies the stated strict crossing condition.

Now suppose that responses are source-neutral and there are two candidates. At (x, x) each candidate receives half of each voter's vote probability, hence share $1/2$. A deviation to y changes the deviator's probability by $1/2$ for a voter with $s_v(y) > s_v(x)$, by $-1/2$ for a voter with $s_v(x) > s_v(y)$, and by zero for an indifferent voter. Weighting and summing gives

$$\mathcal{V}_1(y, x) - \mathcal{V}_1(x, x) = \frac{N(y, x) - N(x, y)}{2\Omega}.$$

The same calculation applies to the other candidate. Requiring this quantity to be nonpositive for every feasible y proves the symmetric equilibrium characterization. If x strictly majority-defeats every other feasible platform under the stated absolute-majority convention, then $N(x, y) > \Omega/2$ and $N(y, x) \leq \Omega - N(x, y) < \Omega/2$ for every $y \neq x$; the equilibrium condition follows. The condition can hold without an absolute-majority winner: if every voter is indifferent between all feasible platforms, every symmetric profile is an equilibrium and no platform strictly majority-defeats another when the menu contains at least two platforms.

For the extension in (10), let $m \geq 2$ source-neutral candidates all choose x . Each initially receives probability $1/m$ from every voter. A deviation to y changes that probability by $(m - 1)/m$ for a strict supporter of y , by $-1/m$ for a strict opponent, and by zero for an indifferent voter. Therefore the share change is

$$\frac{(m - 1)N(y, x) - N(x, y)}{m\Omega}.$$

The symmetric profile is an equilibrium exactly when $(m - 1)N(y, x) \leq N(x, y)$ for every feasible y . This calculation concerns deviations from symmetric profiles; no restriction on asymmetric profiles has been imposed or derived. $\square$

Proof of Proposition 4.5. All comparisons take place at the common stance, so the acquisition bounds force that stance to remain unchanged in every regime. If $b \preceq c$, every raw-message witness to conflict between the fixed prior P_v and b remains a witness for c at the same class. Thus the conflict set for b is contained in that for c , and its class count weakly increases. The fixed thresholds give

$$\mathbf{1}\{\zeta_v(c) = +\} \leq \mathbf{1}\{\zeta_v(b) = +\}, \quad \mathbf{1}\{\zeta_v(c) = -\} \geq \mathbf{1}\{\zeta_v(b) = -\}.$$

Filtering at the common stance preserves inclusion, so $A_v(b) \preceq A_v(c)$. Meet with the fixed anchor and the valuation are order-preserving. Since the same empty understanding is subtracted from both recognition terms, $R_v(c) \geq R_v(b)$. Union with the fixed prior also preserves inclusion, giving $D_v(c) \geq D_v(b) \geq 0$.

The actual direct gain equals $D_v(b)$ on incorporation and zero on the two identity branches. Consequently

$$s_v(b) = R_v(b) + \lambda_v \mathbf{1}\{\zeta_v(b) = +\} D_v(b) - \delta_v \mathbf{1}\{\zeta_v(b) = -\}.$$

Subtract the corresponding formulas for c and b , and write the gain difference as

$$\begin{aligned} & \mathbf{1}\{\zeta_v(c) = +\}D_v(c) - \mathbf{1}\{\zeta_v(b) = +\}D_v(b) \\ &= \mathbf{1}\{\zeta_v(c) = +\}[D_v(c) - D_v(b)] - [\mathbf{1}\{\zeta_v(b) = +\} - \mathbf{1}\{\zeta_v(c) = +\}]D_v(b). \end{aligned}$$

This is (12), with every bracketed quantity nonnegative. The argument requires only monotonicity of the valuation, not additivity.

If the larger speech incorporates, both speeches incorporate and neither is rejected, so both loss terms vanish. More generally, both loss terms vanish whenever the two regimes agree. These observations give the stated weak score comparisons. Applying the incorporation-indicator inequality separately to each voter also shows that the set of incorporators weakly shrinks with raw content.

For completeness, suppose that the component speeches b, d and their pointwise union $z = b \cup d$ are all feasible. Then

$$\begin{aligned} s_v(z) &\geq \max\{s_v(b), s_v(d)\} \\ &\quad - \lambda_v \max\{\mathbf{1}\{\zeta_v(b) = +\}D_v(b), \mathbf{1}\{\zeta_v(d) = +\}D_v(d)\} \\ &\quad - \delta_v \mathbf{1}\{\zeta_v(z) = -\}. \end{aligned} \tag{24}$$

Choose a component attaining the larger score and apply the decomposition to its inclusion in z . Discard the nonnegative gain terms. Its lost uptake cannot exceed its actual component uptake, which is bounded by the displayed maximum; its increase in the rejection indicator cannot exceed $\mathbf{1}\{\zeta_v(z) = -\}$. These inequalities prove the bound. Feasibility is used to interpret the union as an available strategy; it is not supplied by the pointwise set operation. $\square$

Proof of Proposition 4.6. Let $y \in \mathcal{I}_Q$ and let x be feasible with $\phi(x) \preceq \phi(y)$. For each $v \in Q$, the conflict-count inequality from the proof of Proposition 4.5 gives $d(P_v, \phi(x)) \leq d(P_v, \phi(y)) \leq \tau_{1v}$. Thus every member of Q incorporates x , proving $x \in \mathcal{I}_Q$. This is a down-set property of the preorder induced on feasible platform labels by the broadcast order; no injectivity of ϕ is needed.

Now take $x, y \in \mathcal{I}_Q$ and suppose that $A_v(\phi(x)) \preceq A_v(\phi(y))$ for every $v \in Q$. In this comparison the raw broadcasts need not be ordered. For each member, monotonicity of F_v gives $R_v(\phi(x)) \leq R_v(\phi(y))$ and $D_v(\phi(x)) \leq D_v(\phi(y))$. Both regimes are $+$, so their scores are $R_v + \lambda_v D_v$. Since $\lambda_v \geq 0$, every member weakly prefers y to x . $\square$

Proof of Proposition 4.7. Fix a voter v and a feasible rival platform y . Write $b^* = \phi(x^*)$ and $b = \phi(y)$. The admitted-content hypothesis and monotonicity of F_v imply $R_v(b^*) \geq R_v(b)$ and $D_v(b^*) \geq D_v(b) \geq 0$. Because v incorporates x^* ,

$$\begin{aligned} s_v(x^*) &= R_v(b^*) + \lambda_v D_v(b^*) \\ &\geq R_v(b) + \lambda_v D_v(b) \\ &\geq R_v(b) + \lambda_v \mathbf{1}\{\zeta_v(b) = +\}D_v(b) - \delta_v \mathbf{1}\{\zeta_v(b) = -\} = s_v(y). \end{aligned}$$

The last inequality uses $\lambda_v, \delta_v \geq 0$ and $D_v(b) \geq 0$. It allows every regime for the rival speech. Thus x^* maximizes every voter's score over the feasible menu.

Source neutrality makes these score comparisons valid for a candidate deviating against a rival at x^* . In particular $N(y, x^*) = 0$ for every y , and the two-candidate criterion in Proposition 4.4 proves equilibrium. For any finite $m \geq 2$, the same equality makes $(m-1)N(y, x^*) - N(x^*, y) \leq 0$. Equation (10) therefore proves that the symmetric profile with all m candidates at x^* is an equilibrium as well. $\square$

Calculation for Example 4.8. Write $P = \{p\}$, $Q = \{q\}$, and $Z = \{p, q\}$ for the three platforms. Without conflict, type 1's ranking is $P \sim Z \succ Q$ and type 2's is $Q \sim Z \succ P$. With r/q conflict, type 1's ranking is $P \succ Z \succ Q$ for every $\delta \geq 0$, while type 2's is unchanged. The inaction variant has the same rankings as the conflict case. With weights 2 and 1, the deviating candidate's shares, with its own choice indexing rows and the rival's choice indexing columns, are therefore

No conflict	P	Q	Z	Conflict or inaction	P	Q	Z
P	1/2	2/3	1/3	P	1/2	2/3	2/3
Q	1/3	1/2	1/6	Q	1/3	1/2	1/6
Z	2/3	5/6	1/2	Z	1/3	5/6	1/2

The first table makes Z the unique best reply to every rival choice, yielding the unique pure equilibrium (Z, Z) . In the second table, the unique best replies to P, Q, Z are respectively P, Z, P . Only (P, P) consists of mutual best replies. The tables cover all nine pure profiles, and their constancy in $\delta \geq 0$ establishes the claim throughout that range.

A.4 Audience and coalition relevance.

Proof of Proposition 5.1. Fix a candidate and voter, suppress their indices, and write P for the prior, U^* for the anchor's semantics, and $A = \alpha \cap b$ for admitted content. Let $b' = b \cap \Lambda$, where Λ contains $(\alpha \cap U^*) \cup O(P)$, and put $A' = \alpha \cap b'$. All sets are at the common stance.

Every raw coordinate in b conflicting with a coordinate of P belongs to $O(P)$ and is retained. Conversely, clipping introduces no new coordinate. The initial conflict witnesses are therefore preserved class by class, giving identical conflict counts and regimes. Also $A \cap U^* = A' \cap U^*$, which preserves recognition, including its common empty-receipt subtraction term.

On either identity branch the endpoints equal P . On incorporation under expansion, intersecting $P \cup A$ and $P \cup A'$ with U^* gives the same result. Under revision, an anchored prior coordinate $p \in P \cap U^*$ can be deleted only by an admitted coordinate $q \in A$ conflicting with it. Such a q belongs to $O(P)$, so it remains admitted in A' . Deletions of anchored prior coordinates therefore agree, as do anchored additions. Thus revision also gives equal endpoint intersections with U^* . The common stance and these equal intersections preserve anchored value for the fixed valuation, hence valued change and the direct score.

For repetition, hold the admitted receipt fixed. The two incorporating maps are idempotent:

$$(P \cup A) \cup A = P \cup A, \quad ((P \setminus O(A)) \cup A] \setminus O(A)) \cup A = (P \setminus O(A)) \cup A.$$

An initially inactive or rejecting encounter repeats from the same prior. After an incorporating encounter, another incorporation returns the same state by idempotence, while either other regime also retains it. Each process therefore keeps its first cognitive output at every later round. Their anchored endpoints and frozen initial evaluative records agree, so their scores agree at every horizon. This argument does not require their later regimes to agree. Taking a union of individual masks preserves each member's conditions; the same applies to a union across candidates. $\square$

Proof of Theorem 5.6. Fix the candidate and suppress its index. Compare the external speech b with $b' = b \cap \Lambda_{Q,H}$; primes mark the latter process. All bounds and report rules are the same in both processes. First, for every participant and every round,

$$P_i^t \subseteq \mathcal{S}_i, \quad P_i'^t \subseteq \mathcal{S}_i.$$

Indeed, the priors satisfy this inclusion, every addition belongs to $\alpha_i \cap \mathcal{B}_i$, and neither identity nor revision creates any other coordinate. The receipt bounds apply in both processes because they cover the external coordinate pool and the full permitted report ranges.

Put $d_i = d(i, Q)$. We prove equality of retained states $P_i^t \cap \mathcal{D}_i = P_i^{t'} \cap \mathcal{D}_i$ and positive-round regimes $\zeta_i^t = \zeta_i^{t'}$ whenever $t \geq 1$ and $t + d_i \leq H$. At round zero all priors and initialization markers agree, including outside this cone.

Consider an included participant at round t . Its own preceding retained state is covered by the induction hypothesis, or by the identical initialization when $t = 1$. For each parent $j \rightarrow i$, $d_j \leq d_i + 1$, so

$$(t - 1) + d_j \leq t + d_i \leq H.$$

Thus the parent's preceding retained state and regime also agree. Since $J_j \subseteq \mathcal{D}_j$, its report rule reads identical arguments and emits the same report. At initialization reports are empty by definition. Pooling therefore gives raw receipts agreeing on Λ_i : every incoming report agrees, and the two external speeches agree there because $d_i \leq H - 1$ implies $\Lambda_i \subseteq \Lambda_{Q,H}$.

Take any conflict witness between a current coordinate $p \in P_i^{t-1}$ and a raw coordinate $q \in b_i^t$. The state and receipt bounds put $p \in \mathcal{S}_i$ and $q \in \mathcal{B}_i$. Their conflict implies

$$p \in \mathcal{S}_i \cap O(\mathcal{B}_i) \subseteq \mathcal{D}_i, \quad q \in \mathcal{B}_i \cap O(\mathcal{S}_i) \subseteq \Lambda_i.$$

Both coordinates are preserved. The same argument starting from the primed process proves the reverse implication. Each class is therefore contested in one process exactly when it is contested in the other. The conflict counts and selected regimes agree.

The identity branches retain the equal portions of the preceding states. On incorporation, every admitted addition into $\mathcal{D}_i$ lies in $\alpha_i \cap \mathcal{D}_i \cap \mathcal{B}_i \subseteq \Lambda_i$, so retained additions agree. This proves the expansion case. For revision, any deletion of a retained current coordinate p is caused by an admitted conflicting q . Since $p \in \mathcal{S}_i$, that q lies in $\mathcal{B}_i \cap O(\mathcal{S}_i) \subseteq \Lambda_i$. Its admission is identical under the fixed filter, so retained deletions agree as well. This completes the induction for either named rule.

In particular, voter $v \in Q$ has identical anchor intersections at round H , since $\mu_v^* \subseteq \mathcal{D}_v$. At the first encounter, admitted anchor coordinates are retained in Λ_v , preserving initial recognition; the initial regime also agrees. The fixed initial rejection indicator, prior value, and endpoint value are therefore identical. Substitution in the horizon score proves the claim without any convergence assumption.

If two feasible broadcasts agree on $\Lambda_{Q,H}$, their clips are identical. Applying preservation to each proves response equivalence. The same proof works for every fixed superset of this mask, which justifies taking unions across candidates.

For an edge $i \rightarrow j$, a report emitted after round t is preserved whenever $t + 1 + d_j \leq H$: then $t + d_i \leq H$, so its arguments agree. Replacing the finite cone by all ancestors allows the same induction at every round. Finally, intersection with the fixed mask is idempotent. With the graph, bounds, and read masks fixed, enlarging the audience or horizon only enlarges the set of included ancestors, and hence weakly enlarges the union defining $\Lambda_{Q,H}$. $\square$

Proof of Corollary 5.7. Fix the modifying candidate c . Outside the stated affected set, the edit changes no coordinate of $\Lambda_{\{v\},H}^c$, so Theorem 5.6 preserves that voter's score for c . Every rival's score is fixed as well. The voter's allocation among candidates consequently remains identical. For an affected voter, the modifying candidate's vote probability can change in absolute value by at most one. The triangle inequality applied to the weighted share gives

$$|\Delta \mathcal{V}_c| \leq \frac{1}{\Omega} \sum_{v: E \cap \Lambda_{\{v\},H}^c \neq \emptyset} \omega_v.$$

The argument uses neither source neutrality nor a two-candidate restriction. $\square$

Proof of Corollary 5.8. Let $Y = \rho(X) \subseteq X$ be the representative menu, with ρ fixing its members. For every profile, replacing each platform by its representative preserves every voter–candidate score. It therefore preserves each voter’s highest-scoring candidate set, uniform tie allocation, and the whole payoff vector.

At a profile in Y^C , every full-game deviation to x has the same payoff as the reduced-game deviation to $\rho(x)$. Conversely, every reduced deviation is available in the full game. The two games consequently have exactly the same profitable-deviation test at such a profile, proving the equilibrium equivalence. If a full-game profile is an equilibrium, its projected profile has the same payoffs; any profitable reduced deviation there would give an equally profitable full-game deviation before projection. Thus projection also carries full-game equilibria to reduced-game equilibria.

Under horizon source neutrality, score preservation gives $N(\rho(x), \rho(y)) = N(x, y)$ and the reverse equality. Both the absolute-majority comparison and the strict-support margin are preserved on each projected step. A strict step cannot collapse onto a single representative, whose scores against itself are equal. These equalities concern the projected path; they impose no restriction requiring the original path to remain in Y . $\square$

Proof of Proposition 5.9. Put $n = |X| \geq 1$. Use one resource and, for each voter v , a fixed block of distinct meanings $M_v = \{m_{v,1}, \dots, m_{v,n}\}$, disjoint across voters. The finite meaning universe is their union. Give everyone the visible stance e , empty prior semantics, anchor M_v , full credibility, no conflicts, and $\lambda_v = 1$. Fix the valuation

$$\mathcal{W}(e, P) = 1 + |P|, \quad \mathcal{W}(u_\perp) = 0.$$

It is strictly increasing on this field’s understanding order. All these primitives are chosen before any ranking profile.

For a specified profile of weak rankings, assign each voter an integer rank $r_v(x) \in \{0, \dots, n-1\}$, equal on indifference classes and strictly increasing with preference. Define the broadcast map at e by

$$\phi(x) = \bigcup_{v \in V} \{m_{v,k} : 1 \leq k \leq r_v(x)\}.$$

An empty index range contributes nothing. Every encounter incorporates, and either named rule takes the empty prior to this broadcast. Its intersection with v ’s anchor has exactly $r_v(x)$ meanings. Recognition is $r_v(x)$, as is valued uptake; the constant one in the valuation cancels in both differences. Hence $s_v(x) = 2r_v(x)$, realizing the specified weak rankings by changing only ϕ in the fixed field. $\square$

A.5 Electoral competition among appeals.

Proof of Lemma 6.1. For source-neutral scores, put

$$m(x, y) = \frac{N(x, y) - N(y, x)}{\Omega}.$$

With two candidates and uniform ties, the share from choosing x against y is $[1 + m(x, y)]/2$. A candidate can obtain 1/2 by copying its rival. Since shares sum to one, both must obtain 1/2 at any pure equilibrium (x, y) . The absence of profitable deviations then requires $m(z, y) \leq 0$ and $m(z, x) \leq 0$ for every feasible z . Thus $x, y \in \mathcal{X}^*$. Conversely, two members of $\mathcal{X}^*$ have $m(x, y) \geq 0$ and $m(y, x) \geq 0$, hence tie. Neither admits a profitable deviation against it. The pure-equilibrium set is therefore exactly $\mathcal{X}^* \times \mathcal{X}^*$. $\square$

Proof of Theorem 6.2. Use normalized shares and write $m(x, y) = N(x, y) - N(y, x)$. First suppose $\delta > 0$. To cover positive coordinate values as well as the unit-value statement, write $r = R_A^{\text{dir}}(f_B) > 0$ and $\kappa = R_A^{\text{dir}}(f_A) - R_A^{\text{dir}}(f_B) > 0$. A's row is

$$(r + \kappa, r, -\delta, r + \kappa - \delta, r - \delta).$$

B's and C's rankings and ties remain those in the displayed table. Let $t = \text{sgn}(\delta - \kappa)$. The strict-support margins are given by the following upper triangle; reversing a comparison changes its sign:

$m(x, y)$	f_B	g	$f_A + g$	$f_B + g$
f_A	$a - b$	$a + b - c$	$a - c$	$2a - 1$
f_B	—	$a + b - c$	$ta + b - c$	$a - c$
g	—	—	$-a - b$	$-a - b$
$f_A + g$	—	—	—	$a - b$

For example, A's comparison of f_B with $f_A + g$ contributes ta , B contributes b , and C contributes $-c$. This is the only margin whose sign formula changes at a positive charge boundary.

A platform belongs to $\mathcal{X}^*$ precisely when all its margins are nonnegative. The first row gives f_A exactly when $a \geq 1/2$. For f_B , the requirements are

$$b \geq a, \quad a \geq c, \quad a + b \geq c, \quad ta + b \geq c.$$

At $t = -1$ these reduce to $b \geq 1/2$ and $a \geq c$; at $t = 0$ or 1 they reduce to $b \geq a \geq c$. For $f_A + g$ the requirements are $c \geq a$, $a \geq b$, and $c \geq ta + b$. They reduce to $c \geq a \geq b$ at $t = -1, 0$, and to $c \geq 1/2$ with $a \geq b$ at $t = 1$. For $f_B + g$ they reduce to $a \leq \min(b, c)$. The appeal g loses to both combined appeals because $a + b > 0$. These reductions use $a + b + c = 1$ and positivity of all three shares.

At $\delta = 0$, $f_A + g$ weakly improves on f_A for everyone and strictly improves it for C; similarly $f_B + g$ defeats f_B . Both combined appeals defeat g . Their mutual margin is $a - b$, so the zero-charge column follows. Taking $\kappa = 1$ gives the four columns of the theorem; Lemma 6.1 supplies all pure profiles, including asymmetric ones.

On $b > a > c$ with $b < 1/2$, the positive-charge columns give no equilibrium below κ and only f_B at or above it. On $c > a > b$ with $c < 1/2$, only $f_A + g$ survives through $\delta = \kappa$, including zero, and none survives above. On $a < \min(b, c)$, only $f_B + g$ survives at every nonnegative charge. The boundary counts follow by substitution: at zero charge with $a = b$ there are two equilibrium speeches; at equal shares and $\delta = \kappa$ there are three, while the other positive-charge regimes have two. Squaring these sets gives the stated numbers of pure profiles. $\square$

Proof of Proposition 6.3. For $\delta > 0$, restrict attention to $f_B, f_A + g, f_B + g$. A uniquely ranks $f_B + g$ last: its deficits against the other two are δ and κ . B uniquely ranks $f_A + g$ last, and C uniquely ranks f_B last. Any ordering puts one of these three between the others, creating a strict valley for the voter who ranks it last.

At $\delta = 0$, A and B uniquely rank g last, so it must be an endpoint of a common weakly single-peaked order. Reverse the order if necessary to put it first. C ties $g, f_A + g, f_B + g$ at the top and puts both factual appeals strictly below them. Both combined appeals must therefore precede both factual appeals; otherwise C has a strict valley between two top-ranked appeals. If $f_A + g$ precedes $f_B + g$, A has a valley at $f_B + g$ between $f_A + g$ and the later f_A . If their order is reversed, B has a valley at $f_A + g$ between $f_B + g$ and the later f_B . Neither order works. The argument uses only the stated positive-value rankings, so it covers all $\delta \geq 0$ at that scope. $\square$

Proof of Proposition 6.4. Fix the candidate and a block I_j . States start in E_j and remain there: both named rules add only filtered meanings in E_j , while deletion and identity preserve containment.

Compare two external speeches with the same component z_j . Suppose block states and regimes agree through round t . Every parent of a block member is in the block, so its declared state and regime reads agree and its delayed report is identical. Other external components are filtered out and, by $E_j \cap O(E_k) = \emptyset$, cannot conflict with a held meaning. The raw conflict count and admitted receipt are therefore identical in the two processes. Either named update gives the same next state and regime. Identical priors and empty initial reports start this induction, including when the graph has feedback.

Each block voter's initial recognition, initial rejection indicator, and endpoint depend only on z_j . Fixed anchor evaluation consequently gives the asserted score dependence; the anchor need not lie inside E_j .

Against fixed rivals, each voter's probability of choosing the candidate is a function of its score and the fixed rival scores. Summing with coefficients ω_v/Ω over voters in I_j gives the block's payoff contribution, a function of z_j alone. Contributions sum to total share. Product feasibility permits any selection of their maximizers, and such a selection maximizes the sum. Conversely, a component that fails to maximize its contribution could be changed alone to increase total share. This proves the best-response identity. The argument allows source-dependent primitives, any finite candidate count at least two, and arbitrary fixed content valuation; it does not require convergence. $\square$

Proof of Corollary 6.5. By isolation and product feasibility, the selected components jointly maximize every voter's score. Under horizon source neutrality all candidates attain those same maxima at the symmetric profile. A deviation cannot improve any voter's allocation to the deviator, so cannot raise its expected share. If voters in two distinct blocks receive nonempty filtered components, those receipts occupy disjoint coordinate domains and are different. This is polysemy in the stated reception sense. $\square$

Proof of Proposition 6.6. Let $W_Q = \sum_{v \in Q} \omega_v > \Omega/2$ and $W_{V \setminus Q} = \Omega - W_Q$. If $x \notin \mathcal{X}_Q$, every voter in Q strictly prefers every $y \in \mathcal{X}_Q$ to x . Deviating from (x, x) to y earns at least $W_Q/\Omega > 1/2$, so such an x cannot be an equilibrium speech.

Fix $x \in \mathcal{X}_Q$. A deviation outside this set loses every voter in Q and earns at most $W_{V \setminus Q}/\Omega < 1/2$. For a deviation to $y \in \mathcal{X}_Q$, all voters in Q remain indifferent. Section 4's deviation calculation therefore reduces to

$$\frac{N_{V \setminus Q}(y, x) - N_{V \setminus Q}(x, y)}{2\Omega}.$$

Nonpositivity for every such y is necessary and sufficient. The same indifference makes the majority's contribution equal for both candidates whenever both choose inside the set. Applying Lemma 6.1 gives the square of the set defined by these minority inequalities as the full pure-equilibrium set. It is empty exactly when there is no unbeaten member of $\mathcal{X}_Q$. $\square$

Calculation for Example 6.7. At horizon 2 the supporters' score difference, provocative minus calm, is $\eta - 4$; the opponent's is $-\delta_{\text{opp}}$. Their normalized strict-support margin is therefore

$$m(\text{provocative}, \text{calm}) = \frac{2 \operatorname{sgn}(\eta - 4) - \mathbf{1}\{\delta_{\text{opp}} > 0\}}{3}.$$

It is negative for $\eta < 4$, positive for $\eta > 4$, and at $\eta = 4$ negative exactly when $\delta_{\text{opp}} > 0$. With two feasible appeals, the strict margin winner is the only member of $\mathcal{X}^*$; when the margin is zero, both belong. Lemma 6.1 gives the stated unique pure equilibria and the four-profile zero-charge tie. At horizon 1 the supporters' difference is instead $2 - 6 = -4$, while the opponent's comparison is unchanged. Calm therefore has the strict margin advantage for every $\eta > 0$ and $\delta_{\text{opp}} \geq 0$ before the report arrives.

Proof of Proposition 6.8. Full credibility at the common stance makes the initial receipt the broadcast itself, so recognition is $F_v(b) - F_v(u_{e,0})$. Initial incorporation makes the rejection indicator zero. When $\lambda_v = 0$, no endpoint term remains; this proves the first formula at every horizon for any allowed incorporation rule.

For an empty prior and a rule satisfying $\Sigma^+(\emptyset, A) = A$, the direct endpoint equals the broadcast. Its valued change is the same difference $F_v(b) - F_v(u_{e,0})$ as recognition, giving the multiplier $1 + \lambda_v$. Both named rules have this empty-prior property. The common additive constant and positive multiplier preserve benchmark rankings and ties at the stated respective scopes. $\square$

Proof of Proposition 6.9. Suppose a strict Condorcet winner has broadcast b and there is a strictly richer feasible broadcast b' . Monotonicity of meet and valuation gives $F_v(b') \geq F_v(b)$ for every voter. Agreement of rankings and ties with the benchmark means no voter strictly prefers b to b' . The winner therefore cannot strictly majority-defeat the platform expressing b' , a contradiction. Its broadcast is maximal in the feasible image; the argument makes no comparison with incomparable broadcasts. $\square$

Proof of Proposition 6.10. Order the voters' feasible peaks along the given injective ordering, with repetitions when voters have the same peak, and let x^* be their median. For any y to its right, at least $(|V| + 1)/2$ voters have peaks at or left of x^* . Strict single-peakedness makes each strictly prefer x^* to y . For a y to its left, the voters with peaks at or right of x^* supply the corresponding strict majority. Thus x^* is a strict Condorcet winner. The source-neutral two-candidate criterion makes its symmetric profile an equilibrium. This argument uses the actual rankings and the odd unit-weight electorate, not a particular reception construction. $\square$

Proof of Corollary 6.11. A feasible broadcast containing u_A^* attains the majority's maximal benchmark value $\mathcal{W}(u_A^*)$. Strict increase of $\mathcal{W}$ makes equality possible exactly when $b \wedge u_A^* = u_A^*$, equivalently $u_A^* \preceq b$. Benchmark ranking-and-tie agreement therefore identifies the majority's common maximizing set. By Proposition 6.6, every symmetric-equilibrium broadcast contains u_A^* .

For any broadcast containing u_A^* and minority anchor u_B^* ,

$$u_A^* \wedge u_B^* \preceq b \wedge u_B^* \preceq u_B^*.$$

Apply $\mathcal{W}$ and subtract from $\mathcal{W}(u_B^*)$ to obtain the deficit bounds. Strict increase makes the deficit positive exactly when b does not contain u_B^* . The argument applies separately to each minority anchor if they differ. A feasible broadcast containing all anchors gives every voter maximal benchmark value, hence maximal actual score under the ranking agreement, and supports symmetric equilibrium with zero deficits. $\square$

Proof of Proposition 6.12. Since $u_v^* \preceq U$ for every voter, associativity of meet gives

$$(b \wedge U) \wedge u_v^* = b \wedge u_v^*.$$

Thus clipping preserves every anchored value, and repeated clipping has no further effect.

Write $P(b) = (b \vee L) \wedge U$. Since $L \preceq U$, we have $L \preceq P(b) \preceq U$. Moreover,

$$P(b) \wedge u_v^* = (b \vee L) \wedge u_v^* \succeq b \wedge u_v^*.$$

This proves weak improvement for every anchored value. If $L \not\preceq b$, the new meet contains L , whereas $b \wedge u_v^*$ cannot contain L because it lies below b . The comparison is therefore strict for every voter; strict increase of $\mathcal{W}$ preserves that strictness. These arguments use lattice identities and order alone, without distributivity or a common stance.

Under the stated electoral bridge, if $L \not\preceq b$ and $P(b)$ is feasible, every voter strictly prefers the completion. From the source-neutral symmetric profile at b , a candidate deviating to it receives every vote, increasing share from $1/|\mathcal{C}|$ to one. Hence the original profile is not an equilibrium. A feasible U gives every voter maximal anchored value and thus a common actual-score maximum under the bridge, supporting symmetric equilibrium. These weak maxima need not yield strict majority victories over all other feasible platforms. $\square$

A.6 Communication.

Proof of Proposition 7.1. For $u = (e, \mu)$, the rank

$$\ell(u) = |\mathcal{R}_e / \sim_e| + \sum_{C \in \mathcal{R}_e / \sim_e} |\mu(C)|$$

lies between zero and $n(m+1)$, where $n = |\mathcal{R}|$ and $m = |M|$. A proper refinement increases the number of visible classes, and transport preserves every old assignment in each visible child. If the stance stays fixed, a strict semantic increase raises the second summand. Strict richness therefore strictly increases this integer rank. Expansion is extensive; inaction and rejection preserve the prior. Participant i therefore has at most $n(m+1) - \ell(u_i^0)$ strict changes. At a common stance, count unused coordinates instead. These arguments apply along every realized input sequence, including history-dependent reporting.

Meeting with a fixed anchor preserves the order, and $\mathcal{W}$ is increasing. The only horizon-varying term in Section 4's evaluator is the endpoint value, multiplied by $\lambda_v \geq 0$. Each score is consequently nondecreasing and eventually constant. Taking a maximum of settling horizons over the finitely many candidate/platform processes makes all scores, ties, vote shares, payoffs, and best responses constant together. The resulting pure-equilibrium set may be empty.

For the autonomous protocol, put $B_U = \sum_i [n(m+1) - \ell(u_i^0)]$. After round 1, each cognitive profile admits at most $3^{|I|}$ regime profiles. Leaving a cognitive profile by strict increase prevents returning to it, and at most $B_U + 1$ cognitive profiles occur. Thus a complete state repeats within $1 + (B_U + 1)3^{|I|}$ rounds. Determinism makes that repetition recurrent. A recurrent orbit cannot contain a strict cognitive increase, so its period is at most $3^{|I|}$. At a common stance one may replace B_U by $\sum_i (|\mathcal{Z}_e| - |P_i^0|)$. This calendar requires the complete autonomous state. An unrecorded clock can delay the last new meaning arbitrarily long without violating the bound on numbers of changes. $\square$

Proposition A.1 (A fixed input). *At a common stance, fix a raw receipt and all encounter parameters. Under either named rule, $T_b^2 = T_b$, and the understanding/regime pair is constant after at most two encounters, without a coherence assumption. The same conclusions hold with changing stances under maximal acquisition. For Section 3's deterministic stance-only partial acquisition, the pair is constant after at most $n + 2$ encounters, where $n = |\mathcal{R}|$; the limiting stance need not be the join.*

Proof. An initial inaction or rejection leaves the understanding unchanged, so the identical encounter repeats. After incorporation at a common stance, the admitted receipt A remains fixed and both semantic maps are idempotent:

$$(P \cup A) \cup A = P \cup A, \quad (((P \setminus O(A)) \cup A) \setminus O(A)) \cup A = (P \setminus O(A)) \cup A.$$

A subsequent nonincorporating encounter also preserves the output. The second regime is computed from that output and is thereafter constant. Conflicts within A do not alter these identities. Maximal acquisition reaches $e \vee f$ on the first incorporation, after which the stance and admitted receipt are fixed, giving the same argument.

For partial acquisition, each incorporation either strictly refines the stance or reaches a stance fixed by Γ . There are at most n strict refinements. At a fixed stance, one semantic application and one regime confirmation suffice. An earlier inaction or rejection stops the process sooner. At a common stance every admissible acquisition rule fixes that stance. Finally, the main text's ambivalent receipt gives regimes $+, -, -, \dots$ with unchanged cognition after the first encounter, proving sharpness of the two-encounter bound. $\square$

Proof of Theorem 7.2. Expansion makes $P_i^{t+1} \supseteq P_i^t$. Raw conflict counts are nondecreasing in both held meanings and received content. The second receipt contains the original speech, while its recipient holds at least the original prior; hence $\zeta^2 \leq \zeta^1$. If $\zeta^t \leq \zeta^{t-1}$, antitone sign-only reporting makes the receipt at round $t+1$ contain the receipt at round t . Exposure monotonicity therefore gives $\zeta^{t+1} \leq \zeta^t$. This induction begins after the direct encounter and places no order on the marker.

After round 1, every strict augmented transition consumes an unheld coordinate or a downward regime step. The total budget is at most

$$\sum_i (|\mathcal{Z}_e| - |P_i^0|) + 2|I|.$$

Equality between consecutive ordinary states is permanent because the state determines all future inputs. Adding the first round gives the claimed settling calendar. Different participants can finish in different regimes.

For the changing-stance extension, keep the full understanding in the state and require the actual composed receipts to be monotone in full richness along this induction. Section 3's exposure monotonicity then supplies the same argument, with the cognitive rank budget in place of the coordinate budget. Persistence of token labels alone does not supply that premise. $\square$

Proposition A.2 (Pooling with a persistent speech). *Fix a common stance, expansion, unlimited tolerance, full semantic reads, and a complete observation graph. Reports are empty at the marker and transmit the parent's held semantics at every ordinary regime. Write $A_i = \alpha_i \cap b$ for the admitted base speech and define*

$$T(P)_i = P_i \cup \left(\alpha_i \cap \bigcup_{j \neq i} P_j \right), \quad F_b(P)_i = T(P)_i \cup A_i.$$

Both maps are monotone, extensive, and idempotent on $(2^{\mathcal{Z}_e})^I$. From the cold start, $P^1 = P^0 \cup A$ and $P^2 = T(P^1) = F_b(P^0)$, with $P^t = P^2$ for $t \geq 2$.

Proof. Put $G = \bigcup_i P_i$. Then $T(P)_i = P_i \cup (\alpha_i \cap G)$, and $\bigcup_i T(P)_i = G$. Reapplying T adds nothing. For F_b , the global stock can grow to $G \cup \bigcup_i A_i$, but every newly available coordinate admissible to i already belongs to $A_i = \alpha_i \cap b$. Reapplying F_b therefore also adds nothing. Monotonicity and extensivity follow directly from the formulas. The same observation gives $T(P^0 \cup A) = F_b(P^0)$, establishing the delayed protocol calculation. All ordinary encounters incorporate. $\square$

Corollary A.3 (The pooling fixed-point lattices). *For either $\Phi = T$ or $\Phi = F_b$, $\Phi(P)$ is the least fixed profile above P . Fixed profiles form a complete lattice: meets are ambient intersections and joins are Φ applied to ambient unions. Moreover,*

$$\text{Fix}(F_b) = \{P \in \text{Fix}(T) : P_i \supseteq A_i \text{ for all } i\}.$$

The extremes are the empty and full profiles for T , and A and the full profile for F_b .

Proof. If Q is fixed and $P \subseteq Q$, monotonicity gives $\Phi(P) \subseteq \Phi(Q) = Q$; idempotence makes $\Phi(P)$ fixed. For any family of fixed profiles, monotonicity and extensivity make its ambient intersection fixed, and applying the least-extension result to its union gives its least fixed upper bound. This includes empty families. Finally $T(A) = A$, and the formula for F_b gives the displayed fixed-point identity and extrema. These are the explicit lattice operations in the monotone-map fixed-point conclusion. $\square$

Proposition A.4 (Acyclic observation). *At a common stance, an acyclic observation graph of depth d , counting participants on a longest directed path, settles by round $2d$ under either named rule from any initial augmented state. Reports may read any declared parent information. In a changing-stance extension, include full understandings in the state and require fixed parents to supply fixed raw receipts. The bounds are $2d$ under maximal acquisition and $(n + 2)d$ under deterministic stance-only partial acquisition.*

Proof. A root has fixed input and settles by round 2 by Proposition A.1. If all parents settle by round $2h$, their child's receipt is fixed from round $2h + 1$. Two fixed-input encounters settle the child by $2h + 2$. Induction proves the bound; initial regimes affect only the transient. Substituting the changing-stance fixed-input bound at each depth proves the extensions. $\square$

Proposition A.5 (Finite retained records). *At a common stance, supplement each expansion understanding with a record $L_i \subseteq E_i$, where E_i is finite. Let $Y^t = ((P_i^t, L_i^t))_i$. After the cold direct encounter, receipts depend deterministically only on the fixed speech and declared parent components of Y^t . Following the new encounters, update*

$$L_i^{t+1} = L_i^t \cup \eta_i(Y^t, \text{new encounters}),$$

where each fixed deterministic η_i takes values in 2^{E_i} and reads no clock, unrecorded history, or independent previous regime. Put

$$B = \sum_i (|\mathcal{Z}_e| - |P_i^0|) + \sum_i (|E_i| - |L_i^0|).$$

There are at most B strict retained transitions. Retained state is constant by round $B + 1$, and regimes by round $B + 2$.

Proof. The retained state is extensive on a finite product of power sets. Every strict transition consumes at least one unit of the displayed budget. From round 1 its transition is autonomous, so an identity transition repeats forever. At most B further strict transitions are possible, proving the retained-state calendar. Regimes are computed from the preceding retained state and hence become constant at most one round later. Reports in the cold first round are separately suppressed, so its retained-state transition need not be the ordinary map. $\square$

The offset can matter. One self-observing participant holds both $\mathbf{p}, \mathbf{q}$ at each of two visible classes, admits nothing, has only $\mathbf{p}/\mathbf{q}$ conflict and thresholds $(0, 1)$. The base supplies $\mathbf{q}$ at the first class. After the marker, a report supplies $\mathbf{q}$ at the second class while the record is empty. The sole event, *objected*, is retained on rejection. Here $B = 1$: cognition cannot grow, and one event can be added. Regimes are $\diamond, 0, -, 0, 0, \dots$; the event appears at round 2. Retained state settles at $B + 1$, regimes at $B + 2$. The retained state still becomes constant if its cumulative updates read arbitrary hidden history, but that assumption alone provides neither a calendar nor eventual constancy of current regimes.

Ring parity in Example 7.3. For k participants let $E = \sum_i \varepsilon_i \bmod 2$. After k applications of $F(x)_i = x_{i-1} \oplus \varepsilon_i$, each bit has traversed every edge, so

$$F^k(x) = x \oplus (E, \dots, E).$$

The map is invertible. At even parity every period divides k , and solving the fixed-point equations successively gives exactly two solutions. At odd parity, F^k complements every bit, so no period divides k , while F^{2k} is the identity. The cold encounter selects the round-1 profile; it is not itself an application of the Boolean map. $\square$

Proposition A.6 (Endorsement with fixed understandings). *Suppose understandings are fixed throughout the common-stance protocol, and every report is monotone in its parent's ordinary regime, with its semantic argument fixed. From the cold direct encounter, the eventual augmented period is at most two. This holds under either named rule.*

Proof. More favorable preceding regimes give larger receipts, hence weakly greater conflict and lower next regimes. The induced map G on $\{-, 0, +\}^I$ is antitone. Every receipt contains the base speech, so $G(z) \leq \zeta^1$ for every ordinary profile z . Antitonicity and this range bound yield

$$\zeta^2 \leq \zeta^4 \leq \dots, \quad \zeta^1 \geq \zeta^3 \geq \dots.$$

Each subsequence stabilizes on the finite product order. Their limits are exchanged by G , giving period one or two. The marker is outside G 's ordinary domain. $\square$

Empty filters ensure fixed cognition under both rules. Already-held admitted content suffices under expansion only: revision of $\{\mathbf{p}, \mathbf{q}\}$ by $\{\mathbf{p}\}$ removes $\mathbf{q}$ when those meanings conflict and the threshold permits incorporation.

Example A.7 (An acquisition transient before a six-cycle). *Use one common visible class, expansion, zero thresholds, empty repeated speech, and three participants with priors, filters, and anchors*

	P^0	α	μ^*
1	$\{\mathbf{a}\}$	$\emptyset$	$\emptyset$
2	$\emptyset$	$\{\mathbf{E}_3\}$	$\{\mathbf{E}_3\}$
3	$\emptyset$	$\{\mathbf{E}_1\}$	$\emptyset$

Only $\mathbf{a}/\mathbf{E}_3$, $\mathbf{E}_1/\mathbf{E}_3$, and $\mathbf{E}_1/\mathbf{E}_2$ conflict. Each following edge reports its listed token exactly when its parent incorporates, and otherwise reports nothing:

$$3 \rightarrow 1 : \mathbf{E}_3, \quad 1 \rightarrow 2 : \mathbf{E}_1, \quad 3 \rightarrow 2 : \mathbf{E}_3, \quad 1 \rightarrow 3 : \mathbf{E}_1, \quad 2 \rightarrow 3 : \mathbf{E}_2.$$

The cold first encounter gives $(+, +, +)$. Round 2 gives $(-, +, +)$ and the permanently retained profile $(\{\mathbf{a}\}, \{\mathbf{E}_3\}, \{\mathbf{E}_1\})$. Thereafter each participant rejects exactly when the preceding participant in the ring $3 \rightarrow 1 \rightarrow 2 \rightarrow 3$ incorporates. The regimes therefore cycle

$$(-, +, +) \rightarrow (-, +, -) \rightarrow (+, +, -) \rightarrow (+, -, -) \rightarrow (+, -, +) \rightarrow (-, -, +) \rightarrow (-, +, +).$$

Starting instead with those settled understandings gives the cold response $(+, +, +)$ followed by $(-, -, -), (+, +, +), \dots$, of period two. With unit content values and $\lambda = 1$, participant 2's baseline score in the original process is 1 at every $H \geq 2$. Eventual cognitive constancy thus cannot replace fixedness throughout in Proposition A.6.

Proposition A.8 (A sharp recurrent bound with self-observation). *Under expansion in the autonomous protocol, recurrent orbits have constant understandings and period at most $3^{|I|}$. If self-observation is explicitly permitted, sufficiently rich finite fields and chosen thresholds allow per-edge reporting to realize any map of the ordinary regime profiles. In particular the period bound is attained from the cold start.*

Proof. The bound follows from the proof of Proposition 7.1. Fix a desired map $g : \{-, 0, +\}^I \rightarrow \{-, 0, +\}^I$ and set $\text{ord}(-) = 0$, $\text{ord}(0) = 1$, $\text{ord}(+) = 2$. For every recipient i and profile a , create $w_i(a) = 2 + \text{ord}(g_i(a))$ visible classes, indexed by (i, a, ℓ) . Put $c_i = \sum_a w_i(a)$. At one common stance containing all these classes, participant i holds $\mathbf{p}$ on precisely its own classes, admits nothing, and has thresholds $(c_i - 4, c_i - 3)$. Only $\mathbf{p}/\mathbf{q}$ conflict, and the repeated base is empty.

Include every edge $j \rightarrow i$, including $i \rightarrow i$. At an ordinary parent regime z_j , its report places $\mathbf{q}$ in class (i, a, ℓ) exactly when $z_j \neq a_j$; at the marker it reports nothing. Each edge reads only its own parent's regime. Union pooling conflicts with every held class except those indexed by the actual profile z . Consequently

$$d_i(z) = c_i - w_i(z) = c_i - 2 - \text{ord}(g_i(z)).$$

The thresholds produce exactly $g_i(z)$. They are nonnegative because $c_i \geq 2 \cdot 3^{|I|}$; empty filters keep cognition fixed. The direct empty encounter gives all plus, after which a ternary counter visits all $3^{|I|}$ profiles.

The construction uses at most $4|I|3^{|I|}$ visible classes; the counter uses $|I|3^{|I|+1}$. This is sharpness over sufficiently rich fields with self-observation and selected thresholds, not a claim for every fixed field or a loopless graph. No edge performs a joint-profile read. $\square$

Example A.9 (Recurring electoral phases under revision). *Take two source-neutral candidates and five unit-weight voters: drivers X, Y and three identical receivers. All use one common visible class and revision. Their primitives are*

	P^0	μ^*	α	(τ_1, τ_2)
X	$\{\mathbf{k}_X\}$	$\{\mathbf{k}_X\}$	$\emptyset$	$(0, 0)$
Y	$\{\mathbf{k}_Y\}$	$\{\mathbf{k}_Y\}$	$\emptyset$	$(0, 0)$
each receiver	$\emptyset$	$\{\mathbf{p}, \mathbf{z}\}$	$\{\mathbf{p}, \mathbf{q}, \mathbf{z}\}$	$(1, 1)$

Only $\mathbf{k}_X/\mathbf{O}_Y$, $\mathbf{k}_Y/\mathbf{E}_X$, $\mathbf{k}_Y/\mathbf{z}$, and $\mathbf{p}/\mathbf{q}$ conflict. Use $\lambda = \delta = 1$ and $\mathcal{W}(e, \mu) = 1 + \sum_{m \in \mu} w_m$ at the visible stance, with $\mathcal{W}(\perp) = 0$, $w_z = 1/4$, and all other meaning weights one. The menu is $x = \emptyset$, $y = \{\mathbf{z}\}$. The candidate's chosen speech is delivered every round.

With the protocol's one-round delay, X receives $\mathbf{O}_Y$ exactly when Y rejected; Y receives $\mathbf{E}_X$ exactly when X incorporated; each receiver receives $\mathbf{p}$ when X incorporated and $\mathbf{q}$ when X rejected. All other reports, including those at the marker, are empty. Every delivered receipt is coherent.

Under x , the drivers follow the four-cycle of Example 7.4. Receivers incorporate at every round and retain $\emptyset, \{\mathbf{p}\}, \{\mathbf{p}\}, \{\mathbf{q}\}, \{\mathbf{q}\}, \dots$. Their score is 1 at $H \geq 2$ with $H \bmod 4 \in \{2, 3\}$, and zero otherwise. Under y , Y rejects from round 1 and X from round 2. Receivers retain $\{\mathbf{z}\}$, then $\{\mathbf{z}, \mathbf{p}\}$, then $\{\mathbf{z}, \mathbf{q}\}$ permanently. Their scores are $1/2$ at $H = 1$, $3/2$ at $H = 2$, and $1/2$ thereafter. Only initial rejection is charged: X scores zero on both speeches, while Y scores zero on x and -1 on y .

For $H \geq 3$, the score table is

	x in high phase	x in low phase	y
X	0	0	0
Y	0	0	-1
each receiver	1	0	$1/2$

where the high phase has $H \bmod 4 \in \{2, 3\}$. Against y , x receives share 9/10 in the high phase and 3/10 in the low phase: X splits, Y supports x , and the three receivers switch. The unique pure equilibrium is therefore (x, x) at the high-phase horizons $H \geq 3$, and (y, y) otherwise, including $H = 1, 2$. The winning speech's symmetric profile admits no profitable deviation; every other pure profile allows a losing candidate to improve by choosing it.

The comparisons persist when $w_p > 2w_z > 0$ and $\delta_Y > 0$, with the other structure fixed. Changing only the receivers to expansion makes them retain p under x and both p, z under y from round 2. Their scores are then 1 and 3/2, respectively, at the stated weights; (y, y) is uniquely equilibrium at every horizon. These are recurring finite-horizon game comparisons, not a specified process of candidate adjustment.

A.7 Extensions.

Proof of Proposition 9.2. Suppose $u \preceq_{\mathcal{U}} v$. Intersecting their marked partitions with $S \cup \{\star\}$ preserves refinement. Every surviving visible fine class inherits its original meanings, including all those assigned to its surviving coarse ancestor. Hence $\rho_S(u) \preceq_{\mathcal{U}_S} \rho_S(v)$.

Invisible extension preserves and reflects refinement: the added resources belong to the marked class in both extended partitions. It also preserves and reflects the semantic inclusions, since visible classes and their meanings are unchanged. Thus ι_S is an order embedding. Its image consists exactly of understandings with visible scope contained in S , each of which is recovered by restriction followed by extension. The greatest such understanding is t_S . Conversely, an understanding below t_S has visible scope contained in S , since richness preserves visible-scope inclusion. The image is therefore $\downarrow t_S$. The ambient binary meet and join of two states in this ideal remain in the ideal. The order isomorphism onto it consequently preserves both operations.

Restriction after extension recovers every original class and meaning, giving $\rho_S \iota_S = \text{id}_{\mathcal{U}_S}$. For an original visible class C , applying the windows T and then S leaves $C \cap T \cap S$, when nonempty, with the same meanings. Every other resource is invisible. This proves $V_S V_T = V_{S \cap T}$ and the stated scope identity. Composition gives monotonicity, and $T = S$ gives idempotence. Every understanding supported on S is fixed; conversely, the scope identity shows that every fixed understanding is supported there. Monotonicity preserves an ordered pair under the same view.

For the category-boundary cautions, suppose a visible class C meets both S and its complement. Choose $a \in C \cap S$ and $b \in C \setminus S$. The view separates a and b , whereas the original partition does not; thus the latter cannot refine the former, ruling out $V_S(u) \preceq_{\mathcal{U}} u$. The lost visibility of b rules out the reverse order. More generally, $u \preceq_{\mathcal{U}} V_S(u)$ would force $\mathcal{R}_e \subseteq S$ and hence $V_S(u) = u$: no view is strictly richer than its original. If every visible class is retained or removed whole, the view merely merges the removed classes into the invisible class and preserves the others' meanings. It then lies below u .

The principal-ideal characterization gives

$$w \preceq_{\mathcal{U}} u, \quad \mathcal{R}_{e_w} \subseteq S \quad \Longleftrightarrow \quad w \preceq_{\mathcal{U}} u \wedge t_S.$$

Thus $u \wedge t_S$ is the greatest understanding below u supported on S . Its partition merges every cut class wholly into the invisible class and leaves every wholly retained visible class unchanged, with its original meanings.

Restriction need not preserve meets. On $\{a, b\}$, let u have marked partition $\{\{\star\}, \{a, b\}\}$ and let v have $\{\{\star, b\}, \{a\}\}$, with empty meanings on their visible classes. Their ambient meet has no visible resources. Restriction to $S = \{a\}$, however, makes both states equal with a visible, so their restricted meet differs from the restriction of their ambient meet. $\square$

Proof of Proposition 9.3. By Proposition 9.2, the visible scopes of w_1, w_2 are A_1, A_2 . Richness requires visible-scope inclusion. Under condition 1, each scope contains a resource absent from the other, ruling out both orders.

Under condition 2, $w_1 = u_1$. Choose $b \in \mathcal{R}_{e_1} \setminus S_2$. It is visible in w_1 and invisible in w_2 , so $w_1 \not\preceq_{\mathcal{U}} w_2$. If $w_2 \preceq_{\mathcal{U}} w_1$, monotonicity of restriction and its left-inverse identity would give

$$\rho_{S_2}(u_2) = \rho_{S_2}(\iota_{S_2}(\rho_{S_2}(u_2))) = \rho_{S_2}(w_2) \preceq_{\mathcal{U}_{S_2}} \rho_{S_2}(u_1).$$

This contradicts the stipulated strict comparison in the opposite direction. Neither order holds. $\square$